AWK51309.1
CP029329.1:2217467-2233261
Clostridium beijerinckii

ABC_tran PhoU PhoU PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C Pep_deformylase

CAG9706388.1
CAKJVE010000004.1:2289466-2305817
Clostridium neonatale

PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C Glycos_transf_2 ABC_membrane ABC_tran ABC_membrane ABC_tran SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C zf_Tbcl ? ? YicC-like_N YicC-like_C RemA-like Guanylate_kin

CCX89184.1
HF987799.1:64-5732
Clostridium sp. CAG:590

Aminotran_1_2 LysR_substrate HTH_1 ? Amidase_2 ? HTH_3 PBECR3 ?

CCY00660.1
HF988282.1:18800-28987
Enterocloster bolteae CAG:59

? ? HTH_23 Terminase_2 ? SPP1_portal ? Phage_min_cap2 PBECR2 Tad2-like ? ? ? ? ? Minor_capsid_2

CCZ32343.1
HF995324.1:347-11526
Firmicutes bacterium CAG:646

? ? ? TIR Terminase_2 ? SPP1_portal Phage_min_cap2 PBECR2 ? Tad2-like Tad2-like Phage_scaffold ? MRP-S32 ? ?

CDA20688.1
HF999648.1:111009-122160
Ruminococcus sp. CAG:488

? ParB_N DUF4417 DUF4998 ? DUF5681 Terminase_6N SPP1_portal Phage_min_cap2 PBECR2 Phage_scaffold ? ? DUF6751 Minor_capsid_2 ? ?

CDE35330.1
FR899377.1:27274-42277
Eubacterium sp. CAG:38

DHH DHHA1 CLP_protease Radical_SAM MFS_2 Alpha-amylase TPRs HAD_2 PBECR2 Anth_synt_I_N Chorismate_bind GATase IGPS PRAI PALP Trp_syntA ?

CDF02823.1
FR902322.1:71-3044
Ruminococcus sp. CAG:624

PBECR2 Phage_scaffold Phage_capsid_4 DnaT_2 ?

DAE03823.1
BK015374.1:25350-31074
Myoviridae sp. ct2Pw37

? ? ? HTH_3 SpoOE-like ? PBECR3 ? ? ? ? ? SH3_3 Peptidase_C39_2 ?

DAE10213.1
BK015505.1:20985-26206
Myoviridae sp. ctzS633

? ? ? ? ? ? HTH_19 PBECR3 ? ? PRA1 Phage_integrase ? ? ?

DAE24048.1
BK015766.1:12100-20805
Siphoviridae sp. ctg6Y13

dUTPase_2 DUF722 ? Terminase_2 Terminase_3 Terminase_3C SPP1_portal ? Phage_Mu_F PBECR3 ? DUF4355 ? ? ? HK97-gp10_like Phage_tail_terminator_8

DAF21030.1
BK056360.1:19850-26306
Caudoviricetes sp.

? ? ? ? HTH_3 ? BRCT PBECR3 ? HTH_26 Phage_int_M Phage_integrase ? DUF7841 ? ?

DAF83341.1
BK017323.1:23612-29726
Caudoviricetes sp.

? ? ? ? ? HTH_3 ? PBECR3 Phage_integrase ? DUF7841 ? ? ? DUF134

DAG66077.1
BK018475.1:8109-17452
Caudoviricetes sp.

DUF1492 ? DUF4417 ? GP3_package Terminase_6N Tli4_N Phage_min_cap2 PBECR2 ? ? ? YjcQ Phage_scaffold ? DnaT_2

DAH03191.1
BK016859.1:2802-12903
Caudoviricetes sp.

? Phage_terminase Terminase_2 Terminase_3 SPP1_portal FecR_C ? ? Phage_min_cap2 PBECR2 ? ? ? Cys_rich_KTR Phage_scaffold SU10_MCP ?

DAH75143.1
BK024095.1:34930-40499
Caudoviricetes sp.

? ? ? ? ? ? HTH_3 PBECR3 Peptidase_M78 ? ? DUF7479 ? Abi_2 ?

DAI28120.1
BK030401.1:45401-52637
Caudoviricetes sp.

? ? Intein_splicing SPP1_portal Phage_min_cap2 PBECR2 ? ? DUF4355 ? ? ? ?

DAI97639.1
BK020747.1:1-7191
Caudoviricetes sp.

? ? ? GP3_package Terminase_3 Terminase_3C SPP1_portal ? Phage_min_cap2 PBECR2 ? ? ?

DAJ77301.1
BK049708.1:1-10978
Caudoviricetes sp.

PhoH PTS-HPr PhageMin_Tail I-HmuI_NUMOD-like ? ? ? PBECR3 ? ? ?

DAK27804.1
BK035857.1:3072-11877
Caudoviricetes sp.

? ? ? ? Terminase_2 Terminase_6N ? Phage_Mu_F PBECR2 ? ? Peptidase_Prp DUF4355 ? ? Phage_connect_1

DAL11104.1
BK031346.1:1-7413
Caudoviricetes sp.

? ParB_N DUF4417 ? DUF8181 DUF5681 Terminase_6N SPP1_portal Phage_min_cap2 PBECR2 ? ?

DAL89400.1
BK021548.1:23695-28090
Caudoviricetes sp.

? HTH_26 UvsY ? HTH_3 Peptidase_S24 ? ? ? PBECR3 ? Phage_integrase ?

DAM26521.1
BK022286.1:17593-27958
Caudoviricetes sp.

Phage_int_SAM_4 Phage_integrase ? ? ? Phage_terminase Terminase_6N ? Phage_min_cap2 PBECR2 ? Cys_rich_KTR Phage_scaffold ? APG6_N ? DUF6751

DAM97219.1
BK053028.1:2573-13682
Caudoviricetes sp.

Phage_connect_1 MTES_1575 ? ? ? TerL_ATPase TerL_nuclease ? PBECR3 ? ? DUF4407 TMP_3 DUF948 ? Sipho_Gp37 YadA_head YadA_head

DAN18341.1
BK031041.1:4260-12738
Bacteriophage sp.

? HTH_Tnp_1 Terminase_6N Terminase_6C ? ? ? ? Phage_min_cap2 PBECR2 ? ? ? Phage_scaffold ? Phage_cap_E Rho_N

DAN52583.1
BK022948.1:1-8518
Caudoviricetes sp.

Sigma70_r4 ? Terminase_2 Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F ? DUF4355 ? ? ? HK97-gp10_like Phage_tail_terminator_8

DAO37410.1
BK049433.1:19231-38120
Caudoviricetes sp.

PhageMin_Tail Sipho_tail SPP1_Dit Collagen Collagen Collagen BppU-like_C Collagen Peptidase_S74 ? ? Phage_holin_5_2 Amidase_2 SH3_3 PG_binding_3 PBECR2 ? Zn_ribbon_recom Recombinase Resolvase ? Lipoprotein_Ltp Lipoprotein_Ltp Lipoprotein_Ltp ? ? ?

DAP13675.1
BK044082.1:12645-20931
Caudoviricetes sp.

? HVO_0416 ? Terminase_3 ? ? ? Phage_min_cap2 PBECR2 ? ? ? ? DUF6751 Minor_capsid_2 ?

DAP68345.1
BK045162.1:27178-39661
Caudoviricetes sp.

? Phage_TAC_12 ? bPH_3 SHOCT ? ? ? PBECR2 ? ? ? ? Peptidase_S74 ? ATG16

DAP86682.1
BK045547.1:35496-42137
Caudoviricetes sp.

LRR_EndoS ? ? ? ? ? ? PBECR3 ? ? SH3_3 Peptidase_C39_2 TRAP-gamma ? ? ?

DAQ07740.1
BK053671.1:12579-21815
Caudoviricetes sp.

? ? ? ? Phage_Mu_Gp27 Terminase_6N Portal_Mu Phage_Mu_F PBECR3 ? ? Mu-like_Pro ? Mu-like_gpT FluMu_N Phage_Mu_Gp36

DAQ65550.1
BK058182.1:24792-30825
Caudoviricetes sp.

? ? ? ? HTH_31 ? HTH_3 Peptidase_S24 PBECR3 ? ? Fic ? Phage_int_SAM_3 Phage_integrase ? ?

DAR05125.1
BK058944.1:10005-17607
Caudoviricetes sp.

HTH_3 ? ? ? Arm-DNA-bind_4 Phage_integrase ? ? PBECR3 ? SocA_Panacea RelA_SpoT Phage_r1t_holin XhlA Peptidase_M15_3 PG_binding_1 PG_binding_1 ?

DAR68663.1
BK033402.1:8843-19475
Caudoviricetes sp.

DUF722 SocA_Panacea ? ? ? Terminase_6N Terminase-T7_RNaseH-like SPP1_portal Phage_Mu_F PBECR2 ZnR PH_gly_zipper RHH_1 Bro-N ? Phage_scaffold ? ?

DAS95669.1
BK042654.1:16210-26101
Caudoviricetes sp.

Myb_DNA-binding DUF3310 ? DUF722 ? Terminase_2 Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F ? ? Phage_pRha ? DUF4355 ? ?

DAU06887.1
BK050479.1:29-8967
Caudoviricetes sp.

? ? ? Terminase_3C SPP1_portal Phage_min_cap2 ? PBECR2 ? ? ? ? ? ? ?

DAU25753.1
BK050820.1:1-4799
Caudoviricetes sp.

? Iron_transport ? HTH_3 HTH_26 PBECR3 ? HTH_26 ? ParE_toxin ? Resolvase Recombinase Zn_ribbon_recom Polyoma_agno

DAU67740.1
BK051618.1:31893-37302
Caudoviricetes sp.

? Collar Gp10_C ? Phage_holin_6_1 Glyco_hydro_25 LysM LysM ? ? PBECR3 ? ? ? ? ? ? ?

DAY42947.1
BK041568.1:21628-25745
Caudoviricetes sp.

? ? ? DUF948 ? Amidase_2 SH3_3 SH3_3 SH3_3 ? PBECR3 ? SpoOE-like ? ?

DAY78359.1
BK027715.1:2369-11102
Caudoviricetes sp.

? Head-tail_con ? ? ? HTH_3 ? PBECR3 ? ? Peptidase_M78 ? Phage_int_M Phage_integrase ? Phage_integrase

DAZ36894.1
BK026654.1:32867-37957
Caudoviricetes sp.

? ? ? ? ? ? ? PBECR3 ? ? DUF3800 ? ? Phage_int_SAM_3 Phage_integrase ?

DAZ41574.1
BK026799.1:994-8517
Caudoviricetes sp.

? Phage_integrase Phage_int_SAM_3 ? ? ? ? Arc PBECR3 ? DUF7836 Terminase_3C Terminase_3 UPF0253 ? DnaB_C DnaB DUF6291

EOS35936.1
KE159611.1:672748-684099
Lachnospiraceae bacterium M18-1

COesterase ? DDE_Tnp_1 DDE_Tnp_1 DUF3450 LIP-1 DDE_Tnp_IS66 PBECR2 Thioredoxin Peptidase_Prp ? ? ? UTRA GntR ?

ETP73164.1
ALYD01000006.1:66904-83124
Lachnospiraceae bacterium JC7

Esterase Esterase BPD_transp_1 BPD_transp_1 SBP_bac_8 His_kinase His-Kinase Receiver HTH_18 PBECR3 Creatininase ECF_trnsprt ? ? DUF2964 ? His-Kinase His-Kinase HAMP

GFI30871.1
BLME01000408.1:1329-7711
Lachnospiraceae bacterium

? ? ? Uma2 Gp49 HTH_3 HTH_3 PBECR3 PemK_toxin HTH_26 ? HicA_toxin HicB_lk_antitox

GHU90796.1
BNZO01000052.1:224-8192
Clostridia bacterium

Terminase_3 SPP1_portal Phage_min_cap2 PBECR3 Phage_scaffold Phage_capsid Topoisom_bac Toprim_Crpt ? Recombinase

GKH49619.1
BQNU01000001.1:709314-719816
Eubacteriales bacterium

? ? ? ? HTH_3 HTH_3 Peptidase_M78 PBECR3 Zn_Ribbon_1 Phage_integrase DUF2752 FakA-like_C FakA-like_M Dak2 Asp23 ? PseudoU_synth_2 Voltage_CLC

HAL02469.1
DMLF01000109.1:65-4753
Lachnospiraceae bacterium

PFK MFS_1 DUF1810 ? DUF3006 PBECR3

HAN03548.1
DMNU01000111.1:38-8928
Lachnospiraceae bacterium

N-OB tRNA-synt_2 ACPS ABC_membrane ABC_tran SMC_N ABC_membrane ABC_tran ? Glyco_hydro_3 PBECR2

HAS06684.1
DMVE01000014.1:80668-98355
Eubacterium sp.

Peptidase_A25 SpoIIP Peptidase_M20 DUF5688 S4 PseudoU_synth_2 Nlig-Ia DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_2 BRCT HTS PBECR3 ? ? Polysacc_synt Aminotran_5 Alpha-amylase Alpha-amylase_C_2 PHP Na_Pi_cotrans Na_Pi_cotrans PhoU PhoU

HBC84890.1
DNKG01000163.1:1-7187
Clostridiales bacterium

? UPF0102 PBECR2 Desulfoferrodox Methyltransf_31 NTPase_I-T DUF3242 MFS_1 MFS_1 ?

HBK60453.1
DNZO01000096.1:21-10986
Bacillota bacterium

? ? SdpI ? Methyltransf_11 Acetylase PBECR3 DUF2178 HTH_3 Polbeta ABC_tran ? ABC_tran NUDIX

HCG85223.1
DPGE01000009.1:5877-21399
Lachnospiraceae bacterium

Aldo_ket_red Fe-ADH ADH_Fe_C UvrD-helicase UvrD_C UvrD_C NUDIX DUF436 GntR Aminotran_1_2 ? PBECR2 NUDIX SpoU_methylase DUF1622 DegV Aldedh ADH_Fe_C Fe-ADH MarR_2

HCX63115.1
DPPW01000428.1:1-6136
Clostridiales bacterium

ABC_tran_CTD Abhydrolase_1 DNA_methylase RE_HaeIII PBECR3 ? DNA_methylase

HCY5602920.1
DAKVTN010000034.1:9994-18721
Staphylococcus aureus

? ? Cu_amine_oxidN1 DUF5704 ? HTH_36 HTH_54 PBECR3 ?

HDR7625183.1
DAOJEJ010000049.1:37-6834
Bacillus mycoides

? CC_CEP250 Mob_Pre PBECR3 ? ? DUF3967 DUF3967 MerR_1 HTH_3

HEL2011950.1
DARZHF010000059.1:243-3475
Streptococcus suis

Gp58 ? ? Phage_holin_4_1 CHAP SH3_5 PBECR3

HET7578783.1
DASSNY010000061.1:11550-26126
Bacillales bacterium

ACR_tran ACR_tran TetR_N Sigma70_r2 Sigma70_r4_2 ? Peptidase_C39_2 LplA-B_cat Isochorismatase PBECR3 Acetylase Zn_dep_PLPC HTH_DeoR DeoRC PP2C_2 adh_short ABC_tran UPF0014

HHZ06091.1
DURY01000020.1:5160-21172
Clostridiales bacterium

ABC-3 ABC_tran SecD_SecF_C SecD_SecF_C SecDF_P1_head PBECR2 Asp_aminotransf OAD_beta OAD_gamma DUF6017 RepA_N SipL SipL LysM SpoIVA_ATPase SpoIVA_middle SpoIVA_C PG_binding_1 PG_binding_1 PG_binding_1 NLPC_P60

HIQ82410.1
DVFZ01000049.1:52-8766
Candidatus Pullichristensenella stercorigallinarum

Mac Hexapep Peptidase_S24 HTH_3 HTH_31 DUF722 HTH_Tnp_1 Terminase_6N Terminase_3C SPP1_portal PBECR2 Phage_scaffold ?

HIR59746.1
DVHC01000064.1:1-17476
Candidatus Onthousia excrementipullorum

SWIM SNF2_assoc SNF2-rel_dom Helicase_C LMWPc HATPase_c_3 HSP90 HSP90 Sulfatase DUF6057 AbiEii HTH_Tnp_4 DDE_Tnp_4 PBECR3 ABC_tran ABC_membrane AA_kinase PPDK_N PPDK_N PPDK_N PEP-utilizers PEP-utilizers_C ? ? DUF8429 Lipase_GDSL_2

HIT10037.1
DVKE01000007.1:263-13367

Candidatus Onthousia faecigallinarum

Cation_ATPase_N E1-E2_ATPase Cation_ATPase Hydrolase_3 Cation_ATPase_C DUF6056 E1-E2_ATPase Hydrolase ? PBECR3 S4 PseudoU_synth_2 ATP_bind_3 Sigma70_r2 Sigma70_r4 PRC Spore_YunB YutD-like DUF1292

HIU11195.1
DVMD01000013.1:1-6836

Candidatus Onthocola stercorigallinarum

Peptidase_C39 ABC_membrane ABC_tran ? Bacteriocin_IIc ? ? Ribosomal_S21 YqeY PBECR3 DUF1016_N DUF1016_N YhcG_C ? YabP YqfD

HIW31194.1
DXHF01000009.1:1-5718
Candidatus Paenibacillus intestinavium

PCRF RF-1 PrmC_N MTS Spore_II_R PBECR3 Sua5_yciO_yrdC Sua5_C Mntp

HIX67722.1
DXEM01000019.1:21937-34463
Candidatus Anaerostipes excrementavium

ATP-synt_DE_N ATP-synt_DE Uma2 CYTH Methyltransf_4 Thioredoxin ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_CTD AI-2E_transport PBECR2 ? DUF1848 CN_hydrolase DUF554 C_GCAxxG_C_C MecA_N PBP_dimer Transpeptidase DUF6070

HJA81020.1
DWYM01000045.1:13354-24057
Candidatus Mediterraneibacter intestinipullorum

? MethyltransfD12 ? Sigma70_r4 Terminase_2 Terminase_6N Terminase-T7_RNaseH-like SPP1_portal Phage_Mu_F PBECR2 ? ? Phage_scaffold Phage_capsid_4 Phage_connect_1 Phage_HCP HK97-gp10_like

HJB89216.1
DWXL01000185.1:1-7793

Candidatus Blautia excrementigallinarum

Phage_integrase PBECR3 AAA_14 DUF4143 ? DDE_Tnp_1

HJB95510.1
DWXI01000135.1:1-9645

Candidatus Mediterraneibacter intestinigallinarum

? ? ? PBECR2 RHH_10 ORF6N NMT1 Lar_restr_allev

HJJ11749.1
DAJPHB010000001.1:57070-74324
Clostridiaceae bacterium

? TMP_3 CsmB ? ? ? Gp10_C Phage_holin_4_1 PBECR2 Amidase_2 ? ? SecB ? WHD_AraR SpaA SpaA MucBP DUF11

HNX63936.1
DAOPYP010000011.1:161-10582
Oscillospiraceae bacterium

AA_kinase ACT_9 Acetylase HI0933_like Pyr_redox_2 FAD-depend_C PBECR3 Collar Gp10_C ? Glucosaminidase LysM ? ? MCPsignal FeS Fe_hyd_lg_C Fer4_7 sCache_3_3 MCPsignal

HWL23356.1
DATMFE010000106.1:42289-53937
Ureibacillus sp.

? Cu_amine_oxidN1 ParA-Soj-NTPase DUF5388 ? HTH_36 PBECR3 ? Phage_int_SAM_2 PriCT_1 ? MerR_1 DUF3967 Zn_ribbon_Elf1 ?

HYF82483.1
DASPIR010000018.1:455-7499
Clostridia bacterium

PALP Hydrolase_2 ? ? ? PBECR3 ? ? DNA_alkylation ?

KOS61578.1
LIYL01000044.1:73636-97039
Lysinibacillus sp. FJAT-14222

? ? DUF6471 DUF4969 ? PemK_toxin CCDC93_CC PhageMin_Tail PBECR3 PhageMin_Tail Peptidase_M23 Phage_lysozyme2 DUF6558 ? DUF6558_C YOMG_N Prophage_tail_N2 Prophage_tail ?

KXB45916.1
KQ959644.1:378-11396
Tissierellia bacterium KA00581

DUF6892 LRRs LRRs Oxidored_FMN NMT1 NMT1 NMT1_2 BPD_transp_1 ABC_tran PBECR3 Adenyl_transf DUF2179 YitT_membrane

KXB64342.1
KQ959917.1:28982-41869
Gemella haemolysans

ABC_membrane ABC_tran ? ABC_membrane ABC_tran ABC_membrane ABC_tran Herpes_UL92 TetR_N ABC2_membrane_3 ? PBECR3 MepB-like PspC PhdYeFM_antitox ? Fic NanE UTRA GntR

KXH86914.1
LPUT01000003.1:24527-37427
Sporosarcina sp. HYO08

? ? DUF7010 ? DUF188 Chal_sti_synt_N Chal_sti_synt_C ? PBECR3 Na_H_antiporter Na_H_antiporter Excalibur SNase CRCB CRCB UvrD_C UvrD-helicase Peptidase_S26 Hydrolase_4

MAO91979.1
NZSA01000053.1:1179-16599
Rhodospirillales bacterium

Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Acyl-CoA_dh_C PBP_like HTH_17 BPL_LplA_LipB_2 DUF6352 DUF6494 ? Na_H_Exchanger CorC_HlyC PBECR3 B12-binding MM_CoA_mutase PDH_N TYRAAT1_C ? Sigma70_r4_2 PhyR_sigma2 DUF4168 Amidase PALP

MBC5723091.1
JACOPO010000006.1:6168-17438
Flintibacter hominis

? ? ? HTH_23 ? DUF739 HTH_3 Peptidase_S24 PBECR3 ? Bro-N Phage_int_SAM_3 Phage_integrase CitG CitX LysR_substrate HTH_1 ? ACP

MBC9710089.1
JACUTJ010000495.1:813-5871
Enterococcus sp.

MerR_1 DUF8221 CdiI_4 ? SUKH_5 DUF2207_C Phage_integrase HTH_36 PBECR3

MBD5082256.1
JAAVIE010000141.1:2644-15927
Oscillospiraceae bacterium

DUF2705 VanY DUF2705 ABC_tran DUF6985 Receiver Trans_reg_C His-Kinase His-Kinase PBECR2 MFS_1 DUF4284 ? ? HTH_3 ? GyrI-like HTH_18

MBD5103640.1
JAAVIG010000039.1:1-6237
Oscillospiraceae bacterium

S-AdoMet_synt_C S-AdoMet_synt_M S-AdoMet_synt_N PBECR2 Mg_chelatase_C Mg_chelatase ChlI PRK HAD_2 MCPsignal

MBD5104791.1
JAAVIG010000166.1:175-5772
Oscillospiraceae bacterium

PBECR2 EAL Peptidase_M16 Peptidase_M16_C M16C_assoc PreP_C

MBD5114673.1
JAAVIF010000031.1:131-24149
Oscillospiraceae bacterium

PDDEXK_1 UvrD_C UvrD-helicase PDDEXK_1 PDDEXK_1 UvrD_C ADDB_N Sulfatase ABC2_membrane_5 ABC_tran GntR S-AdoMet_synt_C S-AdoMet_synt_M S-AdoMet_synt_N PBECR2 Mg_chelatase_C Mg_chelatase ChlI PRK HAD_2 MCPsignal HAMP_2 HAMP dCache_1 VTC DUF4956 Gal_mutarotas_2 Glyco_hydro_31_2nd Glyco_hydro_31_3rd

MBD5138874.1
JAAVHL010000074.1:557-10369
Ruminococcus sp.

HEPN BofA DUF2508 ? ? ? Metallophos_2 PBECR2 ? Rotavirus_VP3 Lipase_GDSL_2 Glyco_hydro_3 GH94_b-supersand Glyco_hydro_94 Glyco_hydro_65C

MBD5155524.1
JAAVHD010000065.1:20117-35043
Oscillibacter sp.

PhoH YbeY AAA_18 MMR_HSR1 MMR_HSR1_Xtn KH_2 Acetylase Acetylase RecO_N RecO_C PBECR2 Acetylase MutS_V Smr PTS-HPr Amidohydro_2 TraX GIY-YIG UvrC_M UVR UvrC_RNaseH UvrC_RNaseH_dom HHH_5 MFS_1

MBD5457361.1
JAAVAH010000055.1:254-8690
Lachnospiraceae bacterium

ABC2_membrane_6 ABC_tran Cys_rich_CPCC SCP2 FtsX MacB_PCD FtsX ABC_tran ? PBECR3 ?

MBD5477283.1
JAAUZX010000079.1:3830-9154
Lachnospiraceae bacterium

Mntp ? ? ? ? ? RelB PBECR2

MBD5479671.1
JAAUZU010000020.1:21959-42954
Lachnospiraceae bacterium

tRNA_Me_trans tRNA_Me_trans_M tRNA_Me_trans_C ? ? SBP_bac_8 SBP_bac_8 tRNA_U5-meth_tr NUDIX PBECR3 ? Peptidase_M20 MatE MatE GFO_IDH_MocA_C3 GFO_IDH_MocA AraC_binding HTH_18 Transglutaminase TgpA_N vWA

MBD5493959.1
JAAUZS010000053.1:11387-29548
Lachnospiraceae bacterium

ABC2_membrane_4 AraC_binding HTH_18 ? ? BD-FAE DUF5839 HTH_3 PBECR3 FliG_C HTH_3 ? Methyltransf_11 DUF6179 ? ? MerR_1

MBD5546295.1
JAAUZD010000008.1:17660-34242
Lachnospiraceae bacterium

Trans_reg_C Receiver SseB SseB_C SUFU MgtC HTH_3 FtsX FtsX ABC_tran PBECR3 His-Kinase His-Kinase Trans_reg_C Receiver Aldo_ket_red ? PPC Beta-lactamase zf-HC2

MBE3572391.1
JADBKD010000035.1:383-8705
Moorella humiferrea

? Methyltr_RsmB-F RsmF-B_ferredox NusB DUF116 Formyl_trans_C Formyl_trans_N Pep_deformylase PriA_C Helicase_C Zn_ribbon_PriA DEAD PriA_3primeBD Cu_amine_oxidN1 PBECR3

MBE5821157.1
SVGE01000001.1:188472-203867
Clostridiales bacterium

FeoC PseudoU_synth_2 Peptidase_S11 PBP5_C ? Peptidase_U32 DUF3656 Peptidase_U32 DUF386 Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C PBECR3 ? DUF713 ? tRNA-synt_1c Anticodon_2 SPASM Radical_SAM ? RmuC HN

MBE5881216.1
SVFI01000003.1:99856-110757
Lachnospiraceae bacterium

DUF1349 ? ? ? OTCace OTCace_N Acetylase ICMT PBECR3 Peptidase_S41 Tricorn_C1 DUF2705 ABC_tran Arc ? DUF739 HTH_3

MBE6074202.1
SVBZ01000023.1:34284-45849
Selenomonas ruminantium

Terminase_2 DUF2188 ? Terminase_3 Terminase_3C SPP1_portal Phage_scaffold Gp13-like Phage_Mu_F PBECR2 YjcQ Phage_connect_1 ? HK97-gp10_like Phage_tail_terminator_8 ? Gp18_domIII_N Phage_sheath_domII Phage_sheath_1 Phage_sheath_1C

MBE6919864.1
SVMH01000071.1:1-4706
Oscillospiraceae bacterium

PBECR2 AzlD AzlC Ribosomal_L32p YceD CYTH SpoIIE_N

MBE6946072.1
SVLX01000039.1:371-16785
Oscillospiraceae bacterium

TruB_N TruB_C_2 FAD_syn Flavokinase ? F_bP_aldolase PFK PBECR3 Arc Chromate_transp Chromate_transp SLH SLH SLH fn3_5 Peptidase_S8 MarR tRNA-synt_2 N-OB HemN_C Radical_SAM

MBE6951710.1
SVLP01000041.1:11131-20324
Oscillospiraceae bacterium

HTH_18 UPRTase TsaD TsaE FtsH_ext AAA AAA_lid_3 Peptidase_M41 PHP NUDIX_4 HhH-GPD PBECR3

MBE7063781.1
SVJG01000109.1:1-6430
Oscillospiraceae bacterium

PIG-L ? Arabinose_Iso_C Lyx_isomer Peripla_BP_3 LacI PBECR2 rRNA_methylase

MBK0348750.1
JAEHNW010000061.1:146-9288
Tuanshanicoccus yangjingiae

? Phage_holin_5_2 PBECR3 Glucosaminidase Amidase_5 SH3_5 Fic_N Fic HTH_22 PAS Ribonuc_L-PSP AA_permease_2 Aminotran_1_2 DUF1706

MBO2526710.1
PITX01000010.1:1674-14960
Clostridiales bacterium

Extensin_1 Zn_ribbon_3 FakA-like_C FakA-like_M Dak2 Asp23 Metalloenzyme PBECR2 C_GCAxxG_C_C ? DegV HSP20 SMC_N ABC_tran ABC_membrane ParA ParA-Soj-NTPase Wzz

MBO4510898.1
JAFZZF010000262.1:655-3345
Lachnospiraceae bacterium

PBECR3 WHD_HrcA HrcA GrpE

MBO4609686.1
JAFZWW010000104.1:9416-23657
Lachnospiraceae bacterium

HTH_18 Receiver His-Kinase His_kinase HAMP ? DUF2809 SpoU_methylase Flavodoxin_4 ? PBECR3 TPRs VWA_2 VWA_2 ? vWA AAA_lid_2 AAA_3 Hpt Receiver His-Kinase His-Kinase

MBO5005577.1
JAGATI010000197.1:106-8511
Clostridia bacterium

? PBECR3 Enolase_N Enolase_C ? YlmH_1st YlmH_RBD MTRES1_C THF_DHG_CYH THF_DHG_CYH_C DNA_processg_A DNA_processg_A

MBO5095532.1
JAGAQR010000010.1:69-8305
Bacilli bacterium

SEC-C ? Arm-DNA-bind_4 Phage_int_SAM_3 Phage_integrase PBECR2 SEC-C ? ABC_tran MacB_PCD FtsX ? SLH SLH

MBO5167691.1
JAGAPX010000005.1:58097-70776
Phascolarctobacterium sp.

LysM ? DUF3846 ? ? HTH_26 HTH_3 PBECR3 Resolvase Recombinase Zn_ribbon_recom PGI CitMHS SUFBD_core SufBD_N ABC_tran LGT CoA_transf_3

MBO5207769.1
JAGAPK010000003.1:31932-44663
Lachnospiraceae bacterium

BPD_transp_1 Glyco_hydro_77 LacI Peripla_BP_3 CYTH ? EamA LOR PBECR2 Adenyl_transf DJ-1_PfpI SASP NMT1 BPD_transp_1 ABC_tran TM1266-like Trigger_N FKBP_C Trigger_C

MBO5282380.1
JAGANP010000122.1:7317-15592
Lachnospiraceae bacterium

PBECR2 MecA Wzy_C AI-2E_transport PseudoU_synth_2 DegV Trans_reg_C Receiver His-Kinase His-Kinase HAMP Sigma_reg_N

MBO5336239.1
JAGAMG010000019.1:21461-36558
Lachnospiraceae bacterium

? Aminotran_3 AA_kinase ? ArgJ Acetylase Semialdhyde_dhC_1 Semialdhyde_dh PBECR3 Arginosuc_synth Arginosuc_syn_C DUF1294 Receiver MBG_2 YDG MBG_2 MBG_2 MBG fn3

MBO5605432.1
JAGAGI010000401.1:144-2218
Acidaminococcaceae bacterium

HTH_3 Peptidase_M78 PBECR3 Arm-DNA-bind_4

MBO6161860.1
JAGBNH010000020.1:13714-20745
Eubacterium sp.

HTH_3 Peptidase_S24 DUF6472 Asp_Glu_race DUF362 Phage_integrase Phage_int_SAM_5 HTH_17 HTH_3 PBECR3 YycI

MBO6178340.1
JAGBMX010000051.1:224-11173
Selenomonadaceae bacterium

PGI ? ACR_tran YknX_C Beta-barrel_RND_2 BSH_RND PBECR2 PBECR3 ? Lactamase_B Autotransporter

MBO6178607.1
JAGBMX010000066.1:35850-53516
Selenomonadaceae bacterium

AdoHcyase_NAD AdoHcyase Lysine_decarbox His-Kinase SpoIIE HAMP dCache_1 Asparaginase_C Asparaginase DUF4127 DUF2225 CbiK PBECR2 ? DUF4160 SSF Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C DUF6384 ? ?

MBO6178750.1
JAGBMX010000077.1:71-12496
Selenomonadaceae bacterium

PBECR2 MmeI_C MmeI_TRD MmeI_Mtase MmeI_hel MmeI_N Sirtuin Na_Ala_symp ? Sigma70_r2 Sigma70_r4_2 Pyruv_OxRed_insertion Fer4_16 Fer4_7 EKR POR PFOR_II POR_N

MBO6243721.1
JAGBLS010000364.1:216-1518
Clostridia bacterium

Radical_SAM_C Radical_SAM PBECR3

MBO6308106.1
JAGBLB010000103.1:191-3973
Oribacterium sp.

TnpB_IS66 PBECR3 rve Mu-transpos_C_2 IstB_IS21

MBO8128240.1
JACDOK010000005.1:54102-67006
Peptococcaceae bacterium

DUF192 ? Receiver CbiA MqsR_toxin MqsA_antitoxin SocA_Panacea Cu_amine_oxidN1 PQQ_2 PBECR3 ? T2SSE T2SSF ? ? ? ?

MBP3330577.1
JAGBHV010000075.1:207-4488
Clostridia bacterium

? CCDC89 ? Lar_restr_allev ? ? ? PBECR2

MBP3502613.1
JAGBDU010000024.1:4414-18865
Clostridia bacterium

? DUF3560 ? S-AdoMet_synt_M S-AdoMet_synt_N ? tRNA_U5-meth_tr TRAM Hydrolase PBECR3 Acyl-ACP_TE_C Acyl-ACP_TE ABC_tran ABC-ATPase SUFBD_core Fe_hyd_SSU Fe_hyd_lg_C Fer4 Fer4_6 Rubredoxin HHH_5 UvrC_RNaseH_dom UvrC_RNaseH UVR UvrC_M GIY-YIG DUF5684 Transglutaminase

MBP3506742.1
JAGBDX010000107.1:61540-79280
Lachnospiraceae bacterium

PP_kinase_C PP_kinase_C_1 PP_kinase PP_kinase_N HD Abi_2 DUF2383 Flagellin_C Flagellin_N HlyIII Myo5a HTH_23 PBECR3 CN_hydrolase DUF6240 DUF6240 Band-7 Flot Peptidase_M23 G5 LysM SKI Flavin_Reduct Big

MBP3568097.1
JAGBCP010000028.1:1-6060
Lachnospiraceae bacterium

GH94_b-supersand MFS_2 ? PBECR2 ? PEPCK_ATP

MBP3616976.1
JAGBBP010000014.1:195514-206273
Lachnospiraceae bacterium

Isochorismatase P5CR_dimer F420_oxidored TraX Flavin_Reduct MarR AlkZ-rel Lactamase_B PBECR3 NUDIX Acetylase ? ? ABC2_membrane_3 ABC_tran ?

MBP3691077.1
JAGAZS010000280.1:1-1411
Schwartzia sp. (in: firmicutes)

SBP_bac_3 DUF3781 PBECR3

MBP3798018.1
JAGAXR010000069.1:655-11151
Ruminococcus sp.

? ? ? ? ? HNH Terminase_4 PBECR2 TerL_ATPase TerL_nuclease Phage_portal Ntox50 ? ? Peptidase_S78 Phage_capsid ?

MBP3907973.1
JAGAVT010000142.1:237-7903
Turicibacter sp.

Aldose_epim PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV ? ? DUF6693 ? NA37 PBECR3

MBP5309196.1
JAGCZV010000043.1:1-7264
Lachnospiraceae bacterium

GTP_EFTU MgtE CBS MgtE_N Transgly Transpeptidase Zn_Ribbon_1 LEA_4 Phage_integrase PBECR3 EAL

MBP6890204.1
JAGOPJ010000006.1:34942-46407
Veillonella sp.

PBECR3 DHH DHHA2 UvrD-helicase UvrD_C PcrA_UvrD_tudor Nlig-Ia DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_5 HHH_2 BRCT GatC Amidase GatB_N GatB_Yqey ?

MBQ0014965.1
JAGHVH010000025.1:7967-19274

Candidatus Limimonas egerieequi

ParE_toxin PhdYeFM_antitox PNPOx_N PBECR3 MshEN T2SSE Transgly Transpeptidase RuvX UvrD-helicase UvrD_C PcrA_UvrD_tudor FTSW_RODA_SPOVE TsaE TsaD

MBQ1274232.1
JAFOHG010000028.1:266-12221
Cellulosilyticum sp.

Trans_reg_C Receiver DUF2178 HTH_3 PAP2 HTH_3 Acetylase WD40_RLD WD40_RLD PBECR3 Abi_2 GmrSD_C GmrSD_N Cys_Met_Meta_PP

MBQ1658708.1
JAFOQX010000121.1:1-8151
Clostridia bacterium

RepA_N PBECR3 AAA_14 DUF4143 Mob_Pre HNH GmrSD_N FAM76 RepL ?

MBQ1954683.1
JAFPCC010000147.1:1130-8382
Clostridia bacterium

PBECR3 BNR_2 BNR_2 ? Ribosomal_L28 DapB_C DapB_N DHDPS Semialdhyde_dhC Semialdhyde_dh

MBQ2741217.1
JAFQVE010000008.1:58-9893
Oscillospiraceae bacterium

Eco57I MTS PrmC_N DUF1385 His-Kinase His-Kinase HAMP Cache_WalK Trans_reg_C Receiver ? TspO_MBR PBECR3 ParG HTH_3 YhcG_C DUF1016_N NTP_transf_2 UxaC

MBQ2777045.1
JAFQVW010000338.1:43-3578
Peptococcaceae bacterium

Ribosomal_S21 DcpS_C TRAM Radical_SAM UPF0004 PBECR2 DUF5105

MBQ2986441.1
JAFQRJ010000013.1:2309-17019
Tyzzerella sp.

adh_short_C2 KduI Glyco_hydro_88 Hepar_II_III_C DUF624 Bac_rhamnosid6H Bac_rhamnosid PHP PBECR3 ATPase_2 ? RelA_SpoT His-Kinase His-Kinase Trans_reg_C Receiver PTS_EIIC_2 ?

MBQ2986907.1
JAFQRJ010000021.1:259-10264
Tyzzerella sp.

Pept_tRNA_hydro UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF SurA_N_3 Rotamase_2 YqeY PTAC adh_short Nucleos_tra2_C Gate Nucleos_tra2_N PBECR3

MBQ2994319.1
JAFQOU010000096.1:337-2963
Peptococcaceae bacterium

Ribosomal_S21 ? TRAM Radical_SAM UPF0004 PBECR2

MBQ2994511.1
JAFQOU010000143.1:1-5813
Peptococcaceae bacterium

Peptidase_M56 PBECR3 Peptidase_M56 ?

MBQ3057547.1
JAFQMQ010000007.1:40900-47957
Clostridia bacterium

PYC_OADA HMGL-like OAD_beta Biotin_lipoyl OAD_gamma CotJA CotJB Mn_catalase PBECR3 Nitroreductase Nitroreductase ArsC

MBQ3145364.1
JAFQIK010000018.1:2655-18210
Clostridia bacterium

? PALP Acetylase Acetylase Acetylase Rubrerythrin VIT1 VIT1 Radical_SAM_C Radical_SAM DNA_processg_A WHD_DprA PBECR2 YlmH_RBD S4 MreD ? PBP_dimer Transpeptidase Transpeptidase MinC_C ?

MBQ3510555.1
JAFQEO010000039.1:4620-19417
Peptococcaceae bacterium

? LiaI-LiaF-TM_PspC EamA EamA Transketolase_N Transket_pyr Transketolase_C ? ? PBECR2 Cthe_2159 GDP_Man_Dehyd PGI MFS_1 MFS_1 HTH_3 LRR_EndoS LRRs

MBQ3540505.1
JAFQCQ010000066.1:1-8544
Phascolarctobacterium sp.

TonB_dep_Rec_b-barrel Plug Pyridox_ox_2 cNMPBD HTH_Crp_2 Sulf_transp ? Calcineurin PBECR3 ? ?

MBQ3686569.1
JAFPYM010000123.1:1-1687
Treponema sp.

AglB_L1 FGS PBECR3

MBQ4235909.1
JAFPQE010000128.1:183-4228
Treponema sp.

DUF3800 Peptidase_M20 PBECR3 DUF2141 Glycos_transf_4

MBQ4504195.1
JAFQAL010000059.1:1-3077
Bacillota bacterium

Fer4_10 HD PolyA_pol_RNAbd PolyA_pol PBECR2 MFS_1

MBQ4606145.1
JAFQFN010000369.1:73-6353
Clostridia bacterium

PBECR2 4HBT DegV Dak2 FakA-like_M FakA-like_C Hydrolase AAA_18 ? Ribosomal_S9

MBQ5588019.1
JAFNUV010000234.1:5042-8606
Selenomonadales bacterium

? ? ? ? HTH_3 HTH_3 ? PBECR3

MBQ5673995.1
JAFNXD010000168.1:1-2050
Lachnospiraceae bacterium

CxxCxxCC PBECR3 tRNA-synt_2c tRNA_SAD

MBQ5908187.1
JAFOCI010000078.1:58846-75205
Treponema sp.

? HI_1054_N ResIII YafQ_toxin RelB ? PBECR3 ? DUF2798 JetA DUF4194 AAA_29 SbcC_Walker_B JetD_C CDP-OH_P_transf

MBQ6347690.1
JAFRJF010000192.1:6-8856
Clostridia bacterium

? ? Sigma70_r4_2 ? Terminase_3 Terminase_3C SPP1_portal Phage_min_cap2 PBECR2 DUF2188 ?

MBQ6752310.1
JAFSPP010000011.1:3262-18832
Clostridia bacterium

Iso_dh Aconitase_C Aconitase BacA FtsK_4TM FtsK_alpha FtsK_SpoIIIE FtsK_gamma Lactamase_B Ribosomal_S2 EF-Ts_N EF_TS PBECR3 Rce1-like ? Wzy_C PBP CitMHS Gp49 HTH_3

MBQ6786451.1
JAFSGY010000027.1:1-6904
Lachnospiraceae bacterium

Glyco_hydro_3 PBECR2 MarR PNPOx_N Abhydrolase_1 PNPOx_N HTH_3 Methyltransf_25 ? ABC_tran ABC_tran_Xtn ABC_tran

MBQ6900977.1
JAFSCP010000045.1:69167-84491
Bacillota bacterium

? BPL_C BPL_LplA_LipB HTH_11 CNNM CBS CBS CorC_HlyC SecA_DEAD P-loop_SecA SecA_SW SEC-C MatE MatE B3_4 WCX PBECR2 HTH_3 DUF6442 ABC2_membrane_5 ABC_tran GntR AA_permease_2 GntR

MBQ7031163.1
JAFSAI010000015.1:22332-34939
Bacilli bacterium

GATase_6 SIS SIS ? MMR_HSR1 N_methyl MMR_HSR1 MMR_HSR1 KH_dom-like ? ? PBECR3 Zip Zip N_methyl N_methyl N_methyl N_methyl AP_endonuc_2 ?

MBQ7046859.1
JAFSDC010000152.1:1-2657
Oscillospiraceae bacterium

Phage_Mu_F PBECR2 Phage_scaffold ?

MBQ7428633.1
JAFTFS010000014.1:30144-37828
Butyrivibrio sp.

Methyltransf_11 HTH_3 ? PBECR3 EAV_GS MerR_1 CheW CheW Lipase_GDSL_2 His_Phos_1 PTS_EIIC_2

MBQ7595311.1
JAFTCM010000061.1:1-2692
Clostridia bacterium

? Phage_min_cap2 PBECR2

MBQ7838202.1
JAFUNA010000024.1:1-10874
Clostridia bacterium

? Bps Sigma70_r4_2 Sigma70_r2 Polysacc_deac_1 HTH_18 MFS_2 Glyco_hydro_2_N2 Glyco_hydro_2 Glyco_hydro_2_C DUF4982 PBECR2 ?

MBQ7863415.1
JAFULK010000031.1:1-13007
Lachnospiraceae bacterium

PEP-utilizers_C CBM30 CelD_N Glyco_hydro_9 Metallophos_2 PBECR3 Zein-binding HTH_7 ? ? ? ? HTH_18 AraC_binding Pectinesterase LysM

MBQ7889337.1
JAFULN010000028.1:166-13271
Erysipelotrichaceae bacterium

TM1586_NiRdase ? HTH_3 MatE MatE DAP_DH_C GFO_IDH_MocA AraC_binding HTH_18 Acetyltransf_10 PBECR3 FMN_bind FAD_binding_2 TetR_N EndIII_4Fe-2S HhH-GPD DnaB_2 DnaD_N EcdD_BsdD_detox PedC-like

MBQ7890575.1
JAFULQ010000001.1:49186-63668
Erysipelotrichaceae bacterium

Flavin_Reduct Peptidase_M3 Pep_deformylase ? ABC_tran Acetylase Acetylase PBECR3 ABC_tran ABC_membrane ABC_tran ABC_membrane MarR_2 GyrI-like PmbA_TldD_3rd PmbA_TldD_3rd P-II

MBQ7895260.1
JAFULV010000080.1:1-2014
Oscillospiraceae bacterium

DUF4379 MTES_1575 HTH_26 PBECR3 CNP_C_terminal

MBQ7962372.1
JAFUIY010000092.1:4057-16880
Clostridia bacterium

Cass2 HTH_18 AAA_18 Acetylase Acetylase Acetylase NUDIX Phage_int_SAM_1 PBECR3 ? Phage_integrase Phage_int_SAM_3 AAA_33 Calcineurin Acetylase Glyco_hydro_39 Acetylase

MBQ8117744.1
JAFUDM010000029.1:3174-15882
Lachnospiraceae bacterium

MatE MatE ? ? ? TnpV PBECR2 ? DUF6020 Metallophos_2 Ham1p_like UPF0020 THUMP RlmL_1st ? Band_7_1

MBQ8354212.1
JAFTRR010000012.1:34152-47311
Oscillospiraceae bacterium

Semialdhyde_dhC Semialdhyde_dh ACT_9 AA_kinase Homoserine_dh NAD_binding_3 Thr_synth_N THR4_C GHMP_kinases_N GHMP_kinases_C ? HTH_3 PBECR2 MatE MatE VanZ TK ? DeoC 23S_rRNA_IVP PNPase

MBQ8510722.1
JAFUQU010000045.1:38787-52291
Clostridia bacterium

PGA_cap Mem_trans Toprim Topoisom_bac Toprim_Crpt FAD_oxidored FAD_oxidored Amidase_3 HAD_2 BD-FAE PBECR3 His-Kinase His-Kinase ? HTH_3 ? YchF-GTPase_C MMR_HSR1 Phage_integrase Tn916-Xis

MBQ8518442.1
JAFUPI010000021.1:45149-57008
Agathobacter sp.

? ? HTH_3 HTH_3 MazE_antitoxin EryCIII-like_N UDPGT PBECR3 TGS tRNA_SAD tRNA-synt_2b HGTP_anticodon Acetyltransf_9 LPG_synthase_C DUF1540 DUF1540 LGT SpoOE-like MraZ MraZ Methyltransf_5

MBQ8538342.1
JAFUOM010000022.1:1194-20325
Ruminococcus sp.

DUF6062 PALP Zn_ribbon_3 IreB Mntp DUF554 GATase_7 Asn_synthase PBECR3 Peptidase_M14 FAD_syn Flavokinase Bac_transf ? LytR_cpsA_psr ? ParA-Soj-NTPase Wzz

MBQ8616194.1
JAFUNX010000015.1:5953-11643
Clostridia bacterium

Phage_holin_6_1 ? ? ? ArdcN ? Arc PBECR3 N6_N4_Mtase N6_N4_Mtase Sigma70_r4_2 ? ? ? ? ?

MBQ8616248.1
JAFUNX010000015.1:39721-42920
Clostridia bacterium

Phage_holin_6_1 ? ? ? ArdcN ? Arc PBECR3 ?

MBQ8688012.1
JAFURN010000097.1:1-11881
Ruminococcus sp.

Thioredoxin Tad2-like ? Phage_Mu_F ? ? HTH_26 PBECR3 Dockerin_1 DUF5620 DUF5620 Cellulase AMP-binding AMP-binding_C POR_N TPP_enzyme_C Fer4 POR ?

MBQ9198652.1
JAFUDG010000003.1:22487-42260
Lachnospiraceae bacterium

TPM_phosphatase PETase HTH_24 AsnC_trans_reg AAA_14 DUF4143 ELFV_dehydrog_N ELFV_dehydrog Acetylase HTH_3 PBECR3 dCache_1 HAMP MCPsignal Bmp Calcineurin Citrate_synt Methyltrans_SAM TetR_N TetR_C_8 MMPL MMPL

MBQ9360319.1
JAFTAP010000048.1:12568-32170
Lachnospiraceae bacterium

? LRRs LRRs LRRs LRRs LRRs AAA-ATPase_like PDDEXK_9 LRRs LRRs ? Iso_dh ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_Xtn ABC_tran_CTD PBECR3 ? YjeF_N Carb_kinase Ala_racemase_N Ala_racemase_C CBS CBS CorC_HlyC TACO1_YebC_N Transcrip_reg ? Calcineurin

MBQ9361207.1
JAFTAP010000083.1:41831-57157
Lachnospiraceae bacterium

Methyltransf_21 ? Peptidase_M78 Glycos_transf_2 C2185-like_N Radical_SAM C2185-like_N Glycos_transf_2 Radical_SAM C2185-like_N PBECR3 ? DUF4422 GHMP_kinases_N NTP_transferase IspD ADH_N ADH_zinc_N Glycos_transf_2 HTH_31 HTH_31

MBQ9434891.1
JAFTCA010000011.1:1-1359
Bacteroidales bacterium

PBECR3 ? tRNA-synt_2

MBQ9599852.1
JAFSQM010000409.1:22884-32666
Clostridia bacterium

? DUF3846 ? GIY-YIG HTH_26 Peptidase_S24 ? HicB_lk_antitox PBECR3 RepB-RCR_reg ? PhageMetallopep ? Terminase-T7_RNaseH-like SPP1_portal Phage_Mu_F

MBQ9605012.1
JAFSQY010000343.1:1-2528
Bacillota bacterium

ROK TsaD PBECR2

MBQ9658342.1
JAFTFU010000084.1:157-2598
Clostridia bacterium

? Phage_holin_4_1 Glyco_hydro_25 CW_7 LysM PBECR2

MBQ9743952.1
JAFTJE010000012.1:43030-57795
Clostridia bacterium

? HTH_ParB ParB_N ParA-Soj-NTPase HTH_ParB ParB_N ? FUR ? PBECR3 HTH_3 MerR_1 DNA_methylase ? ? DNA_methylase HTH_3

MBQ9768118.1
JAFTJW010000149.1:25553-41010
Lachnospiraceae bacterium

ABC_tran ABC2_membrane_6 ABC2_membrane_6 DAGK_cat YegS_C Methyltransf_11 ATPase_2 DUF234 Mrr_cat HTH_3 PBECR3 HTH_3 Glyco_hydro_65m DUF2508 ? T2SSE T2SSF aBig_2 T2SSF

MBR0464542.1
JAFRLF010000127.1:1-2190
Clostridia bacterium

ArsC PBECR2 ParE_toxin PhdYeFM_antitox HMA

MBR1537424.1
JAFUWB010000192.1:1-11685
Treponema sp.

DUF6273 PBECR3 DUF4160 ? DUF4325 Fer4 Fer4 Fe_hyd_lg_C NMT1 Glycos_transf_2 TdpA_C

MBR1670247.1
JAFUYO010000019.1:10578-19793
Butyrivibrio sp.

PBECR3 ? Glyco_hydro_88 MatE Glyco_transf_11 PS_pyruv_trans Glycos_transf_2 Glycos_transf_2 DUF6056

MBR1694489.1
JAFUZC010000261.1:1-3247
Selenomonas sp.

PBECR3 Phage_integrase Ribosomal_S9 Ribosomal_L13 Epimerase

MBR1702825.1
JAFUZJ010000050.1:126024-138986
Lachnospiraceae bacterium

? Penicillinase_R Peptidase_M56 Peptidase_M56 WD40_RLD ? Sigma70_r2 Sigma70_r4 ? ? PBECR3 HTH_38 Sigma70_r2 Sigma70_r4_2 ? zf-HC2 ? SRAP HTH_3

MBR1708927.1
JAFUZM010000146.1:15249-28689
Clostridia bacterium

DUF3783 LGT Acetyltransf_4 AP_endonuc_2 GFO_IDH_MocA GFO_IDH_MocA_C3 Lyase_1 FumaraseC_C DDE_Tnp_1 PBECR3 PTS-HPr DUF4364 DUF6020 S4 IspD YgbB Mrr_cat TnpV

MBR1742574.1
JAFVAA010000538.1:1-9482
Lachnospiraceae bacterium

? Acylphosphatase zf-HYPF zf-HYPF Sua5_yciO_yrdC HypF_C HypF_C_2 PBECR2 HSDR_N_2 Methyltransf_31 RelB PIN Acetylase ? ?

MBR1883666.1
JAFVCY010000017.1:15764-31257
Clostridia bacterium

GatC Amidase GatB_N GatB_Yqey PBECR2 RuvB_N AAA_lid_4 WHD_RuvB QueH ? ? Glycoamylase GH94_b-supersand GH94_b-supersand GH94_b-supersand GH94_b-supersand

MBR2037310.1
JAFVOT010000176.1:1-9719
Lachnospiraceae bacterium

? Trypan_PARP PBECR3 Cthe_2159 Peptidase_S41 PDZ_6 CtpB_N-like Peptidase_M23 SlyX CC_PcsB CENP-F_leu_zip FtsX FtsX_ECD ABC_tran

MBR2311553.1
JAFVQF010000282.1:573-8451
Oscillospiraceae bacterium

NTP_transferase tRNA-synt_1c ? PBECR2 HD KH_1 RNase_Y_N DUF6809 ? Zn_Ribbon_1 ?

MBR3601018.1
JAFXGP010000021.1:17-6674
Lachnospiraceae bacterium

PBECR2 Multi_Drug_Res ? MarR_2 ABC_membrane ABC_tran ABC_membrane ABC_tran

MBR4152654.1
JAFYNH010000050.1:1-12069
Selenomonadaceae bacterium

Cj0814 IF3_N IF3_C Ribosomal_L35p Ribosomal_L20 MgsA_C AAA_assoc_2 AAA PBECR2 dUTPase Hfq IPPT IPPT IPT Lactamase_B_2 MCPsignal HAMP 4HB_MCP_1 MCPsignal HAMP 4HB_MCP_1 MCPsignal HAMP 4HB_MCP_1

MBR5292999.1
JAFYOP010000030.1:1-2704
Clostridia bacterium

P-II PBECR3 DUF2806 Alpha_L_fucos

MBR5400773.1
JAFZOJ010000034.1:140464-151672
Treponema sp.

Peptidase_M20 PNPase Macro IU_nuc_hydro ? PIN HTH_3 PBECR3 ? DUF4417 ? ? HTH_3 EcoRI_methylase GmrSD_N HNH

MBR5795930.1
JAFZDT010000140.1:1-16600
Erysipelotrichaceae bacterium

? Synaptojanin DUF2207 DUF2207_C ? VanY PseudoU_synth_2 Methyltrans_SAM PBECR3 UvrD-helicase UvrD_C PcrA_UvrD_tudor Nlig-Ia DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_2 BRCT GatC Amidase GatB_N GatB_Yqey ? TRAM

MBR6228438.1
JAFYBS010000027.1:3505-15216
Eubacterium sp.

Methyltransf_11 Acetylase PBECR2 Peptidase_C11 CBM9_2 Big ? PP-binding AMP-binding AMP-binding_C_3 LUD_dom ABC_tran

MBR6394510.1
JAFYGE010000057.1:42708-55370
Ruminococcus sp.

Met_gamma_lyase Acetyltransf_7 Peptidase_M42 AMPT Peptidase_M42 Transpeptidase Dockerin_1 Cel124_C MIF PBECR2 ? Glycos_transf_2 PAP2 ? GSHPx TSCPD Staph_reg_Sar_Rot

MBR7926928.1
JAGSSS010000002.1:158914-173414
Tuanshanicoccus yangjingiae

SBP_bac_5 BPD_transp_1_N BPD_transp_1 OppC_N BPD_transp_1 ABC_tran oligo_HPY ABC_tran oligo_HPY ADH_N ADH_zinc_N Phage_holin_5_2 PBECR3 ROK HTH_6 SIS SBP_bac_8 BPD_transp_1 BPD_transp_1 NanE ?

MBS4760958.1
JAHALR010000014.1:55021-65905
Clostridium sp.

Cep3 ? Peptidase_M78 ? ? Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F PBECR3 DUF4355 ? ? DUF6093 HK97-gp10_like Phage_tail_terminator_8 Phage_sheath_1C

MBS4761810.1
JAGYYA010000004.1:31673-49212

Carnobacteriaceae bacterium zg-ZUI252

HTH_6 SIS PTS_EIIC PTS_EIIB Glyco_hydro_4 Glyco_hydro_4C MFS_1_like BPD_transp_2 ABC_tran HI0933_like PBECR3 Mur_ligase Mur_ligase_M Mur_ligase_C ? RelB YafQ_toxin tRNA-synt_2b HGTP_anticodon ProRS-C_1 Pyr_redox_2 DUF212

MBS5064533.1
JAGZKF010000042.1:14504-31181
Hungatella hathewayi

? Sortase GramPos_pilinD1 Sortase DUF7601 DUF11 BTAD Elong-fact-P_C EFP EFP_N PBECR3 DHquinase_II SKI SDH_C Shikimate_DH Shikimate_dh_N Hydrolase_like ATP-cone Zn_ribbon_NrdR ArAE_1_C ArAE_1 Rubredoxin

MBS5064604.1
JAGZKF010000044.1:9624-29277
Hungatella hathewayi

SSF ? Glucosamine_iso Peripla_BP_3 LacI Pribosyltran DZR_2 UvrD_C_2 SH3_13 AAA_19 HHH_RecD2 HHH_5 OB_YrrC MreB_Mbl PBECR3 ABC_tran ABC_tran_Xtn ABC_tran ABC_trans_CmpB Apolipoprotein Gly_kinase GntP_permease BD-FAE HTH_30 Diacid_rec PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I

MBS5131393.1
JAGZJD010000134.1:1-4394
Lachnospiraceae bacterium

DUF3987 TrmB HTH_3 HTH_3 PBECR3 Arm-DNA-bind_4 Phage_int_SAM_3

MBS5166365.1
JAGZIN010000042.1:92-4386
Butyricicoccus pullicaecorum

? ? HTH_17 Phage_pRha HTH_3 HTH_26 PBECR3 Phage_pRha HTH_3 HTH_3 Peptidase_M78

MBS5257831.1
JAGZHI010000003.1:22894-33138

Lachnospira eligens

MFS_2 Radical_SAM PBECR3 Glyco_hydro_25 DHHA1 DHH ABC_tran BPD_transp_1 DUF5721 ? Peptidase_S26

MBS5317204.1
JAGZGL010000014.1:48146-63053
Clostridiales bacterium

PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran PhoU PhoU DUF4300 ? PBECR3 Xan_ur_permease MGS AICARFT_IMPCHas ? Cu_amine_oxidN1 EPSP_synthase ACT PDH_C PDH_N DAHP_synth_1 DAHP_snth_FXD

MBS6025813.1
JAHAHB010000040.1:166-7367
Paraclostridium sordellii

Phage_pRha FAM81B TrmB HTH_36 PriCT_1 Stc1 HTH_49 Resolvase PBECR3 ? ? ?

MBS6266296.1
JAHAEC010000055.1:94-6883
Clostridium sp.

SPP1_portal Phage_min_cap2 PBECR2 DUF4406 ? Phage_scaffold Phage_capsid CarboxypepD_reg DnaT_2 DUF6751 ?

MBS6599619.1
JAGZYN010000002.1:59092-71983
Clostridium sp.

DUF676 TerD vWA-TerF-like Fe-ADH ADH_Fe_C DUF188 Isochorismatase DUF2809 Adenyl_transf PBECR3 Guanylate_kin Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

MBS6832933.1
JAGZUX010000340.1:1-2317
Clostridiales bacterium

? PBECR2 GTP_EFTU_D3 GTP_EFTU_D2 GTP_EFTU

MBS7194667.1
JAGZPA010000001.1:16060-30638
Eubacterium sp.

ABC_sub_bind BPD_transp_2 ABC_tran Flavodoxin_2 POTRA_1 FtsQ_DivIB_C Tubulin FtsZ_C DUF1699_N CarD_TRCF_RID PBECR3 ? Transglutaminase ? Cu_amine_oxidN1 AF_1845 Cu_amine_oxidN1 MatE MatE SepQ_C

MBS7210833.1
JAGZOR010000131.1:1-3927
Lachnospiraceae bacterium

FAD_oxidored HTH_1 LysR_substrate ABC_tran ABC_membrane DH_ALS2 HTH_20 PBECR3

MCB5069248.1
JAJBEV010000023.1:5635-13313
Streptococcus mutans

CodY HTH_CodY Isochorismatase N-OB tRNA-synt_2 GatC Amidase GatB_N GatB_Yqey ? PBECR3

MCB6989782.1
JAJCKF010000004.1:134237-153431
bacterium 210820-DFI.6.38

Aminotran_5 Polysacc_synt ? Pribosyltran ? RQC Zn_ribbon_Top1 UvrD_C UvrD-helicase UvrD-helicase Cys_Met_Meta_PP PBECR3 NMT1 PBP5_C Peptidase_S11 Polysacc_deac_1 DUF1980_C cobW CobW_C cobW ABC-3 ABC_tran

MCB6992207.1
JAJCKG010000001.1:329714-346016
bacterium 210820-DFI.6.37

Phage_integrase Phage_int_SAM_1 SfsA SfsA_N ? LGT MMR_HSR1 LED_rpt YchF-GTPase_C Phage_holin_1 HTH_3 Peptidase_S24 PBECR3 Big Big-like ? Phage_integrase Ammonium_transp ? MMPL MMPL Amidohydro_3

MCC8044691.1
JAJBVE010000048.1:19714-42895
Clostridiales bacterium

Na_Ala_symp LRRs LRRs LBP_C LBP_M Lact_bio_phlase Lact_bio_phlase Glyco_hydro_88 SBP_bac_1 BPD_transp_1 BPD_transp_1 PBECR3 Glyco_hydro_31_2nd Glyco_hydro_31_3rd Lipase_GDSL_2 Lipase_GDSL_2 CE2_N Radical_SAM NUDIX tRNA_NucTran2_2 PolyA_pol_RNAbd PolyA_pol Trigger_C

MCC8051740.1
JAJBVD010000077.1:149049-164990
Clostridiales bacterium

Bac_transf Epimerase UDPG_MGDP_dh_C UDPG_MGDP_dh UDPG_MGDP_dh_N DUF3368 Peptidase_M78 HTH_3 MatE MatE PBECR3 DUF1540 DUF1540 ABC2_membrane_2 ABC_tran NPCBM_assoc NPCBM_assoc HGTP_anticodon tRNA-synt_2b tRNA_SAD TGS ? LytR_cpsA_psr

MCC8160854.1
JAJBTN010000070.1:172-7690
Oscillospiraceae bacterium

PBECR2 NUDIX NDPSase_N Methyltransf_32 dCMP_cyt_deam_1 GARS_C GARS_A GARS_N Cu_amine_oxidN1 AICARFT_IMPCHas DUF559

MCD7743733.1
JAJQHZ010000004.1:144-10463
Lachnospiraceae bacterium

? ABC_tran ABC_membrane ABC_tran ABC_membrane tRNA_edit DUF6449 ABC2_membrane_5 ABC_tran HTH_3 PBECR3

MCD7749940.1
JAJQHT010000301.1:131-3123
Oscillospiraceae bacterium

Dus PBECR2 Sigma70_r4 Sigma70_r2 Peptidase_U4

MCD7751896.1
JAJQHY010000060.1:1-5820
Lachnospiraceae bacterium

? Gln-synt_C Gln-synt_N ParA DUF134 Asparaginase Asparaginase_C ? PBECR3

MCD7776998.1
JAJQHH010000020.1:1-8432
Clostridiales bacterium

SpoIID SPOR DHquinase_II SDH_C Shikimate_DH Shikimate_dh_N PGP_phosphatase PBECR2 Cu_amine_oxidN1 MatE MatE ?

MCD7841100.1
JAJQGC010000064.1:1-3310
Lachnospiraceae bacterium

? PBECR3 DUF6198 Orn_DAP_Arg_deC Orn_Arg_deC_N

MCD7841180.1
JAJQGC010000084.1:1-5011
Lachnospiraceae bacterium

PBECR3 His-Kinase His-Kinase sCache_like Trans_reg_C Receiver PhoU PhoU ABC_tran BPD_transp_1

MCD7862395.1
JAJQFS010000023.1:8847-17567
Lachnospiraceae bacterium

? ? SWIM ? ? ? HTH_3 PBECR3 Phage_int_SAM_3 Phage_integrase UPF0102 Peptidase_S26 MMR_HSR1 RNase_HII Peptidase_S26 Ribosomal_L19

MCD7866168.1
JAJQFQ010000195.1:184-14115
Clostridiales bacterium

Topoisom_bac ? NikA-like MobC ? ? PBECR2 Relaxase DUF4230 DUF6591

MCD7881080.1
JAJQFJ010000043.1:14819-25823
Clostridiales bacterium

? Terminase_6N Terminase_6C ? ? DUF4417 Phage_Mu_F Resolvase HTH_28 PBECR2 DUF4355 Phage_capsid_4 ? Phage_connect_1 ? HK97-gp10_like ?

MCD8014428.1
JAJQDT010000105.1:18049-39834
Lachnospiraceae bacterium

DUF4143 AAA_14 AAA-ATPase_like PDDEXK_9 HAUS-augmin3 HTH_17 Phage_int_SAM_5 Phage_integrase GH85_C GH85_C Methyltranf_PUA RsmF_methylt_CI Methyltr_RsmB-F Methyltr_RsmB-F Methyltr_RsmF_N PBECR3 SpoU_methylase Aminotran_1_2 AsnC_trans_reg AIRS_C VanW PG_binding_4 DUF3377 ROK

MCD8046210.1
JAJQDD010000112.1:4525-14525
Clostridiales bacterium

? ? ? ? ? HTH_3 HTH_3 Peptidase_S24 PBECR3 ? DUF4160 Phage_int_SAM_3 Phage_integrase SpoIVA_C SpoIVA_middle SpoIVA_ATPase HxlR PTA_PTB PTA_PTB GntR

MCD8051135.1
JAJQCW010000039.1:134-8622
Clostridiales bacterium

PBECR3 ? Zn_Ribbon_1 Phage_integrase Glyco_transf_4 Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 Stealth_CR1 Stealth_CR2 Stealth_CR4 CpsB_CapC Wzy_C

MCD8069418.1
JAJQCU010000366.1:398-6167
Lachnospiraceae bacterium

GHMP_kinases_N SipL SipL PBECR2 DHO_dh DHODB_Fe-S_bind ?

MCD8090122.1
JAJQCI010000190.1:7466-17529
Clostridiales bacterium

HMGL-like HCS_D2 LeuA_dimer Aconitase Aconitase Aconitase_C Iso_dh SLH ILVD_EDD_N ILV_EDD_C TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C PBECR3

MCD8119813.1
JAJQBQ010000044.1:157071-173629
Lachnospiraceae bacterium

Lectin_C ? AAA_33 ? PhdYeFM_antitox Zn_ribbon_Top1 HKD Helicase_C ResIII AEP-TOTE PBECR3 ? ? ? HTH_26 DUF2975 NTP_transf_2 ?

MCD8130605.1
JAJQBK010000021.1:1-14303
Lachnospiraceae bacterium

Lon_C Lon_lid AAA LON_substr_bdg ClpB_D2-small AAA_2 zf-C4_ClpX CLP_protease Trigger_C FKBP_C Trigger_N SASP CstA CstA DUF8052 PBECR3 Peptidase_C39_2 ? CAP SH3_3 DUF3950 ?

MCD8161884.1
JAJQBD010000001.1:10753-23077
Synergistaceae bacterium

? ? DUF6198 ? Fer4 Flavodoxin_5 MatE MatE ? PBECR3 DUF1980_C ABC-3 ABC_tran ? FUR NAGPA RlmN_N Radical_SAM

MCD8238447.1
JAJPZT010000040.1:14712-20799
Clostridiales bacterium

Ribosomal_L11_N Ribosomal_L11 Ribosomal_L1 Phage_VG64 Phage_VG64 ? ? HTH_3 HTH_26 PBECR3

MCD8331408.1
JAJPYC010000089.1:58-1908
Oscillospiraceae bacterium

PBECR2 AA_kinase

MCD8341946.1
JAJPXV010000083.1:196-7045
Clostridiales bacterium

His-Kinase HAMP Trans_reg_C Receiver Peptidase_S11 PBP5_C PBECR2 PP_kinase_N PP_kinase PP_kinase_C_1 PP_kinase_C

MCD8381031.1
JAJPXG010000152.1:4735-11433
Lachnospiraceae bacterium

? PBECR3 tRNA_edit LysR_substrate HTH_1 SpoVAC_SpoVAEB ? SpoVAD DegT_DnrJ_EryC1 FdtA

MCD8395640.1
JAJPWX010000003.1:11955-29917
Lachnospiraceae bacterium

CbiD DUF6363 Patatin Bmp ABC_tran Bac_DnaA ABC_tran BPD_transp_2 BPD_transp_2 HhH-GPD PBECR3 ? HMA E1-E2_ATPase Hydrolase HMA MatE MatE DUF2229 ? BcrAD_BadFG Aldo_ket_red

MCF0163452.1
JABUTM010000022.1:7966-21640
Fusobacterium necrophorum

TFIIE_alpha Tox-REase-5 AVEN ? DUF7438 DUF722 RadC DUF4116 DUF4116 ? ArdcN MPTase-PolyVal PBECR3 DNA_methylase DNA_methylase ? Zn_ribbon_DnaG DNAG_N Toprim_2 ArdA MORN_2 ? ? ParA-Soj-NTPase

MCH4157673.1
JAKVKY010000009.1:44529-57192
Acidaminococcaceae bacterium

? ? DUF2513 ? ? ? HTH_3 PBECR3 ? ? AAA_23 ABC-ATPase MethyltransfD12 NYN ? ?

MCH5186225.1
JAFLUR010000253.1:1-3514
Oscillospiraceae bacterium

GGDEF His-Kinase His-Kinase Receiver PBECR3 Hpt Receiver

MCH5203975.1
JAFLUE010000036.1:133-9879
Oscillospiraceae bacterium

RrnaAD PBECR2 ABC_tran ABC_membrane ABC_tran ABC_membrane MarR_2 NAD_binding_1 DHODB_Fe-S_bind Fer4_20 Pyr_redox_2 ? Trns_repr_metal

MCH8189463.1
JADFIC010000546.1:493-2302
Proteobacteria bacterium

? RmuC PBECR3

MCI1951802.1
JALCPQ010000002.1:14755-28553
Clostridiales bacterium

Radical_SAM SPASM HTH_Crp_2 cNMPBD Fer4_21 AF_0155 Cupin_2 Fer4_13 ? Glyco_hydro_25 PBECR3 Phage_holin_4_1 Glyco_hydro_25 ? ? ? Gp10_C Sipho_Gp37 SPP1_Dit

MCI5502379.1
JALERT010000225.1:7063-21562
Lachnospiraceae bacterium

ABC2_membrane Glycos_transf_2 Glycos_transf_2 Glycos_transf_1 Glyco_trans_4_2 Bac_transf FCD GntR Wzy_C PBECR3 LuxS PNPase FlgN FliS FliD_C FliD_N FlaG CsrA

MCI5579561.1
JALEQE010000185.1:2634-9405
Oscillospiraceae bacterium

Transposase_mut Phage_Mu_F PBECR2 ? PALP PolyA_pol PolyA_pol_RNAbd tRNA_NucTran2_2 ? Spore_III_AA Spore_III_AB SpoIIIAC

MCI5741558.1
JALENE010000071.1:1-7643
Lachnospiraceae bacterium

PBECR2 HNH_4 SfsA_N SfsA MatE MatE SBP_bac_3 BPD_transp_1 ABC_tran

MCI5945310.1
JALEIW010000001.1:37661-46891
Oscillospiraceae bacterium

Acetylase DUF3810 ? AASDHPPT_N ACPS DUF4349 His-Kinase His-Kinase Receiver PBECR3 ? PAS ?

MCI6653292.1
JALFTC010000045.1:37590-51676
Ruminococcus sp.

PBP5_C Peptidase_S11 Spore_YtfJ DUF2953 SMC_ScpB SMC_ScpA Peptidase_M50 Ribosomal_L28 PBECR2 DHHA1 tRNA_SAD tRNA-synt_2c Peptidase_U32_C Peptidase_U32 YceG 5-FTHF_cyc-lig DJ-1_PfpI Acetyltransf_9 LPG_synthase_C

MCI6654871.1
JALFTF010000056.1:4326-19995
Clostridium sp.

PK PK_C Pur_ac_phosph_N LytTR LiaF-TM LiaF-like_C EIF3I Wzy_C HD PBECR3 ? Acetylase ABC_tran ? ? AgrB ?

MCI7631617.1
JALFXK010000070.1:3267-13820
Collinsella sp.

? ? HTH_17 ? ? DUF739 HTH_3 PBECR3 ? Phage_int_SAM_3 Phage_integrase ? DUF134 Nitro_FeMo-Co Gln-synt_C Gln-synt_N Transposase_20

MCI7717803.1
JALFVW010000021.1:25555-33224

[Pasteurella] aerogenes

PBECR3 Phage_tail_S ? BacA ABC_tran FecCD Peripla_BP_2 TonB_dep_Rec_b-barrel Plug

MCI7760149.1
JALFUX010000250.1:1-9624
[Eubacterium] saphenum

Phage_Mu_F PBECR2 Acetylase L_biotic_typeA LANC_like DUF4135 DUF4135 Trans_reg_C Receiver His-Kinase His_kinase ABC_tran UvrA_DNA-bind Viral_helicase1

MCI7798732.1
JALEVX010000028.1:34105-46050
Spirochaetia bacterium

DUF6788 Rrf2 ? PNPOx_N PBECR3 ANKs ? Radical_SAM Sirtuin Macro KdpD DUF4118 His-Kinase His-Kinase Receiver Trans_reg_C

MCI8274775.1
JAAVXU010000032.1:1-10125
Lachnospiraceae bacterium

HTH_3 DUF4352 PBECR3 Hol_Tox Phage_integrase ABC_tran oligo_HPY ABC_tran BPD_transp_1 OppC_N BPD_transp_1 BPD_transp_1_N SBP_bac_5

MCI8352423.1
JAAVZA010000019.1:16196-24729
Clostridia bacterium

Na_Ca_ex Na_Ca_ex Sigma70_r2 Sigma70_r3 Sigma70_r4 Sigma70_r4 Sigma70_r2 Peptidase_U4 Peptidase_U4 Phage_integrase ? PBECR3 HTH_3 ? ? ? ? Baculo_PEP_C DUF5931

MCI8352963.1
JAAVZA010000041.1:1-11214
Clostridia bacterium

Filament CPSase_sm_chain GATase PBECR2 YgjP-like YgjP-like Sos7 Cas_Cas6_C Cas_CXXC_CXXC DevR Cas_Cas5d ?

MCI8396996.1
JAAVZW010000034.1:411-12144
Clostridia bacterium

Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C PBECR2 DUF2651 Gate Gate ? ParE_toxin RNase_Y_N KH_1 HD Sigma70_r2 Sigma70_r4

MCI8414565.1
JAAWAE010000032.1:1-9335
Ruminiclostridium sp.

EFG_III EF-G_D2 GTP_EFTU Dus Voltage_CLC Voltage_CLC ? ? Coat_F PBECR3 Flavokinase FAD_syn TruB_C_2 TruB_N DHHA1 DHH RBFA GTP_EFTU_D2

MCI8420461.1
JAAWAF010000227.1:8391-17695
Oscillospiraceae bacterium

Phage_terminase Terminase_3 Terminase_3C SPP1_portal PBECR2 Peptidase_Prp ? Phage_capsid ? DUF7297 Minor_capsid_2 ?

MCI8426914.1
JAAWAI010000016.1:1-7090
Lachnospira sp.

? ? ? HTH_3 HTH_26 PBECR3 ? DUF739 HTH_3 RelB REase PH_gly_zipper ?

MCI8509889.1
JAAWBV010000095.1:9752-18688
Lachnospiraceae bacterium

Gly_radical HTH_Crp_2 cNMPBD HTH_3 PBECR3 HAD_2 HTH_3 PLDc_N ABC_tran ABC2_membrane_2 MerR_1 MerR-DNA-bind Spermine_synth TgpA_N

MCI8532540.1
JAAWCA010000157.1:1-9000
Lachnospiraceae bacterium

HTH_6 SIS Glyco_hydro_1 PTS_IIB PTS_EIIC PTS_IIA ? Peptidase_M78 PBECR3 ELFV_dehydrog ELFV_dehydrog_N His-Kinase His-Kinase

MCI8578319.1
JAAWCU010000167.1:242-11899
Lachnospiraceae bacterium

HTH_1 LysR_substrate PBECR3 Peptidase_M41 AAA AAA_2 Hydrolase_3 ATPgrasp_Ter PylC-like_N PELOTA_1 StiP TRSP PRTase_2 C-C_Bond_Lyase TerD

MCI8646768.1
JAAWED010000042.1:2886-15960
Bacillota bacterium

? Phage_rep_org_N DUF6291 ? ? HTH_26 HTH_26 PBECR3 Phage_int_SAM_3 Phage_integrase FtsX MacB_PCD FtsX ABC_tran His-Kinase His-Kinase Trans_reg_C Receiver Peptidase_S49 HI0933_like

MCI8687703.1
JAAWEX010000059.1:1-9909
Lawsonibacter sp.

dUTPase LuxC LuxE Peptidase_U32 DUF3656 Peptidase_U32 ZapA DnaB_2 DnaB_2 IstB_IS21 PBECR2 Peptidase_A8 S4 PseudoU_synth_2 SecG

MCI8828886.1
JAAWGZ010000023.1:5485-21138
Ruminiclostridium sp.

DUF554 ECH_1 3HCDH_N 3HCDH MCPsignal HAMP_2 HAMP CZB MatE MatE Ribosomal_S20p Peptidase_A25 Peptidase_A25 PBECR2 ? ? ? Beta_helix YcxB ZapA Peptidase_U32 DUF3656

MCI8833038.1
JAAWHY010000021.1:1-10129
Clostridia bacterium

Peptidase_M23 DUF5711 Colicin_V Colicin_V PBECR3 HmpF MMPL MMPL TetR_N DsbD_TM Seadorna_VP11 AhpC-TSA FeThRed_B Thioredoxin

MCI8972885.1
JAAWJB010000013.1:283-12272
Lachnospiraceae bacterium

HTH_3 PBECR3 ELFV_dehydrog ELFV_dehydrog_N PhoU PhoU Na_Pi_cotrans Na_Pi_cotrans Cys_rich_CPCC TerD AIM24 Hydrolase_3 Cation_ATPase_C Hydrolase_3 Cation_ATPase E1-E2_ATPase Cation_ATPase_N

MCI8993567.1
JAAWJJ010000057.1:6184-22907
Eubacterium sp.

LysR_substrate HTH_1 URO-D DUF1266 Receiver Trans_reg_C His-Kinase ADH_Fe_C Fe-ADH adh_short PBECR3 HTH_1 LysR_substrate URO-D MFS_1 ATP-synt_D ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_N ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_Xtn ATP-synt_ab_N ATP-synt_F

MCI9002717.1
JAAWJP010000074.1:329-11099
Oscillibacter sp.

Ribul_P_3_epim ? GTP_EFTU GTP_EFTU_D2 EFG_C BipA_C PBECR2 Fer4_12 Peptidase_S8 Serpin DHHA2 DHH DUF2992

MCI9097847.1
JAAWKY010000104.1:1-12939
Lachnospiraceae bacterium

HTH_ParB ? HTH_26 ? Sigma70_r4_2 ? PBECR3 Acetylase GntR FCD ILVD_EDD_N ILV_EDD_C F420_ligase Glyco_hydro_4C Glyco_hydro_4 AraC_binding HTH_18

MCI9190451.1
JAAWML010000070.1:1664-18142
Lachnospiraceae bacterium

? ? FMN_bind MCPsignal CCDC39 HAMP_2 HAMP dCache_1 DUF2461 Sigma70_r4_2 Sigma70_r2 HTH_11 DUF1970 WYL WCX PBECR3 DUF3795 MarR BNR_2 ABC_tran ABC_tran YjbR BrxC_BrxD

MCI9310459.1
JAAWOL010000062.1:215-10103
Lawsonibacter sp.

HTH_26 ? Cation_ATPase_N E1-E2_ATPase Cation_ATPase HAD Hydrolase_3 Cation_ATPase_C PBECR3 HTH_3 RHH_5 Malt_amylase_C Alpha-amylase PRK Radical_SAM HTH_3 ?

MCI9326712.1
JAAWOR010000030.1:124-9550
Lachnospiraceae bacterium

PBECR3 DDE_Tnp_1 DUF772 Sde182_NH-like His_kinase His-Kinase Receiver HTH_18 SBP_bac_1 DUF3502 BPD_transp_1 BPD_transp_1

MCI9418192.1
JAAWQA010000055.1:148-6193
Eubacterium sp.

? SLT DUF5688 ? ? HTH_3 HTH_26 PBECR3 ? PemK_toxin ? ?

MCK3895834.1
JAASBT010000061.1:475-7377
Streptococcus suis

? NikA-like EcoRI_methylase GmrSD_N HNH ? PBECR3 TraG-D_C TrwB_AAD_bind

MCL1819351.1
WRJE01000005.1:28622-47514
Oscillospiraceae bacterium

? Sigma70_r2 Sigma70_r4_2 DUF3040 ? LiaI-LiaF-like_TM1 Receiver His-Kinase His-Kinase PAS FeS Fe_hyd_lg_C Fer4_10 HD_5 Receiver PBECR2 Helicase_C_2 DEAD_2 DEAD PDDEXK_1 ? Flg_new_2 Flg_new_2 CBM_X2 CBM_X2 PBECR2 Trigger_N FKBP_C Trigger_C CLP_protease zf-C4_ClpX AAA_2 ClpB_D2-small

MCL1820374.1
WRJE01000035.1:15473-24993
Oscillospiraceae bacterium

MerR_1 Ribosomal_S20p PNPOx_N Peptidase_U4 Sigma70_r2 Sigma70_r4 ? PBECR2 QueT DUF4364 HTH_19 Cupin_2 HEPN Polbeta AMP-binding AMP-binding_C Dus

MCL1821122.1
WRJE01000090.1:1-5084
Oscillospiraceae bacterium

Amidase_3 RecG_wedge DEAD Helicase_C RecG_dom3_C GtrA_DPMS_TM PBECR2 PBECR3

MCL1823187.1
WRJM01000022.1:8415-21522
Oscillospiraceae bacterium

? Collagen ? Amidase_3 SPOR Phage_holin_4_1 AgrB AgrB PBECR3 ? ? MCP-like_PDC_1 PAS GGDEF His-Kinase His-Kinase Receiver Isochorismatase ? Qua1 Receiver His-Kinase

MCL1875423.1
WRHH01000009.1:13915-37159

Synergistaceae bacterium

? MCPsignal HAMP 4HB_MCP_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 DNA_pol3_delta_C ? HicB_lk_antitox AAA_14 DUF4143 PBECR3 TIR AP-ATPase TPRs TPRs TPRs TPRs TPRs TPRs TPRs 2Fe-2S_thioredx 2Fe-2S_thioredx Dus Pan_kinase GATase_3 MMS19_N Receiver His-Kinase His-Kinase PAS PAS PAS

MCL1904494.1
WRHO01000049.1:19-2603
Oscillospiraceae bacterium

Phage_cap_E ? PBECR2 ? Phage_tail_X ?

MCL1949960.1
WRGF01000188.1:5244-19971
Turicibacter sp.

? ? HypA ZnuA ABC_tran ABC-3 ATP-cone NRDD PBECR3 HhH-GPD FAD_oxidored ACT_9 AA_kinase Semialdhyde_dhC Semialdhyde_dh Hexapep_2 Hexapep Hexapep DapH_N DapB_C DapB_N DHDPS

MCL2019438.1
WRKM01000014.1:274-14797
Oscillospiraceae bacterium

? YhcG_C DUF1016_N ? Fer4_7 EKR POR PFOR_II POR_N DHH DHHA1 Biotin_lipoyl PYC_OADA HMGL-like Biotin_carb_C CPSase_L_D2 Biotin_carb_N GGDEF PAS HisKA_7TM PBECR3 ?

MCL2019712.1
WRKM01000018.1:14362-27584
Oscillospiraceae bacterium

HTS Hexapep Mac Polysacc_synt MurJ HEPN Polbeta ToxN_toxin Polbeta PBECR3 zf-C2H2 Cohesin_load SH3BP5 HSP70 ? ? ? TatD_DNase

MCL2109264.1
WRMK01000087.1:124-9357
Oscillospiraceae bacterium

Amidase Amidase GatC ? PBECR3 ? MCPsignal dCache_1 Trans_reg_C Receiver His-Kinase His-Kinase 7TMR-DISM_7TM

MCL2109897.1
WRMK01000188.1:1-3554
Oscillospiraceae bacterium

? DUF4153 DUF2262 PBECR3 ? ACT_9 AA_kinase

MCL2195165.1
WQRM01000061.1:1-8703
Oscillospiraceae bacterium

tRNA-synt_1g Anticodon_1 tRNA-synt_1 ? ? ? DZR_2 Pribosyltran MreB_Mbl PBECR2 ?

MCL2300252.1
WQTL01000391.1:271-6901
Bacillota bacterium

? HTH_26 ? ? Bro-N PBECR3 DUF4143 AAA_14 Flg_new Sigma70_r4_2 ?

MCL2378223.1
WQVJ01000010.1:42332-60407
Defluviitaleaceae bacterium

Peptidase_S11 PBP5_C MBOAT ? ? ? Transglutaminase Asp_Glu_race PBECR3 AMP-binding AMP-binding_C PP-binding 2TM UvrD_C UvrD-helicase HhH-GPD RimK Aminotran_1_2 Aminotran_1_2 GntR

MCL2426824.1
WQWL01000098.1:1-10996
Oscillospiraceae bacterium

? ? Rep3_N Rep3_C ? Resolvase HTH_7 ? Rep3_N Rep3_C PBECR3 Rep3_C Rep3_N Rep3_C Rep3_N ? ? ?

MCL2427544.1
WQWL01000222.1:195-2351
Oscillospiraceae bacterium

Rep3_N Rep3_C PBECR3 MazE_antitoxin HTH_26

MCL2446732.1
WQWM01000021.1:27517-37557
Oscillospiraceae bacterium

Glyoxalase_6 Glyoxalase ? MreB_Mbl PBECR2 HTH_18 DUF4830 Zip Zip Fe_dep_repr_C Fe_dep_repress SMC_N YafQ_toxin ?

MCL2628509.1
WQXY01000022.1:23301-36558
Oscillospiraceae bacterium

? Polbeta HEPN ADH_Fe_C Fe-ADH PBECR3 ODP Flavodoxin_1 Flavin_Reduct RUBY_RBDX ? He_PIG Dockerin_1 Cohesin Trypan_PARP Sortase Nitroreductase Virulence_RhuM

MCL2638530.1
WQYC01000029.1:26426-35396
Oscillospiraceae bacterium

? SpoIIIAH DUF348 G5 3D HTH_3 RelE SpoU_sub_bind SpoU_methylase ? PBECR3 DHHA1

MCL2693719.1
WRCK01000026.1:60-6803
Oscillospiraceae bacterium

DUF6273 PBECR3 3D G5 DUF348 DUF348 Receiver CheW Aminotran_1_2 Receiver tRNA-synt_1b

MCL2752649.1
WRBU01000009.1:1265-11577
Defluviitaleaceae bacterium

PrsW-protease NUDIX ? ? DUF5640 ? ? PBECR3 Ribosomal_S18 SSB Ribosomal_S6 ZT_dimer Cation_efflux ABC2_membrane_5 ABC_tran tRNA_SAD tRNA-synt_2b HGTP_anticodon

MCL2819509.1
WQXS01000003.1:28720-49609
Oscillospiraceae bacterium

CtsR_C CtsR Phage_int_SAM_1 Phage_integrase HflX_C MMR_HSR1 GTP-bdg_M GTP-bdg_N Ribosomal_L31 GalKase_gal_bdg GHMP_kinases_N PseudoU_synth_2 ? PBECR3 SWM_repeat fn3 Sortase Receiver LytTR ABC_tran TOBE_2 BPD_transp_1 BPD_transp_1 SBP_bac_8

MCL2821276.1
WQXS01000059.1:5246-14702
Oscillospiraceae bacterium

PBECR2 Glyco_hydro127C Glyco_hydro127M Beta-AFase-like_GH127_cat AP_endonuc_2 AMP-binding AMP-binding_C ? His_Phos_1 dUTPase LPG_synthase_C Beta-lactamase

MCL2857635.1
WRAV01000033.1:13268-20741
Oscillospiraceae bacterium

TM1586_NiRdase DUF2461 DNA_binding_1 DUF3795 Beta_elim_lyase Big_14 Peptidase_M20 PBECR3

MCL2864417.1
WRBB01000004.1:15702-30432
Lachnospiraceae bacterium

BPD_transp_1 Transketolase_N Transket_pyr Transketolase_C MR_MLE_N MR_MLE_C DHDPS NUDIX_4 HhH-GPD DsrE PBECR3 ThiF Aminotran_5 Beta_elim_lyase AAA_14 Zip Zip tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 SBP_bac_11

MCL5289689.1
JAMCWI010000035.1:233-10611
Bacillota bacterium

Phage_P4_cII RecA_C RecA_N Resolvase Recombinase Zn_ribbon_recom DUF4131 SLT HHH_3 SLBB PBECR3 DUF421 YetF_N Anticodon_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 RsfS

MCL6611186.1
JAILZP010000011.1:62908-74124
Peptococcaceae bacterium

? Vut_1 QueF ParB_N DUF6610 ? Terminase_6N Terminase_6C SPP1_portal Phage_min_cap2 PBECR2 ? DUF4355 Phage_capsid_4 DnaT_2 ? HK97-gp10_like Phage_tail_terminator_6

MCM1008671.1
JAMPIL010000016.1:3770-16462
Ruminococcus flavefaciens

? ? ? ArdA ? ? Rab11-FIP3 PBECR3 ORF6N ? ? ? NUMOD4 HNH_3 ? ?

MCM1024182.1
JAMPIC010000020.1:22906-34755
Prevotella sp.

HEPN BofA DUF2508 ? ? ? Metallophos_2 PBECR2 Sigma70_r2 Sigma70_r4_2 DUF4179 ? Lipase_GDSL_2 GH94_b-supersand Glyco_hydro_94 Glyco_hydro_65C LRRs LRRs Flavin_Reduct

MCM1056176.1
JAMPHT010000002.1:62538-78983
Bacillota bacterium

NAD_binding_1 DHODB_Fe-S_bind Fer4_20 Pyr_redox_2 Zn_Ribbon_TF Tetraspanin SPOUT_MTase DUF4253 ? 2H PBECR3 HD_5 DUF6044 DUF6020 Ion_trans TrkA_C TrkA_C HhH-GPD LysM Glyco_hydro_18 Glycos_trans_3N Glycos_transf_3 PYNP_C

MCM1097827.1
JAMPHC010000012.1:28886-38747
Ruminococcus flavefaciens

? ? ? ? ? FitA-like_RHH HTH_3 PBECR3 ParE_toxin ? Polbeta HepT-like Zn_ribbon_3 Tocopherol_cycl EPSP_synthase

MCM1102525.1
JAMPHA010000001.1:55102-73465
Clostridium sp.

Hexapep Hexapep tRNA-synt_1e DALR_2 CysS_C Ribonuclease_3 SpoU_sub_bind SpoU_methylase Sigma70_r2 Sigma70_r4_2 ? MatE MatE PBECR3 DA_C SBP_bac_5 SBP_bac_5 BPD_transp_1_N BPD_transp_1 OppC_N BPD_transp_1 ABC_tran ABC_tran oligo_HPY DUF3899 ?

MCM1102569.1
JAMPHA010000001.1:99263-119799
Clostridium sp.

Phosphorylase ABC_tran Peripla_BP_4 LacI PFK DcrB WD40_RLD Peptidase_M56 Peptidase_M56 Penicillinase_R PBECR3 ? MatE MatE CheC CheC ? ? SbcC_Walker_B AAA_23 SbcD_C Calcineurin

MCM1125353.1
JAMPGL010000004.1:80363-94023
Lachnospiraceae bacterium

Lipase_GDSL_2 ABC_membrane ABC_tran ABC_membrane ABC_tran LRRs ? HTH_3 HTH_26 PBECR2 Peptidase_A25 Ribosomal_S20p HTH_11 Acetylase Cass2 DUF3788 AAA_18

MCM1133987.1
JAMPGS010000001.1:56399-62944
Clostridium sp.

FlaG FliD_N FliD_C FliS FlgN PIN PhdYeFM_antitox Glyco_trans_1_2 PBECR3 Flagellin_N

MCM1144873.1
JAMPGJ010000077.1:1-13971
Lachnoclostridium sp.

Acetylase TsaD TsaE YcxB FtsX MacB_PCD YknX_C Beta-barrel_YknX ABC_tran PBECR3 MCPsignal CCDC39 HAMP ECF_trnsprt S1 HHH_9 HHH_9 HHH_3 Tex_YqgF Tex_central_region Tex_N BsuBI_PstI_RE BsuBI_PstI_RE_N ?

MCM1154420.1
JAMPGI010000006.1:55528-69567
Roseburia sp.

SDH_C Shikimate_DH Shikimate_dh_N Hydrolase_like ? ATP-cone Zn_ribbon_NrdR PRC SH3_3 ZnR Sigma70_r2 Sigma70_r3 Sigma70_r4 PBECR3 Fer4_8 Aldo_ket_red Ada_Zn_binding Lactamase_B GGDEF HD_5 Receiver Sigma70_r4 Sigma70_r2 Peptidase_U4 FtsZ_C Tubulin POTRA_1

MCM1159845.1
JAMPGD010000003.1:38287-52748
Roseburia sp.

DUF3892 PTA_PTB PTA_PTB Acetate_kinase DUF6273 ? Lipase_GDSL_2 ? PBECR3 ETF_alpha ETF ETF Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N 3HCDH 3HCDH_N ECH_1 Thiolase_C Thiolase_N Peptidase_M23

MCM1268334.1
JAMPEN010000059.1:104-15252
Bacteroidales bacterium

GDPD ? BSH_RND BPD_transp_1 BPD_transp_1 SBP_bac_8 PBECR3 Receiver His-Kinase His-Kinase PAS Receiver Trans_reg_C His-Kinase His-Kinase ? ABC_tran ABC-ATPase ABC_tran_Xtn ABC_tran PHP

MCM1298108.1
JAMPDW010000008.1:10379-22588
[Eubacterium] siraeum

CarD_TRCF_RID UvrB_inter ParE_toxin PhdYeFM_antitox PBECR3 ECF_trnsprt PFF1_TM YfhO Pan_kinase Glycos_transf_2 DUF4097 MethyltransfD12 Endonuc-EcoRV

MCM1328805.1
JAMPDU010000025.1:6245-20022
Ruminococcus sp.

FliB_C CxxCxxCC FliB_C SHMT ? ? ? PBECR2 Gln-synt_C-ter Gln-synt_C GSIII_N HTH_3 Sel1 DUF8297 Sel1 GGDEF HD_5 HTH_18 CAP

MCM1329061.1
JAMPDU010000034.1:1987-16743
Ruminococcus sp.

Spore_permease Spore_GerAC_N Spore_GerAC Fic GH85_C vWF_A VWA YfbK_C FTSW_RODA_SPOVE ? GalKase_gal_bdg GHMP_kinases_N PBECR2 Rce1-like DUF2975 DUF2975 DUF2975 DUF2975 HTH_26 PglZ iPGM_N

MCM1334816.1
JAMPDV010000063.1:6761-14049
Eubacterium sp.

NRDD PBECR2 PBECR2 Zn_Ribbon_1 DUF4282 MTS PrmC_N DUF1385 ? YafQ_toxin ? ?

MCM1335197.1
JAMPDV010000094.1:5920-11099
Eubacterium sp.

? REP_ORF2-G2P ? ? HIRAN DUF4160 DUF2442 PBECR3 Fic

MCM1388496.1
JAMPCN010000105.1:1-6440
Bacillus sp. (in: firmicutes)

YoaP PBECR2 Glyco_hydro_3 Glyco_hydro_3_C Fn3-like WD40_RLD ? LytTR Flavi_NS4A ABC2_membrane_3 Phage_holin_3_6

MCO7138144.1
JAMXWQ010000105.1:71-6001
[Clostridium] leptum

Pox_D5 DUF5906 D5_N ? HTH_26 HTH_26 PBECR3 Resolvase Recombinase Zn_ribbon_recom VCA0040-like

MCQ2550311.1
JAKSSF010000039.1:9641-21148
Lachnospiraceae bacterium

HTH_3 PBECR3 Bps Wzy_C MshEN T2SSE T2SSF_N T2SSF T2SSF N_methyl ComP_DUS A24_N_bact Peptidase_A24 PilM_2 PilM_2 PilN

MCQ2968870.1
JAKSUK010000015.1:24-11473
Clostridium sp.

BPD_transp_1 BPD_transp_1 ABC_tran PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease

MCQ9628257.1
JABUOI010000063.1:174-11991
Cetobacterium somerae

PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV NTP_transferase NTP_transferase dTDP_sugar_isom RmlD_sub_bind PBECR3 Epimerase OMP_b-brl_2 Na_Ala_symp Orn_Arg_deC_N

MCR4948946.1
JAILCO010000132.1:1-11234
Treponema sp.

? Glyco_hydro_43 GH43_C2 SASA SIAE_C Pribosyltran GATase_7 TP_0700 PhdYeFM_antitox PIN PBECR3 CitMHS

MCR5103063.1
JAILEL010000034.1:169-10308
Eubacterium sp.

Peptidase_M50 GcpE GcpE_C DNA_pol3_a_NI RNase_T DNA_pol3_alpha DNA_pol3_alpha DNA_pol3_finger HHH_6 PBECR3 HAD_2 HMGL-like

MCR5539669.1
JAILMR010000015.1:60121-75630
Ruminococcus sp.

Radical_SAM GHMP_kinases_N GHMP_kinases_C His_Phos_1 Acetylase ? Acetylase Tyr_Deacylase PBECR3 BipA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU Pro_isomerase Pro_isomerase Pro_isomerase ? HTH_1 LysR_substrate Cys_Met_Meta_PP AAA_33

MCR5831616.1
JAILRJ010000063.1:298-8433
Lachnospiraceae bacterium

? PcrA_UvrD_tudor UvrD_C UvrD-helicase Trp_repressor CheX CheC Na_H_Exchanger BTAD Trans_reg_C ? PBECR3

MCT4611935.1
JAOARL010000006.1:2227-16280
Clostridia bacterium

? NUDIX PilZ Methyltr_RsmF_N Methyltr_RsmB-F RsmF_methylt_CI Methyltranf_PUA EamA EamA RelE HTH_3 PBECR3 PALP DAP_epimerase DAP_epimerase PRS2_N DAHP_synth_1 ? Zn_ribbon_Top1 Radical_SAM_N Radical_SAM DUF3362 fn3

MCW9036026.1
JAPHRK010000072.1:235-13241
Rhodospirillales bacterium

Penicil_amidase PAS PAS GGDEF PBECR3 DnaB DnaB_C Ala_racemase_N Ala_racemase_C MlaE ABC_tran Zn_ribbon_LapB AAA_25 ChlI Colicin_V GATase_7 Pribosyltran

MCX4341127.1
JAPKWF010000045.1:8968-28507
Lachnospiraceae bacterium

FtsX ABC_tran Receiver LytTR SPOB_a HATPase_c_5 Peptidase_S51 ? NUDIX PBECR2 7TMR-DISM_7TM GGDEF Glyco_hydro_2_N2 Glyco_hydro_2 Mannosidase_ig Ig_mannosidase ? SEC-C ? TGS tRNA_SAD tRNA-synt_2b HGTP_anticodon ?

MCX6382887.1
JAPLCN010000007.1:810-3387
Actinomycetota bacterium

DDE_Tnp_1_6 PBECR3

MDA2119323.1
JAQFQR010000025.1:1-7831
Bacillus cereus

MerR_1 DUF3967 DUF8426 HTH_36 PBECR3 Mob_Pre TrmB HTH_3 ? ?

MDD2402176.1
JAQUGH010000084.1:557-8891
Clostridia bacterium

? ? RTX_C ? ? PemK_toxin PBECR3 DDE_Tnp_1 MORN MORN Big ? ? DUF3800

MDD3403707.1
JAQUUC010000033.1:356-7815
Hespellia sp.

PBECR3 C_GCAxxG_C_C Fe-ADH ADH_Fe_C Flavin_Reduct DUF2271 Mac Hexapep Hexapep FMN_red ABC_tran UvrA_DNA-bind ABC_tran ABC-ATPase

MDD3404076.1
JAQUUC010000044.1:4128-18345
Hespellia sp.

? ? DUF2513 DUF739 HTH_3 Peptidase_M78 PASTA DUF7609 PBECR3 Resolvase Zn_ribbon_recom Cation_ATPase_N E1-E2_ATPase Hydrolase ParA HAD_2 Sugar-bind HTH_24 UTP20 PHP FGGY_C FGGY_N

MDD3797508.1
JAQVDT010000396.1:1-2568
Lachnospiraceae bacterium

Phage_Mu_F PBECR2 DUF7302 DUF4355

MDD4834818.1
JAQUTA010000065.1:172-12476
Lutispora sp.

? ? FR47 ? DNA_binding_1 ? zf-HC2 PBECR3 ? AzlC tRNA_edit ABC_membrane ABC_tran ABC_membrane ABC_tran

MDD6078323.1
JAQXLG010000225.1:1-15113
Clostridiales bacterium

Peptidase_M16_C Peptidase_M16 Peptidase_M16_C Glyco_hydro_94 GH94_b-supersand GH94_b-supersand Glycoamylase UPF0122 SRP54_N SRP54 SRP_SPB Ribosomal_S16 KH_KhpA-B PBECR2 RimM PRC_RimM tRNA_m1G_MT

MDD6193488.1
JAQXIY010000018.1:5234-11903
Lachnospiraceae bacterium

ABC_tran ABC_tran ECF-ribofla_trS ? zf-like Pribosyltran HAD_2 HTH_3 PBECR3

MDD6219569.1
JAQXIK010000007.1:5830-16183
Clostridia bacterium

HTH_18 Phage_integrase DUF7307 HTH_26 PBECR3 DUF2179 YitT_membrane Ribonuclease_3 DUF1967 MMR_HSR1 GTP1_OBG Aldolase PfkB Fer4_21 Fer4_9 Fer4_21 FeS Rnf-Nqr

MDD6235406.1
JAQXIF010000027.1:27159-49707
Lachnospiraceae bacterium

CotJB Peptidase_M23 Phage_r1t_holin XhlA LPD3 T4_baseplate HicB_lk_antitox PBECR3 HicB_lk_antitox Resolvase Recombinase Zn_ribbon_recom SPOR Peptidase_M23 Phage_r1t_holin XhlA MuF_C ?

MDD6319279.1
JAQXGE010000003.1:1624-17278
Oscillospiraceae bacterium

DctP DctP HTH_18 Receiver His-Kinase His_kinase HAMP YcxB ? ? PBECR2 Phage_scaffold ? ? BPD_transp_1 BPD_transp_1 SBP_bac_1 Receiver Trans_reg_C

MDD6446711.1
JAQXZV010000018.1:3439-18257
Ruminococcus sp.

? ABC_membrane ABC_tran Virul_fac_BrkB ? PBECR2 Cys_Met_Meta_PP HTH_1 LysR_substrate SCP2 Sel1 ? HTH_1 LysR_substrate DUF2714

MDD6673622.1
JAQXVK010000046.1:3276-13115
Eubacteriales bacterium

? PrgI TrbL_2 PBECR3 DUF4143 AAA_14 TrbC T4SS-DNA_transf ? ArdcN DUF3801 Relaxase

MDD6795093.1
JAQXTC010000119.1:8877-15451
Clostridiaceae bacterium

PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

MDD6845170.1
JAQXSD010000131.1:1-6831
Clostridia bacterium

Gln-synt_N PGA_cap Polysacc_deac_1 Polysacc_deac_2 PBECR3 MFS_1 Glyco_trans_1_2 ParA

MDD7383611.1
JAQYDD010000012.1:3215-12827
Peptoniphilaceae bacterium

? Terminase_2 Terminase_6N Terminase_6C SPP1_portal Phage_Mu_F DUF4258 ? ? PBECR3 DUF4355 Phage_capsid ? Phage_connect_1 DUF3213 ? Tail_completion_gp17

MDD7410143.1
JAQYST010000009.1:4918-17098
Fusobacteriaceae bacterium

? Sigma70_r4 Terminase_2 Abi_alpha ? Terminase_6N Terminase-T7_RNaseH-like Hint_2 BPS1 SPP1_portal Phage_Mu_F PBECR2 ? MSC Phage_scaffold ? ? ? HK97-gp10_like

MDD7647493.1
JAQYOS010000271.1:1-10203
Ruminococcus bromii

TrbC SMC_N ABC_tran MatE MatE TetR_N TetR_C_8 PBECR2 HhH-GPD OGG_N DUF6062 SpoU_sub_bind SpoU_methylase ?

MDE5587117.1
JAIDLX010000197.1:410-1545
Bacilli bacterium

PBECR3 ThiI

MDE5718039.1
JAIDJT010000546.1:54-5922
Lachnospiraceae bacterium

HicA_toxin ? PBECR3 Phage_int_SAM_3 Phage_integrase DUF932 ? ? ? Mor

MDE5741062.1
JAIDIY010000057.1:279-10203
Oscillospiraceae bacterium

Pribosyltran_N Pribosyl_synth Pept_tRNA_hydro UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF ? DUF4179 tRNA-synt_2b HGTP_anticodon ProRS-C_1 Calcineurin PBECR3 Ribosomal_S6 SSB Ribosomal_S18

MDE5939753.1
JAIDFA010000063.1:1-7476
Lachnospiraceae bacterium

PBECR3 RIO1 MatE MatE AraC_binding HTH_18 HTH_3 Lactamase_B PGI Glyco_tran_28_C Glyco_transf_28

MDE6064081.1
JAIDCO010000417.1:269-3275
Lachnospiraceae bacterium

HicB_lk_antitox CcdA PBECR3 Resolvase Recombinase Zn_ribbon_recom GvpG NMT1_2

MDE6837159.1
JAIEKV010000173.1:1-2037
Acutalibacter sp.

HGTP_anticodon PBECR2 ? Zn_ribbon_IS1595 GntR ABC_tran

MDE6964045.1
JAIEJN010000752.1:1-2201
Lachnospiraceae bacterium

? ABC_tran GntR PBECR3

MDE7231182.1
JAIEDJ010000354.1:65-3306
Oscillospiraceae bacterium

HD_5 PBECR2 Cytidylate_kin2 ? HTH_3

MDE7234127.1
JAIEDL010000302.1:1-1906
Ruminiclostridium sp.

PBECR2 Enolase_C Enolase_N

MDE7247373.1
JAIEDN010000308.1:78-4097
Lachnospiraceae bacterium

Mrr_N Mrr_cat PBECR2 GGDEF GATase_7 A2L_zn_ribbon DmmA-like_C

MDF2474072.1
JAQZAX010000109.1:3725-9018
Anaerocolumna sp.

PBECR3 ? Acetyltransf_CG FeoA FeoB_N FeoB_Cyto Gate FeoB_C Gate FeoB_associated ? HD HD_assoc

MDF2540144.1
JARBTJ010000109.1:333-4167
Herbinix sp.

PBECR3 ? Ntox17 DUF3597 TEN_YD-shell RHS_repeat TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat RHS_repeat

MDF2609577.1
JARBUF010000024.1:145-16351
Lachnospiraceae bacterium

NAD_binding_1 DHODB_Fe-S_bind Fer4_20 Pyr_redox_2 GGDEF EAL ? DUF5688 sCache_2 HAMP MCPsignal YetF_N DUF421 PBECR3 ? S4 PseudoU_synth_2 Nlig-Ia DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_2 BRCT HTS

MDF2613058.1
JARBUI010000003.1:84781-100313
Clostridia bacterium

DUF1576 DUF1576 ? VanW PG_binding_4 ? ? ZT_dimer Cation_efflux ? PBECR3 DeoC CheW Receiver tRNA-synt_His HGTP_anticodon N-OB tRNA-synt_2 C_GCAxxG_C_C Acetylase DUF5655 Amino_oxidase

MDF2877158.1
JAQSYC010000013.1:1-12961
Clostridia bacterium

Na_H_Exchanger GTP_EFTU EF-G_D2 RF3_C IF3_N IF3_C Ribosomal_L35p Ribosomal_L20 GGDEF PBECR3 ? ? Zn_ribbon_RPB9 Resolvase HTH_31 HsdM_N N6_Mtase Methylase_S Methylase_S

MDL2250416.1
JAJDHQ010000029.1:18616-27678
Lachnospiraceae bacterium OttesenSCG-928-J05

HTH_38 PBECR3 AAA_15 AAA_15 MethyltransfD12 DpnII N6_N4_Mtase MethyltransfD12 N6_N4_Mtase N6_N4_Mtase HTH_3

MDO4301555.1
JAUNGN010000024.1:22460-32792
Clostridia bacterium

Zn_Ribbon_1 ? ? DUF1492 Terminase_2 Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F PBECR2 Phage_scaffold Gp13-like ? ? HK97-gp10_like Phage_tail_terminator_8 YqzN_YkzM

MDO4318311.1
JAUNGI010000027.1:13333-24552
Lachnospiraceae bacterium

? Sigma70_r4 DUF6731 ? HTH_23 Terminase_6N SPP1_portal Phage_min_cap2 PBECR2 Phage_min_cap2 Peptidase_Prp ? Phage_capsid_4 ? DUF6751 Minor_capsid_2

MDO4554936.1
JAUNBT010000058.1:70-8072
Lachnospiraceae bacterium

ABC2_membrane_2 DrrA1-3_C ABC_tran TetR_N Nitroreductase TrmO_N TrmO_C DUF1275 GntR FCD PBECR2 NifU GntR FCD Methyltransf_11

MDO4671240.1
JAUMZL010000030.1:6379-10631
Aerococcus sp.

Phage_cap_E Rho_N Phage_connect_1 ? HK97-gp10_like ? Phage_holin_Dp1 PBECR3 Amidase_2

MDO4813522.1
JAUMWC010000001.1:61430-72811
Gemella sp.

UPF0014 Rhodanese HIRAN ? HGTP_anticodon tRNA-synt_2b GlyRS_ins_1 GlyRS_ins_1 Zn_ribbon_DnaG DNAG_N Toprim_4 Sigma70_r1_1 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 PBECR3 ROK PhzC-PhzF FlK NUDIX DUF2929 SNase TsaE

MDO4864283.1
JAUMVH010000131.1:20-7655
Ruminococcus sp.

Ribosomal_L13 Ribosomal_S9 Rce1-like Glycos_transf_2 ? ? SBP_bac_8 PBECR2 ?

MDO4953094.1
JAUNNW010000075.1:1-6808
Synergistaceae bacterium

DUF1376 ERF ? PDDEXK_1 ? HTH_3 HTH_3 Peptidase_S24 PBECR3 CRISPR_Cas2 ? rve_2 rve HTH_21

MDO5344408.1
JAUNON010000003.1:91374-104604
Lachnospiraceae bacterium

? ? ? RNHCP HTH_31 HTH_3 Peptidase_S24 Uma2 PBECR3 Resolvase Zn_ribbon_recom GvpG SH3_3 SH3_3 LptM_cons Amidohydro_1 C_GCAxxG_C_C SH3_3 SH3_3 IMS IMS_HHH IMS_C Peptidase_S26

MDO5416083.1
JAUNPZ010000011.1:37858-45356
Lachnospiraceae bacterium

ABC_tran PHP AP_endonuc_2 Band_7_1 Zn_Ribbon_TF DNA_ligase_ZBD TPM_phosphatase Gate FeoB_C Gate FeoB_N PBECR3

MDO5775838.1
JAUNWP010000022.1:14619-30912
Eubacteriales bacterium

Pribosyltran DHHA1 Rab5-bind tRNA_SAD tRNA-synt_2c ? dCMP_cyt_deam_1 4HBT PBECR3 His-Kinase His-Kinase HAMP Trans_reg_C Receiver UvrD_C UvrD_C UvrD-helicase S1 KH_1 RNase_PH_C RNase_PH PNPase RNase_PH_C RNase_PH PGA_cap VTT_dom DUF2442

MDO5805530.1
JAUNXB010000004.1:175573-183736
Eubacteriales bacterium

Xan_ur_permease Pribosyltran ? LMWPc PALP TGT Queuosine_synth PBECR3

MDP2359049.1
JAUXWZ010000201.1:2828-12529
Beijerinckiaceae bacterium

? CbiA ? ? DUF4258 HTH_3 HTH_29 PBECR3 Resolvase Recombinase ? ? ? ? ? DUF5681

MDP4085785.1
JAUZPL010000043.1:17108-27393
Bacillota bacterium

Rep3_C Rep3_N ? ? DUF5763 GIY-YIG HTH_26 ? ParA-Soj-NTPase PBECR3 Phage_int_SAM_1 Phage_integrase Ccdc27 ? DUF2642 ? MTES_1575 HTH_26

MDR0445945.1
JAIRRO010000051.1:9771-19733
Oscillospiraceae bacterium

adh_short PBECR2 RrnaAD MatE MatE Glycos_transf_2 HIGH_NTase1 Acetate_kinase ? GATase Asn_synthase GMP_synt_C

MDR0489699.1
JAIRRA010000015.1:5137-16577
Oscillospiraceae bacterium

PBECR3 fn3 ? fn3 UGSC FxsA DctM NMT1_3 New_glue Peptidase_M24 Creatinase_N Peptidase_M24 Creatinase_N

MDR0603915.1
JAIROC010000176.1:1-3680
Bacteroidales bacterium

DDE_Tnp_IS1 Phage_integrase PBECR3 HTH_17 DDE_Tnp_1

MDR0858394.1
JAISLB010000088.1:91-10142
Oscillospiraceae bacterium

YoeB_toxin Fe-ADH ADH_Fe_C PIN ? ABC_tran TOBE_2 BPD_transp_1 BPD_transp_1 PBECR2 Peptidase_M23 ? RadC UPF0758_N AcnX_swivel_put AcnX_3rd AcnX_2nd AcnX

MDR0884131.1
JAISJQ010000046.1:135-11445
Oscillospiraceae bacterium

GtrA_DPMS_TM PBECR2 PBECR2 ? Acetyltransf_9 NadA ABC_tran ABC_tran YegS_C DAGK_cat YobI-ATPase DUF6273

MDR0897402.1
JAISIY010000094.1:1934-17207
Oscillospiraceae bacterium

Peptidase_M32 Pps Pps Fer4_5 Fer4_5 ? Redoxin Fer4_17 Aldo_ket_red HTH_26 PBECR3 ABC_tran FtsX FtsX FtsX FtsX FtsX NgoMIV_restric DNA_methylase Ribosomal_L17

MDR0925385.1
JAISJE010000369.1:4250-12543

Hungatella sp.

PBECR3 ? DUF364 DUF4213 BATS Radical_SAM MoeA_C MoCF_biosynth MoeA_N Peripla_BP_2 ABC_tran FecCD

MDR1068244.1
JAISEX010000131.1:422-2803
Eubacteriales Family XIII. Incertae Sedis bacterium

Acyltransferase PBECR3

MDR1627512.1
JAISWM010000032.1:1-3037
Oscillospiraceae bacterium

TerL_nuclease PBECR3 ? Phage_portal HTH_Tnp_4

MDR1629239.1
JAISXG010000027.1:151-18506
Oscillospiraceae bacterium

? Phage_holin_5_2 ? YomN PhageMin_Tail ? ? PBECR3 Synaphin Dockerin_1 Gp53-like_C P_C10 DUF6558 DUF6558_C SASP LGT

MDR1703804.1
JAISVU010000101.1:1-7765
Clostridiales bacterium

SEC-C VanY Hpt PAP2 PBECR3 DUF4097 SHOCT-like_DUF2089-C DUF4097 DUF4097 HAAS PadR ? CSD

MDR1915478.1
JAISRE010000033.1:22494-31225
Synergistaceae bacterium

? Helicase_C DEAD Alpha-amylase_C Alpha-amylase Alpha-amylase CBM_48 EamA EamA Int_N Phage_integrase ? HTH_17 PBECR3

MDR2277386.1
JAITLS010000099.1:23284-37376
Vagococcus sp.

YomN Phage_integrase HTH_3 ? PhageMin_Tail ? Zn_Ribbon_1 Ribosomal_L12 PBECR3 DUF6558 DUF6558_C ?

MDR2590597.1
JAITFR010000160.1:1-3313
Oscillospiraceae bacterium

Acetylase TTRAP Topoisom_bac Toprim PBECR3 CD1107-like HTH_36 ?

MDR3354502.1
JAISLO010000145.1:1-11660
Synergistaceae bacterium

? PAPS_reduct PAPS_reduct MarR_2 HTH_36 ? Phage_pRha ANT ? PBECR3 T4SS-DNA_transf KfrB KfrB Helicase_C Helicase_dom

MDU6573202.1
JAWHCX010000006.1:1-11218
Gemella haemolysans

AA_permease_2 sCache_3_2 SPOB_a His-Kinase Receiver HTH_11 ? ? ? PrmA PBECR3 Y1_Tnp

MDU7631965.1
JAWFTO010000013.1:67622-85695
Lachnospiraceae bacterium

FliG_C FliG_M FliG_N YscJ_FliF_C YscJ_FliF FliE Flg_bb_rod SLT PilZ DEDD_Tnp_IS110 PBECR2 Flagellin_C Flagellin_N Resolvase HTH_28 Flagellin_N Flagellin_C Flagellin_N Flagellin_C CxxCxxCC FliB_C Bac_export_2 ?

MDX1949775.1
JANRLM010000062.1:2818-19957
Rickettsiales bacterium

? IMPDH ? Wzz TolA_bind_tri DAHP_synth_1 Aminotran_1_2 ? PBECR3 Plug TonB_dep_Rec_b-barrel TonB_C HemS HemS QueC Methyltransf_23 GDT1 Trigger_N FKBP_C Trigger_C

MDX2103902.1
JANRKE010000126.1:53267-60974
Alphaproteobacteria bacterium

PBECR3 ArsC DUF1203 ANTAR Receiver Glycos_trans_3N TP_methylase CysG_dimeriser Sirohm_synth_M NAD_binding_7 Fer2_BFD Molydop_binding Molybdopterin Molybdop_Fe4S4 Rieske

MDY3817917.1
JAXEKS010000044.1:33308-49767
Lachnospiraceae bacterium

? DUF3592 ? DUF3592 ? ? HTH_26 PBECR3 Calcineurin Bmp NAD_binding_1 DHODB_Fe-S_bind Fer4_20 Pyr_redox_2 Aminotran_1_2 DUF5717_N DUF5717 DUF5716_C

MDY4010192.1
JAXEGQ010000011.1:413-3116
Fusobacterium gastrosuis

? RelB PBECR3 ?

MDY5665296.1
JAXFCI010000240.1:1-8022
Blautia sp.

HTH_3 PBECR2 Sigma70_r1_1 ? TrkA_C TrkA_N TrkH DUF4118 His-Kinase His-Kinase Receiver Trans_reg_C Amidohydro_1 Adenine_deam_C

MDY5986927.1
JAXFQT010000165.1:2-7999
Lachnoclostridium sp.

PBECR2 PALP Nitroreductase Metallophos_2 ABC_tran SUFBD_core DUF2238 Ribonuclease

MDY6064946.1
JAXFOR010000005.1:1-5787
Finegoldia sp.

VG15 DUF4258 DUF1871 PBECR3 YjcQ NAT10_TcmA_helicase Phage_scaffold Phage_capsid PADR1_N Phage_Gp19 ?

MEA4846142.1
JAYFRB010000060.1:72-9427
Clostridiaceae bacterium

Acetyltransf_4 NDK PEMT DUF2776 ? ? ? ? PBECR3 S1 KH_1 RNase_PH_C RNase_PH PNPase RNase_PH_C RNase_PH Ribosomal_S15 Flavokinase FAD_syn PUA TruB_C_2 TruB_N

MEA4882955.1
JAYFQW010000011.1:10944-21938
Clostridia bacterium

DDE_Tnp_1 ? MatE ? PBECR3 NUDIX Thioredoxin_3 ? PemK_toxin Peptidase_M78 HTH_3 DDE_Tnp_1 IstB_IS21

MEB8615834.1
JARKVW010000202.1:7879-15330
Bacillus cereus

Cu_amine_oxidN1 ? VbhA ? RepB_primase SgrR_N DsbD_TM ? PBECR3

MED9966498.1
JAXJKY010000042.1:11539-26412
Blautia sp.

? ? Arc DUF739 ? HTH_3 Peptidase_S24 MORN MORN PBECR3 Peptidase_M78 PBECR3 Resolvase Recombinase Zn_ribbon_recom CotJB CotJA TrkH Radical_SAM Beta_elim_lyase ACT_8

MEE0048060.1
JAXJHQ010000119.1:1-3851
Eubacteriales bacterium

Terminase-T7_RNaseH-like SPP1_portal Phage_Mu_F PBECR2 Phage_scaffold

MEE0515509.1
JAXIZZ010000311.1:1431-12192
Emergencia sp.

? ? AAA_25 Toprim_2 Zn_ribbon_DnaG ? HTH_17 HTH_26 HTH_3 PBECR3 Phage_integrase ? Fer4 Fer4_5 Fer4_5 ? Redoxin BPD_transp_1 ABC_tran

MEE0686507.1
JAXIWY010000174.1:1-1859
Lachnospiraceae bacterium

Polysacc_deac_2 PBECR3 Bact_surface_Ig-like YkuD

MEE0695137.1
JAXIXN010000013.1:80283-101155
Fusicatenibacter saccharivorans

TauE Amidohydro_1 Lys_export DUF3100 cNMPBD HTH_Crp_2 Peptidase_U32 PDDEXK_9 AAA-ATPase_like PBECR3 HKD MZB Helicase_C Helicase_C SNF2-rel_dom DNA_methylase DNA_methylase YagJ ResIII

MEE1137544.1
JAXJTD010000044.1:88355-100316
Acutalibacteraceae bacterium

? HNH_repeat DUF4430 Prenyltrans ? ? Sipho_Gp157 HTH_26 PBECR3 Sigma70_r4 Sigma70_r2 G39P_N ? HTH_17 TubC_N HTH_19 Phage_int_SAM_3 HTH_26 Phage_integrase

MEE1225195.1
JAXJRQ010000188.1:118-4922
Clostridia bacterium

GerA ParA-Soj-NTPase ParB_N HTH_ParB ParB_C DUF4446 XLF_CC PBECR3

MEE1367591.1
JAXJOQ010000017.1:7199-16140
Evtepia sp.

RL10_C Ribosomal_L10 Ribosomal_L1 Ribosomal_L11 Ribosomal_L11_N KOW NusG SecE Ribosomal_L33 Radical_SAM PBECR2 DUF2798 DUF2798 F420_ligase Ig_3 SLH

MEE1552642.1
JAXJMJ010000208.1:2382-14317
Lachnospiraceae bacterium

TraB_PrgY_gumN ? ? PAP2 Phosphorylase Glyco_hydro_77 HlyIII PBECR2 DDE_Tnp_ISL3 HTH_Tnp_ISL3 ABC_tran

MEE3399247.1
JAZAUF010000159.1:1-4868
Eubacterium sp.

DNA_pol3_alpha DNA_pol3_alpha DNA_pol3_finger HHH_6 PBECR3 HAD_2 HMGL-like

MEE3436121.1
JAZASJ010000028.1:15612-29766
Treponema sp.

Epimerase ADH_Fe_C Fe-ADH AraC_binding HTH_18 RE_endonuc ResIII T3RM_EcoP15I_C N6_N4_Mtase HTH_3 HTH_3 PBECR3 Flavodoxin_3 Staph_reg_Sar_Rot DUF1349 Cyclase Acetylase LPG_synthase_C Acetylase Acetyltransf_7

MSR90372.1
VULX01000002.1:95994-112456
Inconstantimicrobium porci

AAA_22 Mu-transpos_C HTH_TnsB TnsB_N Tn7_Tnp_TnsA_C Tn7_TnsA-like_N SIS SIS GATase_6 VanY PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I DUF4004 PBECR3 DUF8106 ? Acetylase PNPase ? Hydrolase Acetate_kinase

NLC43680.1
JAAZEK010000196.1:1-1751
Clostridiales bacterium

Mn_catalase ? PBECR3

NLL66263.1
JAAYNX010000023.1:12124-26057
Clostridiaceae bacterium

BUD22 Polysacc_synt MatE HSP20 PHP PCRF RF-1 Zip Zip Acetylase PBECR3 DUF3298 PdaC CYTH Sod_Cu NTP_transferase GDP_Man_Dehyd Glycos_transf_2 Glyco_transf_7C ?

NLZ35132.1
JAAZIM010000225.1:13808-28154
Clostridiales bacterium

BPD_transp_1 BPD_transp_1 ABC_tran PhoU PhoU MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C

NMC56957.1
JAAZOI010000220.1:108-5424
Eubacteriaceae bacterium

CGGC CGGC DUF554 PBECR3 Abi_C UvrD_C UvrD-helicase

NOJ74025.1
JABFOR010000069.1:933-13567
Paenibacillus alvei

MerR_1 ? ? ? ? PBECR3 Hol_Tox ? ? ? Cu_amine_oxidN1 CARDB PXO2_28_C HTH_3

NUB12292.1
WFKB01000057.1:1-12587
Azospirillum brasilense

? Phage_pRha ? HTH_3 CSD TPRs AP-ATPase ? PBECR3 Phage_YdaY ? Phage_tube_2 HTH_OrfB_IS605 OrfB_IS605 Cas12f1-like_TNB Phage_tail_terminator_5 HK97-gp10_like Bac_DNA_binding

OBS51774.1
LXWX01000106.1:211-15091
Methylosinus sp. 3S-1

? Asn_synthase ? Ribosomal_L35p Ribosomal_L20 PBECR3 Pyr_redox_2 Pyr_redox_dim ? MBOAT Abhydrolase_6 PRC CitMHS MFS_1 MFS_1

OGO77454.1
MGOS01000088.1:1005-15699

Clostridiales bacterium GWB2_37_7

ARM_TBCD_2nd ? Imm63 ? Glyoxalase_3 Metallophos_2 PBECR3 ? ? ? Acetylase Methyltransf_25 ? Methyltransf_31

OGO85619.1
MGOV01000029.1:26504-40336

Clostridiales bacterium GWD2_32_59

HycI HypA cobW Acylphosphatase zf-HYPF zf-HYPF Sua5_yciO_yrdC HypF_C HypF_C_2 HupF_HypC HypD AIRS AIRS_C PBECR3 DUF3592 Thioredox_DsbH Glyco_hydro_88 Glyco_hydro_76 ? ? Ribosomal_L7Ae Ribosom_S12_S23 Ribosomal_S7

OLA14846.1
MNSL01000001.1:38135-49782

Eubacterium sp. CAG76_36_125

? DnaJ ? DUF1492 ? ? CopG_antitoxin PBECR3 ? Resolvase ? ? ? N6_Mtase Methylase_S ?

OUO29315.1
NFIS01000009.1:21714-37913
Lachnoclostridium sp. An298

? ? VTT_dom Glyco_transf_4 Glycos_transf_1 DUF374 Receiver Trans_reg_C His-Kinase His-Kinase PBECR2 AIM24 AAA_3 AAA_lid_2 vWA Transglutaminase BATS Radical_SAM Radical_SAM ?

OZQ60411.1
NFUY01000121.1:211-12697

Paenibacillus taichungensis

HTH_24 DUF5704 Cu_amine_oxidN1 ? Phage_integrase Phage_int_SAM_2 DUF4200 PBECR3 Resolvase AbiEii ?

PID67419.1
PDQC01000006.1:5750-10979
Fusobacteriales bacterium

DUF4355 Phage_capsid_4 ? ? ? Phage_tube_2 ? PBECR2 PhageMin_Tail

PPS80021.1
LQZI01000043.1:313-8881
Clostridioides difficile

Cupin_2 ? Methyltransf_11 tRNA_edit PBECR3 ? DUF948 ? Amidase_3 dCMP_cyt_deam_1 ? ?

PRR84132.1
PVXQ01000003.1:81650-96108
Clostridium vincentii

BPD_transp_1 ABC_tran PhoU PhoU MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C Aminotran_4 PBECR3 YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C ?

QYA34109.1
CP079956.2:126219-141839
Macrococcus psychrotolerans

? ? SMODS-polBeta Prok-E2_B ThiF HTH_38 rve Prok-JAB Cap15_TM Cap15_CD_rec PBECR3 HTH_7 Resolvase Dala_Dala_lig_N Dala_Dala_lig_C ? ? RadC ? ?

RGF36173.1
QTVA01000010.1:60268-73380
Ruminococcus sp. AF43-11

Voltage_CLC ? ? ? FAD_binding_2 QRPTase_N QRPTase_C PHP PBECR2 ? ? His-Kinase His-Kinase HAMP Trans_reg_C Receiver dCache_1 HTH_30 ?

RGF89631.1
QTVU01000094.1:1-4539
Firmicutes bacterium AM55-24TS

AbiH ? ? Tn916-Xis PBECR2 ?

RGT26550.1
QRWM01000005.1:67707-77367
Thomasclavelia ramosa

? ? ? ? ORF6C Bro-N HTH_26 ? PBECR3 ? Arm-DNA-bind_4 Phage_int_SAM_3 Phage_integrase Acyl-ACP_TE_C Acyl-ACP_TE PGA_cap VanY ? ABC_tran

RJW87145.1
QTLT01000008.1:65723-81004
Clostridiales bacterium AF36-10

Imm17 Aminotran_1_2 AI-2E_transport SpoIID PG_binding_1 ? Phage_terminase ? Phage_Mu_F PBECR2 ? ABC_trans_CmpB Phage_holin_4_1 GerE Receiver ? ? ?

RKI22573.1
RAYF01000087.1:830-12899
bacterium D16-36

TetR_N TetR_C_17 Lactamase_B_2 Acetylase ? ? ? ? PBECR2 Gins15_N ? TTRAP PadR DUF2992 HTH_18 Transglutaminase

RYI25094.1
SEHK01000024.1:5446-17058
Bacillus infantis

? ? ? ? ? PhageMin_Tail Baculo_PEP_C HTH_36 PBECR3 Replic_Relax Replic_Relax FtsK_SpoIIIE ? HTH_17 ? MTES_1575 DUF3967

RYI30615.1
SEHK01000003.1:541421-558583
Bacillus infantis

YomX_N YomX_C ? YomV_2nd YomU_Tube YomO YomN Phage_int_SAM_4 PBECR3 RHH_3 ? HNH_repeat PH_gly_zipper PhageMin_Tail PhageMin_Tail SLT Peptidase_M23 DUF6558 DUF6558_C

SCJ70309.1
FMGH01000015.1:2212-13428
uncultured Flavonifractor sp.

? ? DUF1492 Terminase_2 Terminase_3 Terminase_3C ? SPP1_portal Phage_Mu_F PBECR2 ? Bro-N HTH_3 Phage_scaffold ? ? ? HK97-gp10_like

SDZ90678.1
FNRK01000001.1:76198-88790
Eubacterium aggregans

AmiS_UreI ? ? cobW UreD DUF5986 HTH_3 Peptidase_M78 PBECR3 ? Beta_elim_lyase PAS NtrC-AAA AAA_lid_14 HTH_50 Lactamase_B HpaB_N HpaB NifU AF_0334 FabA

TAH69998.1
SHOH01000040.1:4229-23437
Anaerolineaceae bacterium

HAD_2 Glyco_hydro_94 Glyco_hydro_94 Glyco_hydro_94 GH94_b-supersand HTH_18 Receiver His-Kinase His_kinase HAMP SBP_bac_1 DUF6491 BPD_transp_1 BPD_transp_1 PBECR3 Nitroreductase Nitroreductase GH43_C Glyco_hydro_43 Glyco_hydro_43 GFO_IDH_MocA_C GFO_IDH_MocA AP_endonuc_2 GFO_IDH_MocA_C3 GFO_IDH_MocA AP_endonuc_2 HTH_18 AraC_binding

UVM80915.1
OP030747.1:12462-7177

Bacteriophage sp.

? ? ? ? Terminase_6C ? SPP1_portal PBECR3 ? ? Phage_integrase Phage_int_SAM_4 DUF6471 ArdA ? ?

UYJ02454.1
CP107213.1:1100877-1119215
Faecalibacterium prausnitzii

Calcineurin RecG_dom3_C Helicase_C DEAD RecG_wedge ? ? ? DNA_topoisoIV SPOUT_MTase PBECR3 ? GlyRS_ins_1 GlyRS_ins_1 tRNA-synt_2b HGTP_anticodon ? Peptidase_M29 RimP_N DUF150_C NusA_N S1 KH_NusA_1st KH_NusA_2nd YlxR

WP_000642527.1
NC_018489.1:3704-566
Bacillus thuringiensis HD-771

? ? ? ? ? ? MerR_1 PBECR3 ? DUF4013 Phage_pRha DDE_Tnp_Tn3 DUF4158 Phage_integrase Phage_int_SAM_1 Phage_pRha ?

WP_000104828.1
NZ_CM000734.1:5591723-5606540
Bacillus cereus Rock4-2

Sensor XG_FTase HisKA_3 His-Kinase Receiver GerE Abhydrolase_6 ? SdpI DUF1648 Phage_integrase HTH_36 PBECR3 HTH_31 HTH_3 ? ? ? Rep_1 ?

WP_309419149.1
NZ_JARLYV010000054.1:20-7210
Bacillus paranthracis

HTH_3 ? ? RepD-like_N Rep_trans ? Acetylase HTH_36 PBECR3

WP_326269284.1
NZ_JARSXY010000083.1:471-6258
Bacillus thuringiensis

PBECR3 ? DUF4013 ? ? RepA_N ? ?

WP_002579570.1
NZ_AP019716.1:2647159-2661679
Clostridium butyricum

ABC_tran PhoU PhoU ? PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_294371076.1
NZ_CAKVKB010000002.1:12081-28980
uncultured Clostridium sp.

Ppx-GppA PP_kinase_N PP_kinase PP_kinase_C_1 PP_kinase_C MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C Sigma70_r2 Sigma70_r4_2 DUF4179 DUF5643 SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_195970164.1
NZ_CAKVNE010000003.1:9981-23985
uncultured Clostridium sp.

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease 2H YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6

WP_297631760.1
NZ_CAMTIG010000007.1:9043-23537
uncultured Clostridium sp.

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_118001987.1
NZ_JAHOLM010000023.1:15733-23147
[Clostridium] innocuum

? HTH_19 ? ? ? ? ? PBECR3 ? DnaB_C DnaB G39P_N DUF6291 ? HTH_17 TubC_N

WP_014632713.1
NC_017594.1:397068-414925
Streptococcus salivarius 57.I

FtsX MacB_PCD FtsX OppC_N ABC_tran ? YbaB_DNA_bd ? ? E1-E2_ATPase Hydrolase PBECR3 DUF1430 ABC_tran AAA_15 CM_2 Anth_synt_I_N Chorismate_bind GATase Glycos_trans_3N Glycos_transf_3 IGPS

WP_232515431.1
NZ_CP022558.1:1-6785
Bacillus velezensis

RapH_N RapH_N ? Acyl_LnsA DUF3139 Mob_Pre ? DUF3967 MerR_1 HTH_36 PBECR3

WP_003353526.1
NZ_JARLVJ010000169.1:199-1744
Bacillus smithii

? PBECR3

WP_033063452.1
NZ_ABDP01000048.1:122-9896
Clostridium botulinum Bf

Resolvase HTH_3 ? HTH_23 ? ? PBECR3 Resolvase

WP_012451486.1
NC_010723.1:1218472-1233256
Clostridium botulinum E3 str. Alaska E43

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 RsgA_GTPase MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_244361474.1
NZ_CP040410.1:1983466-9557
Dolosigranulum pigrum

CD1375-like CLIP1 Phage_holin_Dp1 Glucosaminidase ? ? ? PBECR3 ABC_tran ? SMC_N HTH_3 Inositol_P GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C BipA_C DUF1507

WP_009245571.1
NZ_BAABXV010000001.1:1869742-1878443
Mediterraneibacter gnavus

HTH_26 HTH_3 ? HTH_26 ? ? ? PBECR3 SatD ? ? ? ? ? DUF6551

WP_173864945.1
NZ_CAXUME010000002.1:10126-22545
Mediterraneibacter gnavus

? Tad2-like HTH_3 ? HTH_3 HTH_3 Peptidase_S24 BRCT PBECR3 Resolvase Recombinase Zn_ribbon_recom CotJB CotJA ABC_tran ABC_tran_Xtn ABC_tran Bact_surface_Ig-like Bact_surface_Ig-like Bact_surface_Ig-like Flg_new LRRs Discoidin DUF4317 DUF4317

WP_144366606.1
NZ_CABHNA010000038.1:11412-24083
[Ruminococcus] torques

? HTH_26 Glyco_hydro_25 ? DUF6814 T4SS_LegC2C7 ? PBECR3 DivIC His-Kinase His-Kinase Receiver ? FtsX FtsX ABC_tran His-Kinase His-Kinase Trans_reg_C Receiver

WP_257847211.1
NZ_JALBWL010000003.1:250728-264244
Veillonella faecalis

Flavodoxin_1 ODP ECF-ribofla_trS Multi_Drug_Res Iso_dh Aconitase_C Aconitase LeuA_dimer HCS_D2 HMGL-like PBECR3 ? ? PSD4 ? ? ? Oxidored_FMN

WP_144678302.1
NZ_JAYFNZ010000009.1:47076-57209
Desulfitobacterium hafniense

HD DUF6366 Acetylase ? DUF4070 Radical_SAM B12-binding Acetylase Acetylase PBECR3 3HCDH 3HCDH_N RibD_C ? Fic ? ? ?

WP_320950092.1
NZ_JAQDAI010000008.1:137462-154213
Fusobacterium polymorphum

ParA DUF454 Peptidase_S8 EamA EamA DUF6290 ParE_toxin ABC_tran ABC_tran_Xtn ABC_tran PBECR3 TIR TctC TctB TctA FGS DcuC GalKase_gal_bdg GHMP_kinases_N GHMP_kinases_C

WP_220308100.1
NZ_CP071099.1:2156011-2173668
Fusobacterium animalis

TctC TctB TctA ? DUF1413 DcuC GalKase_gal_bdg GHMP_kinases_N GHMP_kinases_C PBECR3 GalP_UDP_transf GalP_UDP_tr_C Epimerase ? MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 FDX-ACB tRNA_synthFbeta B5 B3_4 tRNA_bind

WP_285049437.1
NZ_JAMGTK010000031.1:5871-13922
Fusobacterium necrophorum

Terminase_6N Terminase-T7_RNaseH-like SPP1_portal Phage_Mu_F PBECR3 HicA_toxin HicB_lk_antitox ? Phage_pRha ? ? Phage_capsid_4

WP_264919285.1
NZ_CEVBYJ010000025.1:29315-37745
Selenomonas sputigena

HTH_11 BPL_LplA_LipB BPL_C Pan_kinase Dus Peptidase_M18 Cation_efflux ZT_dimer ? PNPase PBECR2 Rrf2

WP_265070345.1
NZ_CP110383.1:1838037-1850097
Selenomonas sputigena

MM_CoA_mutase B12-binding MeaB Glyoxalase_4 Carboxyl_trans ? Biotin_lipoyl PBECR2 ANKs ANKs ? RelB Cas_Cas1 CRISPR_Cas2 CRISPR_Cas6 Pyr_redox_2 Pyr_redox_dim

WP_022416094.1
NZ_JAAILL010000002.1:72571-91453
Dorea longicatena

DUF7577 PK PK_C TrmB Orn_Arg_deC_N Orn_DAP_Arg_deC ? SpaA SpaA TQ TQ TQ TQ ChW ChW ChW ChW PBECR3 ? 5_3_exonuc_N 5_3_exonuc DNA_polI_exo1 DNA_pol_A CoaE Lactamase_B_2 ACT_6 DUF711 HIGH_NTase1

WP_049947100.1
NZ_AUJS01000041.1:2738-17441
Dorea longicatena AGR2136

? ? ? DNA_pol_A ? ? ? PBECR2 ? ? SLFN_AlbA_2 HATPase_c_4 Fic-like_C LacI Peripla_BP_3 PTS_EIIB PTS_EIIC PTS_EIIA_1 Glyco_hydro_32N Glyco_hydro_32C PcrA_UvrD_tudor

WP_006693050.1
NZ_JH376799.1:482804-501075
Selenomonas infelix ATCC 43532

DNA_pol3_delta DNA_pol3_delta_C Radical_SAM Gly_radical PFL-like MFS_1 Flavodoxin_3 YgbA_NO Asparaginase Asparaginase_C PBECR3 TrmO_N Plug TonB_dep_Rec_b-barrel Pyr_redox_2 Glutaredoxin Glutaredoxin GntP_permease Ysc84

WP_006703550.1
NZ_CAUQVI010000012.1:148-15115
Granulicatella elegans

PCRF RF-1 ? ABC_tran FtsX_ECD FtsX tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C Mur_ligase_M Mur_ligase_C UPF0758_N RadC PBECR3 Dala_Dala_lig_N Dala_Dala_lig_C Mur_ligase Mur_ligase_M Mur_ligase_C DEAD Helicase_C

WP_006864308.1
NZ_ACCL02000031.1:195-15868
Marvinbryantia formatexigens DSM 14469

ATP-grasp_3 ATP-grasp_5 Esterase Esterase Peptidase_S11 Phage_integrase Trns_repr_metal RNase_T PBECR3 ABC_tran OB_MalK TOBE_2 Peripla_BP_4 HAMP His_kinase His-Kinase Receiver GGDEF_2 HTH_18 Peripla_BP_4 AAA_14

WP_007785012.1
NZ_CM001441.1:3663737-3681655
Desulfosporosinus youngiae DSM 17734

Sigma70_r4_2 Sigma70_r2 Cys_rich_CPCC Cation_ATPase_N E1-E2_ATPase Cation_ATPase Cation_ATPase_C YoaP Acetylase MFS_1 ? LigD_Prim-Pol DNA_ligase_A_C DNA_ligase_A_M LigD_N PBECR3 NTPase_1 Methyltransf_25 DUF364 DUF4213 ABC_tran cobW FecCD ABC_tran

WP_118589735.1
NZ_QUHB01000004.1:119797-129378
Eubacterium sp. AF16-48

? Mga HTH_3 HTH_26 HTH_3 HTH_3 Peptidase_M78 PBECR3 Arm-DNA-bind_4 Phage_int_SAM_3 Phage_integrase DUF2007 HTH_3 DUF3784 DUF6019 TnpV DUF4200 Mob_Pre DUF5348

WP_120140417.1
NZ_CP023671.1:222120-236750
Clostridium septicum

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 ?

WP_321823049.1
NZ_CAUZXV010000003.1:79038-91310
Pyramidobacter piscolens

? ? Cas_csx3 HTH_17 ? HTH_3 HTH_3 Peptidase_S24 PBECR3 Scy SWIM Phage_integrase Phage_int_M Arm-DNA-bind_3 Fer4_9 Aldo_ket_red Prismane Fer4_13 Flavodoxin_1 ODP DUF2249

WP_072449488.1
NZ_LT635470.1:773851-788258
Blautia sp. Marseille-P3201T

THR4_C PALP Thr_synth_N Hydrolase_2 LeuA_dimer IPMS_D2 HMGL-like ? ABC-ATPase GntR HTH_3 PBECR3 HKD HKD Aconitase_C Aconitase HTH_1 LysR_substrate KARI_C KARI_N ALS_ss_C ACT_AHAS_ss ATP-synt_DE ATP-synt_DE_N

WP_009441638.1
NZ_ALKG01000077.1:95-12456
Selenomonas sp. FOBRC6

TPRs TPRs Glucosaminidase Methyltransf_11 HD_3 PDT ACT DAHP_snth_FXD DAHP_synth_1 Ribonuc_L-PSP PK_C PK PBECR2 Na_Ala_symp DNA_pol3_delta2 DNAX_ATPase_lid DNA_pol3_gamma3 YbaB_DNA_bd RecR_HhH RecR_ZnF Toprim_4 RecR_C BofA

WP_009530011.1
NZ_JH414640.1:405-6109
Peptoanaerobacter stomatis

DUF6877 Rep3_C Rep3_N PhdYeFM_antitox ? ? ? ? PBECR3

WP_212669098.1
NZ_LT891958.1:608539-625545
Selenomonas felix

Na_Ala_symp HTH_1 LysR_substrate Macro VWA_CoxE BamE ? EcoR124_C T1R_D3-like SWI2_SNF2 REase PBECR2 Methylase_S Methylase_S N6_Mtase HsdM_N HepT-like HepT-like Polbeta ACR_tran

WP_010236308.1
NZ_BAEV01000033.1:12706-22057
Clostridium arbusti SL206

Helicase_C ? ? XhlA ? ? MethyltransfD12 PBECR3 ? DUF4236 DUF4077 ? HTH_7 Phage_integrase ? ?

WP_023973346.1
NZ_CP011966.3:1327513-1343376
Clostridium beijerinckii NRRL B-598

ABC_tran PhoU PhoU PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C Pep_deformylase

WP_173714476.1
NZ_JABSWC010000001.1:475195-488639
Clostridium beijerinckii

? XhlA COG5_N Glyco_hydro_25 HicA_toxin HicB_lk_antitox DUF2140 PBECR3 Amidase_3 Fumerase Fumerase_C TPRs TPRs Receiver Trans_reg_C His-Kinase His-Kinase Ribonuc_red_lgN Ribonuc_red_lgC

WP_192950989.1
NZ_CP150167.1:2594726-2611793
Geobacillus sp. FSL W8-0466

Uma2 PIN ? DDE_Tnp_1 Thioredoxin Phage_integrase Phage_int_SAM_1 HTH_36 PBECR3 DDE_Tnp_1_4 RadC ? ? DDE_Tnp_1 ? ?

WP_035067546.1
NZ_CP012152.1:2552617-2565718
Anoxybacillus gonensis

Bacteriocin_IIc ABHD18 ? DUF6444 zf-IS66 DDE_Tnp_IS66 FlxA His-Kinase Receiver GerE ? PBECR3 UPF0236 UPF0236 UPF0236 PIN ? HTH_Rpn_C Transposase_31 ? ?

WP_127723163.1
NZ_CP029758.2:4181731-4196750
Clostridium sp. AWRP

PALP Hexapep Hexapep QueG_DUF1730 Fer4_16 Acyltransferase HhH-GPD EndIII_4Fe-2S CinA CinA_KH MoCF_biosynth PG_binding_4 VanW PBECR3 EamA EamA NT5C Amidase_3 SpoU_methylase DegV Sigma54_AID Sigma54_CBD Sigma54_DBD CggR_N Sugar-bind

WP_217297202.1
NZ_JAHQYW010000007.1:13738-32031
Faecalibacterium prausnitzii

DHH DHHA2 Acylphosphatase ? Peptidase_C39_2 JetA DUF4194 AAA_29 SMC_hinge SbcC_Walker_B PBECR3 MatE MatE His_Phos_1 ? HTH_1 LysR_substrate CstA CstA CstA ? LysR_substrate HTH_1

WP_187002679.1
NZ_JACOQE010000001.1:726992-735644
Blautia intestinalis

? HTH_17 ? ? ? DUF739 HTH_3 PBECR3 MqsA_antitoxin Peptidase_M78 ? Arm-DNA-bind_4 Phage_int_SAM_3 Phage_integrase HTH_30 GGDEF_2 DUF2179 YitT_membrane PqqD

WP_117484779.1
NZ_JAVYJB010000003.1:156524-176498
Pilosibacter fragilis

YitT_membrane DUF2179 SBP_bac_3 ? ? ADDB_N UvrD_C PDDEXK_1 UvrD-helicase UvrD_C PDDEXK_1 ? PBECR2 Lactamase_B DHFR_1 NAD_binding_1 DHODB_Fe-S_bind ? ? GGGtGRT NifU_N

WP_018213415.1
NZ_KB900391.1:760159-776547
Desulfitobacterium hafniense TCP-A

Iso_dh DctM DctQ DctP DUF6530 MoxR-AAA HTH_3 Desulfoferrod_N PBECR3 UPF0236 Methyltransf_23 Methyltransf_25 Radical_SAM SPASM Abhydrolase_1 FMN_bind FAD_binding_2

WP_018393167.1
NZ_KB889752.1:1336654-1350417
Bacillus sp. 37MA

FecCD ? Big_14 ? MIP ? ? PBECR3 ? ? Big_14 DUF6366 EamA EamA DUF6241 DUF8407 Methyltransf_25

WP_018393228.1
NZ_KB889752.1:1384467-1401496
Bacillus sp. 37MA

cobW AmiS_UreI DsbD_2 PAP2 HTH_8 AAA_lid_14 NtrC-AAA Peripla_BP_5 ABC_tran MatE MatE PBECR3 Hydrolase CMD EamA EamA Pirin Flavodoxin_2 CMD Rrf2

WP_018395877.1
NZ_KB889756.1:320-9767
Bacillus sp. 37MA

? Phage_int_SAM_4 ? ? ? PBECR3 DUF3967 Rep_1 HTH_3 ? ?

WP_167693159.1
NZ_BCTR01000035.1:87270-107224
Sphingomonas azotifigens NBRC 15497

Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N MFS_2 MazE_antitoxin Fic ? ICL ICL HTH_3 Peptidase_M78 ScfRs PBECR3 Peptidase_S9 Peptidase_S9_N Phytase OMP_b-brl_3 Plug ? Peptidase_M16_C Peptidase_M16 Peptidase_M16_C Peptidase_M16 ? Vut_1

WP_276666177.1
NZ_QEWB01000014.1:31640-45668
Caldibacillus debilis

? HTH_3 DUF2178 CorC_HlyC CBS CBS CNNM Zn_peptidase_2 MerR-DNA-bind MerR_1 Aldo_ket_red PBECR3 Acetylase DJ-1_PfpI Maf GmrSD_N SLFN_AlbA_2 Mrr_cat

WP_021168133.1
NZ_AUIL01000031.1:38451-49249
Sporomusa ovata DSM 2662

? DUF3268 ? RNase_T BRCT ? HTH_3 HTH_3 Peptidase_S24 PBECR3 ? RNase_T SWIM DUF8426 ? Virulence_RhuM Resolvase Recombinase Zn_ribbon_recom Mn_catalase

WP_261812909.1
NZ_BROK01000020.1:22653-32614
Paraclostridium bifermentans

? Sigma70_r4_2 Sigma70_r2 ? ASCH Lipase_GDSL_2 Mor HTH_59 PBECR3 HTH_3 Amidase_3 Rce1-like DUF1659 DUF2922 YvrJ EcsC

WP_022216065.1
NZ_BLYL01000015.1:62-12143
Coprococcus eutactus

HSP20 AcetylCoA_hydro AcetylCoA_hyd_C Calcineurin CoA_trans CoA_trans HTH_3 PBECR3 ? RelB OAD_beta OAD_beta ? SpaA SpaA SpaA TQ TQ DUF961

WP_249298653.1
NZ_JACRSX010000037.1:901-6626
Jutongia huaianensis

PBECR3 ? HD_5 ? ? N6_N4_Mtase N6_N4_Mtase MethyltransfD12 ICEA

WP_022790378.1
NZ_ATUT01000017.1:19703-33779
Faecalicoccus pleomorphus DSM 20574

MatE MatE DHHW DHHW MBOAT DUF4358 Esterase PhoU PhoU ? PBECR3 Arm-DNA-bind_4 Phage_int_SAM_3 Phage_integrase Phage_integrase SmpB PGA_cap AP_endonuc_2 Transketolase_C Transket_pyr

WP_023353666.1
NZ_KI535367.1:461-9637
Catonella morbi ATCC 51271

GIY-YIG PBECR2 Receiver Trans_reg_C His-Kinase His-Kinase ? SMC_N ABC_tran ABC_membrane ABC_tran ABC_membrane ABC_tran FecCD

WP_314050288.1
NZ_CAUVSD010000005.1:57224-86915
Abiotrophia defectiva

YSIRK_signal CarboxypepD_reg DUF1542 Gram_pos_anchor ? CpXC PepSY PepSY ? Alpha-amylase_N Alpha-amylase Alpha-amylase Bac_A_amyl_C PBECR3 SBP_bac_8 DUF5683 BPD_transp_1 BPD_transp_1 DUF1189 DUF188 DeoRC HTH_DeoR Lactamase_B_2

WP_024880533.1
NZ_AZUO01000001.1:2807982-2824137
Methylosinus sp. LW3

? ChaC ? tRNA-synt_2e CopC CopD Pro_CA PBECR3 EipB_like YjgF_endoribonc GDPD Trypsin Endonuclease_NS DUF1269 Carb_kinase YjeF_N Cu-oxidase_2 Cu-oxidase_3

WP_242883037.1
NZ_FOIP01000001.1:3248031-3262239
Lacrimispora sphenoides

RusA ? YomN HTH_3 ? Phage_int_SAM_1 Phage_integrase PhageMin_Tail PBECR3 HTH_3 Bro-N ANT ? Zn_Ribbon_1 Choline_bind_3 Choline_bind_2 ? ?

WP_037353771.1
NZ_JNJF01000011.1:76305-90930
Selenomonas sp. FC4001

His-Kinase DNA_gyraseB Toprim DNA_gyraseB_C TerC Peptidase_M48 ? PALP Ppx-GppA DNA_topoisoIV DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C PBECR3 Malic_M malic ? ? ? DUF4181 LytR_cpsA_psr CYTH

WP_026859548.1
NZ_AUEF01000027.1:80-10203
Jeotgalicoccus psychrophilus DSM 19085

RHH_5 DUF6290 SpoIIM Resolvase HTH_7 HTH_36 PBECR3 Penicillinase_R MerR_1 Peptidase_S8 Ntox21 ? ?

WP_027097478.1
NZ_AUJC01000004.1:228690-245159
Clostridium paraputrificum AGR2156

dCache_1 HAMP MCPsignal PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C ? SpoIVA_ATPase SpoIVA_middle SpoIVA_C Aminotran_4 PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_027097478.1
NZ_JAWEFR010000002.1:112668-129137
Clostridium sp.

dCache_1 HAMP MCPsignal PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C ? SpoIVA_ATPase SpoIVA_middle SpoIVA_C Aminotran_4 PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_039206976.1
NZ_FQYW01000009.1:53196-72108
Anaerovibrio lipolyticus DSM 3074

PriA_3primeBD ResIII Zn_ribbon_PriA Helicase_C PriA_C Dynamin_N Dynamin_N REase SWI2_SNF2 T1R_D3-like N6_Mtase Methylase_S ? ? PBECR2 DAP_epimerase DAP_epimerase YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP STAS

WP_205088073.1
NZ_JACJLA010000012.1:16803-27690
Veillonella magna

? ? ? HTH_24 DUF739 HTH_3 Peptidase_M78 PBECR3 Resolvase Recombinase Zn_ribbon_recom SHOCT Lipoprotein_9 ? RNase_T OMPdecase Pribosyltran

WP_051589055.1
NZ_JMMA01000002.1:3143418-3160565
Ruminococcus sp. NK3A76

NUDIX NDPSase_N YfhO YfhO ? DUF6077 tRNA-synt_2b Seryl_tRNA_N HTH_ParB ParB_N ParA-Soj-NTPase PBECR3 ParB_N HTH_ParB Aminotran_1_2 CAP IFT57 ? ABC_tran DUF2705 ECF-ribofla_trS

WP_118059003.1
NZ_BHYQ01000003.1:599840-613023
Parvimonas micra

NUDIX Peptidase_M50 SMC_ScpA SMC_ScpB S4 PseudoU_synth_2 SH3_14 rRNA_methylase PBECR3 ? TetR_N DUF1254 DUF1214 Lactamase_B Alkyl_sulf_dimr Alkyl_sulf_C Acyl_transf_3 MMR_HSR1 RNase_HII

WP_320481777.1
NZ_JAXCWK010000004.1:111511-123642
Sphingomonas sp. CFBP9019

Peptidase_S78 Peptidase_S49 GAF_2 ? ? Peptidase_M78 HTH_3 ? PBECR3 ? ? MerR_1 ? Phage_integrase Ppx-GppA FtsJ

WP_031549195.1
NZ_JNJM01000004.1:97424-111122
Oribacterium sp. FC2011

? Glycos_transf_2 PS_pyruv_trans Glycos_transf_2 ? Radical_SAM ? NTP_transf_2 PBECR3 ? NTP_transf_2 ? HEPN HEPN Polbeta ?

WP_032118102.1
NZ_JACOOO010000008.1:58534-73043
Clostridium hominis

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 RsgA_GTPase MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_033154313.1
NZ_JNLH01000001.1:2351344-2365204
Pseudobutyrivibrio ruminis HUN009

? Ion_trans HNH_4 MazG-like SLFN-g3_helicase Lipoprotein_Ltp Lipoprotein_Ltp Lipoprotein_Ltp ? PBECR2 Zn_ribbon_Top1 NERD ? Mem_trans Mem_trans MOFRL DUF4147 GntP_permease HTH_30 Diacid_rec ?

WP_034553626.1
NZ_JRPC02000006.1:58365-71677
Helicobacter apodemus

LysR_substrate HTH_1 UvrD-helicase UvrD_C PcrA_UvrD_tudor ANKs Catalase-rel Catalase AIRS AIRS_C HI0933_like ? PBECR3 ? ? Autotransporter PATR Peptidase_S8 ? GemA HTH_ParB ?

WP_108897407.1
NZ_CP023704.1:469861-481330
Caldibacillus thermoamylovorans

GHMP_kinases_N GHMP_kinases_C FMN_dh FMN_dh Coat_F ? ? Coat_F NAPRTase_N NAPRTase NAPRTase_C PBECR3 CAP LysM ? UPF0176_N Rhodanese Rhodanese_C LrgB LrgA COX4_pro COX3

WP_052017245.1
NZ_DF239011.1:1385-5078
Exiguobacterium sp. OS-77

DUF5710 DUF8370 HTH_36 PBECR3

WP_256219532.1
NZ_FPJA01000004.1:651266-668030
Selenomonas ruminantium

HTH_3 HTH_17 HTH_36 ? ? Amidase_3 ? PBECR2 DUF1659 DUF2922 YvrJ ? ? Phage_integrase RadC

WP_051682649.1
NZ_JMKI01000018.1:168-9679
Synergistes jonesii

Transposase_mut LPD3 ART-PolyVal LPD22 PBECR3 ? T7_gp14 Phage_T7_Gp13 DUF6808 ?

WP_045725626.1
NZ_AP042432.1:1524169-1537317
Clostridium baratii

YlmH_RBD S4 DivIVA S4 PseudoU_synth_2 Pribosyltran RlmL_1st THUMP UPF0020 ? SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_303126988.1
NZ_JAUNLM010000001.1:66131-80609
Clostridium sp.

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_042169783.1
NZ_CBVJ010000129.1:97621-113552
Paenibacillus gorillae

DUF418 Receiver Trans_reg_C His-Kinase His-Kinase SLH SLH Sortase T4SS-DNA_transf HTH_36 PBECR3 ? Glyoxalase YjzC Ribosomal_S6 SSB Ribosomal_S18 DUF1641

WP_042396974.1
NZ_CYYT01000002.1:71468-86637
Clostridium disporicum

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 RsgA_GTPase MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_042396974.1
NZ_CASDPY010000284.1:1-7198
uncultured Clostridium sp.

SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin

WP_291651923.1
NZ_JAGHFX010000075.1:1412-16133
Clostridium sp.

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_317273460.1
NZ_JASCQJ010000018.1:47277-62573
Clostridium lamae

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C

WP_053083312.1
NZ_LFVU01000027.1:357521-369934
Clostridium cylindrosporum DSM 605

WHD_GTF3C1 ? DUF1653 RNase_T Rubrerythrin RUBY_RBDX Lip_prot_lig_C LplA-B_cat ? PBECR3 ArsC ? Sulfate_transp BIG1_C STAS Fer2_BFD Peptidase_S11 PBP5_C Acetylase Competence Lactamase_B

WP_285252795.1
NZ_JAPWBS010000007.1:74967-84049
Granulicatella sp. 20925_1_28

Hydrolase E1-E2_ATPase DUF6110 DUF4336 ? PBECR3 HI0933_like DUF7549 ? ? ? ? Ribonuc_L-PSP

WP_285252795.1
NZ_JASPDG010000002.1:29870-44270
Granulicatella sp. UMB5615A

SIS SIS GATase_6 IMPDH VanY Hydrolase E1-E2_ATPase DUF6110 DUF4336 ? PBECR3 HI0933_like DUF7549 ? ? ? ? ?

WP_050696558.1
NZ_LN868533.1:4799-12871
Anaeromassilibacillus senegalensis

? ? Zeta_toxin HTH_26 ? ? ? PBECR3 DUF2442 DUF4160 ? RepB-RCR_reg ? TerD DUF3795

WP_055292383.1
NZ_CYYT01000059.1:3430-12464
Clostridium disporicum

Phage_terminase Resolvase HTH_29 ? ParE_toxin PhdYeFM_antitox HTH_24 DUF6115 PBECR3 ? ? NikA-like ?

WP_057978510.1
NZ_LKHP01000006.1:123416-134643
Caloramator mitchellensis

Trns_repr_metal Acetylase Acetyltransf_10 SHMT PBECR3 Peptidase_C69 Sigma70_r2 Sigma70_r4_2 Bps PHP ? Phosphodiest ATG22 ATG22 ATG22

WP_096919210.1
NZ_CABJDW020000010.1:16960-28329
Eubacterium maltosivorans

DUF515 SPOUT_MTase Aldedh Aldedh Hydrolase ABC_tran SUFBD_core EamA EamA PBECR2 DegV ? Peptidase_M73 Peptidase_M73 ? Peptidase_S26 ?

WP_059047469.1
NZ_LN909066.1:1266200-1280267
Paenibacillus rubinfantis

BclA_C RlaP RlaP adh_short_C2 SLH SLH HTH_20 MFS_1 HTH_17 PBECR3 FtsX MacB_PCD FtsX ABC_tran His-Kinase His-Kinase Trans_reg_C Receiver SDH_alpha ACT SDH_beta

WP_059086612.1
NZ_CAXSKA010000025.1:36369-52502
Blautia massiliensis (ex Durand et al. 2017)

GLF NAD_binding_8 PS_pyruv_trans FrhB_FdhB_C FrhB_FdhB_N Fer4_7 GlfT2_domain3 Glyco_tranf_2_3 Glft2_N Glycos_transf_2 Glft2_N Glycos_transf_2 REase PBECR2 TatD_DNase ? ? MoaC tRNA_bind Anticodon_3 tRNA-synt_1g tRNA-synt_1g zf-Tim10_DDP ThrE_2

WP_059170220.1
NZ_LMTI01000001.1:279169-293494
Bacillus sp. FJAT-27445

Peptidase_S24 LexA_DNA_bind LysM Resolvase DUF896 Transketolase_N Transket_pyr Transketolase_C_1 SirA UPF0154 PBECR3 ABC_membrane ABC_tran ABC_membrane ABC_tran SpoOE-like DsbD_TM DUF1453 DUF2621 ?

WP_061401036.1
NZ_CP061203.1:2228742-2241273
Lysinibacillus capsici

zf-dskA_traR LysE MFS_1 MFS_1 BpsA_C ? ? ? PBECR3 Phage_integrase LPAM_1 YndA_coil ? Omega_Repress ? ?

WP_237072110.1
NZ_CP016792.1:4818-32586
Priestia flexa

MobA_MobL DUF3847 pre-PUA_NIP7 ? ? MerR_1 DUF3967 Acetylase ? PBECR3 DUF2283 DUF4258 Peptidase_S8 ? MobA_MobL DUF3847 pre-PUA_NIP7 ? ? MerR_1 DUF3967 Acetylase ? PBECR3 DUF2283 DUF4258 Peptidase_S8 ? MobA_MobL DUF3847 pre-PUA_NIP7 ?

WP_101560438.1
NZ_CAJHLG010000003.1:119879-133609
Aerococcus mictus

? Gp58 ? TSC22 CBM_5_12 ? Phage_holin_Dp1 Glyco_hydro_25 CHAP PBECR3 DUF4177 Endonuclea_NS_2 O-antigen_lig Acetylase DUF2382 Peptidase_M42 AMPT ? Abhydrolase_1

WP_317317377.1
NZ_CAUWFI010000006.1:40627-51289
Longibaculum muris

Hydrolase_3 Hydrolase_4 ABC_trans_CmpB Fer4_12 NRDD ATP-cone PTS-HPr ? PBECR3 Aldo_ket_red MerR_1 HxlR ? ? ABM ?

WP_066719988.1
NZ_FJVE01000007.1:2339514-2358524
Clostridium sp. Marseille-P299

Prenyltransf CTP_transf_1 MutS_III MutS_V DXP_reductoisom DXP_redisom_C DXPR_C Peptidase_M50 GcpE GcpE_C DNA_pol3_a_NI DNA_pol3_a_NII N-OB PHP RNase_T PHP DNA_pol3_alpha DNA_pol3_alpha DNA_pol3_finger HHH_6 PBECR3 ? Acetyltransf_7 Polysacc_deac_1 DNA_ligase_A_M SpoIID YybI_C DUF6471

WP_311696027.1
NZ_CAUPPJ010000015.1:16463-28787
Sneathia sanguinegens

PBECR3 ? Peptidase_S8 PATR Autotransporter ? HAD TK ? ABC_tran

WP_277283996.1
NZ_CAMYDT010000011.1:16616-37025
Sneathia sanguinegens

adh_short_C2 Na_H_Exchanger Ribonuc_red_sm Flavodoxin_NdrI RNR_N Ribonuc_red_lgN Ribonuc_red_lgC Rce1-like ABC_tran ABC_tran_Xtn ABC_tran Phage_Mu_F PBECR3 ? Peptidase_S8 PATR Autotransporter ? HAD TK ? ABC_tran

WP_066825702.1
NZ_LTBA01000021.1:26881-40798
Clostridium tepidiprofundi DSM 19306

DDE_Tnp_2 PH_36 NTP_transf_2 AadA_C Glyoxalase Lactamase_B HAD_2 MerR_1 Methyltransf_11 DUF7588 AAA-ATPase_like PDDEXK_9 PBECR3 ABC_tran ? PDGLE CbiM SurA_N_3 DUF1659 DUF2922 Holin_BhlA

WP_136004752.1
NZ_SRYR01000001.1:806786-821296
Clostridium sartagoforme

BPD_transp_1 BPD_transp_1 ABC_tran PhoU PhoU MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_068989320.1
NZ_BMJN01000004.1:40224-58273
Streptococcus himalayensis

Sipho_tail SPP1_Dit Prophage_tail Collagen Collagen DUF859 ? ? Phage_holin_Dp1 ? PBECR3 Pep_deformylase S4 PseudoU_synth_2 HTH_3 DUF6773 Ribosomal_S15 Peptidase_M3 Peptidase_M3_N RNase_PH RNase_PH_C PNPase RNase_PH RNase_PH_C KH_1 S1

WP_175438261.1
NZ_BDHH01000002.1:55751-68567
Fusibacter sp. 3D3

? ? DDE_Tnp_1_6 ? CCDC172 DDE_Tnp_IS66_C DDE_Tnp_IS66 DDE_3 PBECR3 ? GyrI-like MerR_1 AbrB AbrB LacI Peripla_BP_1 PfkB RbsD_FucU ABC_tran ABC_tran

WP_070000874.1
NZ_FMMP01000028.1:56962-73009
Cellulosilyticum sp. I15G10I2

2-thiour_desulf DUF1722 MFS_1 MFS_1 VanW PG_binding_4 ? ? ZT_dimer Cation_efflux ? PBECR3 DeoC CheW Receiver tRNA-synt_His HGTP_anticodon N-OB tRNA-synt_2 C_GCAxxG_C_C ? Acetylase

WP_314692385.1
NZ_CAUTKC010000005.1:23467-39572
uncultured Abiotrophia sp.

RelA_SpoT NAD_kinase NAD_kinase_C PedC-like PseudoU_synth_2 ? CNNM CBS CBS CorC_HlyC AAA_33 PBECR3 QueH Acetylase GFO_IDH_MocA GFO_IDH_MocA_C HAD_2 APH Voltage_CLC TrkA_C AA_permease_2 Melibiase

WP_071177666.1
NZ_CP017832.1:113551-121983
Butyrivibrio hungatei

HTH_31 ? ? NikA-like Bac_DNA_binding ? ? PBECR2 ? HD_5 ? BHD_1 ? DnaB_2 ?

WP_072521935.1
NZ_DBFANU010000159.1:32477-43794
Blautia sp.

PBECR3 ? MshEN T2SSE T2SSE T2SSF_N T2SSF T2SSF N_methyl ? PilM_2

WP_083572651.1
NZ_FRDH01000004.1:68400-89961
Butyrivibrio hungatei DSM 14810

BPD_transp_1 BPD_transp_1 DUF3502 Glyco_hydro_35 BetaGal_ABD_1 BetaGal_ABD_1 BetaGal_gal-bd Glyco_hydro_2_N Glyco_hydro_2 Glyco_hydro_2_C Glyco_hydro2_C5 ? tRNA_U5-meth_tr PBECR3 NTP_transf_2 AadA_C Acylphosphatase Cas9-BH Cas9_REC HNH_4 Cas9_RuvC Cas9_PI Cas_Cas1 CRISPR_Cas2 Cas_Csn2 Myo5a HTH_7

WP_211182040.1
NZ_JABBJH010000024.1:137-8998
Megasphaera elsdenii

? ? Terminase_2 DUF2188 ? Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F PBECR2 DUF4355 Phage_capsid ?

WP_072984831.1
NZ_FQZB01000004.1:7474-20715
Clostridium cavendishii DSM 21758

Lactamase_B ACP_syn_III ACP_syn_III_C PP-binding ? ? Radical_SAM SPASM ACCA PBECR3 DUF2628 HTH_3 PLDc_N ABC_tran DrrA1-3_C ABC2_membrane_2 VanZ His-Kinase His-Kinase

WP_073088228.1
NZ_FRBC01000003.1:107577-123155
Selenomonas ruminantium

Flp_Fap TadE T2SSE T2SSF ? Tad DNA_topoisoIV DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C PBECR3 Nitroreductase Nitroreductase Malic_M malic ? LytR_cpsA_psr CYTH MgtC ACT_4 dCMP_cyt_deam_1

WP_073125375.1
NZ_BAABCH010000101.1:38454-63442
Asaccharospora irregularis DSM 2635

DUF4397 DUF4397 CspB_prodomain Peptidase_S8 Peptidase_S8 Peptidase_S8 Peptidase_S8 Peptidase_S8 Peptidase_S8 Peptidase_S8 Peptidase_S8 RNR_N Ribonuc_red_lgN Ribonuc_red_lgC Ribonuc_red_sm PBECR3 ? MazG Sirtuin SH3_3 SH3_3 SH3_3 SH3_3 Glucosaminidase Lepto_Lipo_YY_C DNA_pol3_a_NI DNA_pol3_a_NII N-OB RNase_T DNA_pol3_alpha DNA_pol3_alpha DNA_pol3_finger HHH_6 SDF

WP_159434055.1
NZ_FQXM01000006.1:119675-133727
Clostridium grantii DSM 8605

BioY Peptidase_M56 Penicillinase_R ArsP_1 ArsP_1 Cupin_2 cNMPBD HTH_Crp_2 PBECR3 ? ? ABC_tran SNF2-rel_dom Helicase_C ? PTA_PTB PTA_PTB Acetate_kinase

WP_073711606.1
NZ_MRWQ01000006.1:380816-399567
Domibacillus mangrovi

TGL SLATT_fungal RibD_C DinB_2 HpaB HpaB_N PAP2 FGGY_C FGGY_N PBECR3 ? Hpt Receiver GGDEF Receiver Receiver SosA 4HB_MCP_1 HAMP PAS His-Kinase His-Kinase Receiver SpoOE-like PBSX_XtrA Zn_ribbon_8

WP_136122087.1
NZ_CP042988.1:291185-302224
Histophilus somni

DUF2644 DUF2681 ArlX DUF1664 DUF1804 ? Portal_Mu Phage_Mu_F HI_1570 PBECR3 Mu-like_Pro Mu-like_gpT Phage_Mu_Gp36 Phage_Mu_Gp37 Phage_gp49_66 DUF2635 Phage_sheath_domII Phage_sheath_1 Phage_sheath_1C

WP_273061415.1
NZ_CAJMGV010000004.1:54625-72122
Megasphaera coli

YbjN Endonuclea_NS_2 ? DUF5758 Pps WG WG Pribosyltran ? PBECR3 DUF5545 HAMP His-Kinase His-Kinase Receiver Trans_reg_C PMT_2 TonB_dep_Rec_b-barrel Plug ? Malic_M malic

WP_077591409.1
NZ_CP019642.1:37949-51311
Planococcus lenghuensis

HTH_17 ADH_N_2 ADH_zinc_N Phage_integrase Ndc10_N TIR ? DUF4145 HTH_36 PBECR3 ? ? RadC ? ASCH DUF3810 ArdcN ?

WP_149026582.1
NZ_CP019698.1:661186-680034
Desulforamulus ferrireducens

ABC_tran His-Kinase His-Kinase Trans_reg_C Receiver MethyltransfD12 Receiver ? DDE_Tnp_1_4 PBECR3 IstB_IS21 Mu-transpos_C_2 rve DDE_Tnp_1 DUF772 DUF7226 DUF6997 DUF6996 MethyltransfD12 DDE_Tnp_1_4 ?

WP_077849626.1
NZ_LZZM01000223.1:9920-28014
Clostridium puniceum

Peptidase_M29 Cthe_2159 Cthe_2159 GGDEF PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C Pep_deformylase

WP_078275834.1
NZ_CAACXO010000006.1:1-8395
Moraxella caviae

SIS CBS CBS Hydrolase_3 LptC LptD_N ABC_tran BCA_ABC_TP_C HicB_lk_antitox HicB_lk_antitox Raco_middle PBECR3 ? HigB-like_toxin ? LptF_LptG Dus

WP_229143746.1
NZ_JAFLPI010000003.1:218-11300
Bacillus paranthracis

HTH_36 DDE_Tnp_ISL3 HTH_Tnp_ISL3 Zn_ribbon_ISL3 MarR_2 ? PBECR3 Phage_integrase ? ? ? ? ? ?

WP_205703621.1
NZ_CP178353.1:970045-989699
Butyricicoccus porcorum

RnfC_N Complex1_51K SLBB Fer4_10 NQR2_RnfD_RnfE FMN_bind Rnf-Nqr Rnf-Nqr FeS Fer4_10 Fer4_7 Fer4_7 Fer4_16 DUF4015 Phage_min_cap2 PBECR2 Receiver Trans_reg_C HAMP His-Kinase His-Kinase Trypsin PDZ_2 FGAR-AT_linker AIRS_C GATase_5 Ribul_P_3_epim SIS SIS GATase_6 DUF445 DUF445

WP_087247893.1
NZ_NFHK01000022.1:14917-25674
Faecalibacterium sp. An77

Band-7 Sigma70_r2 Sigma70_r4 ? ? Terminase_6N ? Phage_min_cap2 PBECR2 Phage_scaffold ? ? ? DUF6751 Minor_capsid_2 ?

WP_204245016.1
NZ_NFJQ01000002.1:152216-172077
Cloacibacillus sp. An23

PhdYeFM_antitox Fic ? Collagen Collagen Collagen PBECR3 LPD22 RelB PBECR3 ? T7_gp14 ? ? ? ? Bbp16

WP_089967957.1
NZ_FNJM01000003.1:139632-154914
Clostridium gasigenes

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C

WP_090012473.1
NZ_FPBY01000053.1:3293-20983
Clostridium sp. DSM 8431

ABC_tran PhoU PhoU MMR_HSR1 RsgA_GTPase MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C Aminotran_4 PBECR3 Asparaginase Asparaginase_C Xan_ur_permease ? Fib_alpha LytTR Peptidase_C11

WP_090090181.1
NZ_FOMG01000008.1:33921-49341
Clostridium uliginosum

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_279166590.1
NZ_CANBHC010000007.1:12717-26048
Thomasclavelia cocleata

? ? ? ? ? ? HTH_26 PBECR3 ? tRNA_edit CobA_CobO_BtuR PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV ? CBM_48 Alpha-amylase Alpha-amylase Alpha-amylase_C Phosphorylase

WP_092449332.1
NZ_FOJI01000001.1:6811-14951
[Clostridium] fimetarium

? ? Sirtuin DUF951 Ribosomal_S6 SSB Ribosomal_S18 PBECR3 ? ? HTH_17 HTH_3 HTH_3 ? DUF6462

WP_242871688.1
NZ_FNID01000009.1:10771-28587
Acetanaerobacterium elongatum

Spore_GerAC Spore_GerAC_N GerA AdoMet_dc OKR_DC_1 OKR_DC_1_C Spermine_synt_N Spermine_synth Arginase PALP PBECR3 Fer4 Peptidase_M56 Penicillinase_R Na_Ala_symp Methyltransf_11 S-methyl_trans DUF6434 DUF6434 SAP

WP_094545748.1
NZ_NPML01000002.1:95914-104974
Petroclostridium xylanilyticum

FokI_D3 WYL DUF429 ? ? ? ? nSTAND3 PBECR3 ? Peptidase_C39_2 Zn_Tnp_IS91 Y2_Tnp ? ? ?

WP_095267425.1
NZ_NPBY01000074.1:47667-52094
Paenibacillus campinasensis

PBECR3 ? ? ? SLAC1 Fer4_12 DUF2997 DUF1257

WP_098672793.1
NZ_NVEG01000002.1:84-7622
Priestia megaterium

? HTH_36 PBECR3 ? ? Peptidase_M78 HTH_3 DUF5986 FitA-like_RHH GerE PRP9_N

WP_316102203.1
NZ_JAWEFR010000028.1:3173-13836
Clostridium sp.

Sigma70_r2 Sigma70_r4 ? ? B1 ? ParA-Soj-NTPase ? PBECR3 HTH_3 FtsK_SpoIIIE ? ? ? ? HTH_17

WP_172692014.1
NZ_CP045599.1:292-7265
Enterococcus faecalis

HTH_10 L_lactis_RepB_C Rep3_C Rep3_N Resolvase EntA_Immun ? PBECR3

WP_099448799.1
NZ_PEDL01000004.1:68042-81629
Sporanaerobium hydrogeniformans

? Glyco_hydro_3 MacB_PCD FtsX MacB_PCD FtsX ABC_tran BSH_YknX HH_MFP_RND Beta-barrel_RND_2 2-thiour_desulf 4HBT PBECR3 DUF1858 adh_short_C2 Radical_SAM NMT1 BPD_transp_1 ABC_tran PadR

WP_100543367.1
NZ_PHQY01000612.1:42184-56429
Lysinibacillus xylanilyticus

? ? Phage_integrase ? RNase_T Abi_2 NPCBM PBECR3 ? Phage_integrase ? Acyl_LnsA DUF3139 ? Phage_integrase

WP_101771092.1
NZ_PKIC01000014.1:50-5128
Streptococcus parasanguinis

? Rep_trans FtsK_SpoIIIE PBECR3 NikA-like ? ? ?

WP_105406711.1
NZ_PKQK01000001.1:1702843-1722855
Paenibacillus sp. PCH8

LrgB LrgA HTH_1 LysR_substrate ABM DEAD DEAD_2 Helicase_C_2 Alpha-amylase Alpha-amy_C_pro ? PBECR3 ? Anti-Pycsar_Apyc1 UxaC BTAD Trans_reg_C Receiver His-Kinase His-Kinase PAS HisKA_7TM SLH SLH SLH Big Big MarR

WP_108307063.1
NZ_QAFW01000018.1:1-15324
Metalysinibacillus jejuensis

RAMPs RAMPs Cas10-Cmr2_palm2 DUF6602 DUF6602 CRISPR_Cas6 AF_1859_N Sulfate_transp Sulfate_transp STAS ? MUG113 PBECR3 Cas12f1-like_TNB

WP_118657849.1
NZ_JAWHPU010000005.1:95382-117163
Faecalibacterium prausnitzii

PEP-utilizers_C PEP-utilizers PPDK_N PPDK_N PPDK_N GMPPB_C Hexapep ? Pept_tRNA_hydro UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF ? Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N2 PBECR3 ABC_tran ABC_membrane ABC_tran ABC_membrane MarR_2 TOBE_2 OB_MalK ABC_tran HTH_18 MatE MatE PfkB

WP_112663052.1
NZ_QMKH01000003.1:107383-122544
Microvirga flavescens

? CreA Glyoxalase DUF2865 GST_C GST_N_3 Anticodon_2 tRNA-synt_1c NAD_synthase CN_hydrolase PBECR2 DAHP_synth_2 Pyr_redox_dim Pyr_redox_2 Wzy_C TctC TctB TctA DUF2059

WP_115639811.1
NZ_CATNXJ010000022.1:27218-38048
Clostridium perfringens

? ? ? ? ? ? CEP89 PBECR3 TIR GP3_package Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F DUF4355 Phage_capsid_4

WP_118276686.1
NZ_AP019695.1:44641-56494
Amedibacterium intestinale

ALAD HEM4 TP_methylase Porphobil_deamC Porphobil_deam GlutR_dimer Shikimate_DH GlutR_N Aldo_ket_red DUF2583 ? Phage_min_cap2 PBECR2 ? PGI S1 Hexapep LbH_EIF2B LUD_dom NifU DUF212

WP_116063532.1
NZ_QRDZ01000024.1:106614-125787
Cohnella phaseoli

MMPL MMPL TetR_N TetR_C_24 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 P5CR_dimer F420_oxidored HHH_3 Cys_Met_Meta_PP dCMP_cyt_deam_1 PBECR3 DUF4131 Competence Lactamase_B Sigma70_r2 Sigma70_r4_2 zf-HC2 DNA_pol3_delta DNA_pol3_delta_C Ribosomal_S20p Peptidase_A25 Peptidase_A25 SpoIIP

WP_116192902.1
NZ_QTTN01000082.1:373-1679
Paenibacillus taihuensis

HTH_36 PBECR3

WP_116354194.1
NZ_QUQV01000016.1:24348-33118
Bacillus sp. HNG

? ToxN_toxin Phage_integrase Phage_int_SAM_1 ? ? ? HTH_36 PBECR3 HTH_17 ? RH3_dom ParA-Soj-NTPase Sda ? ?

WP_144369234.1
NZ_CABHNB010000030.1:232320-244850
Blautia obeum

SBP_bac_11 BPD_transp_1 ABC_tran HTH_30 GGDEF_2 TOBE_2 OB_MalK ABC_tran ? ? PBECR2 HTH_3 NMT1 MatE MatE AraC_binding HTH_18 EamA EamA ? ?

WP_270831874.1
NZ_JAQEDG010000002.1:124520-145608
Faecalibacillus intestinalis

Phage_int_SAM_1 Phage_integrase Metalloenzyme DUF7888 ? Phage_integrase Phage_integrase ? PBECR3 ? ? ? Phage_integrase Phage_integrase PTS_2-RNA Metallophos_2

WP_119973145.1
NZ_CP032416.1:2106887-2122056
Clostridium fermenticellae

HTH_26 T1RH-like_C T1R_D3-like SWI2_SNF2 REase DUF4325 ? Fic Methylase_S Methylase_S N6_Mtase HsdM_N PBECR3 LD_cluster3 TIR TIR ? TIR TIR PG_binding_1 Amidase_2

WP_120465414.1
NZ_KZ987725.1:7823-20230
Paenibacillus aceti

MobA_MobL ? ? ? HTH_16 Sigma70_r4_2 YvrJ PBECR3 PBECR3 HTH_3 ? HTH_3 ? PBECR4 ParG Lmo0850

WP_120486365.1
NZ_SRZB01000045.1:1291-9007
Hominisplanchenecus murintestinalis

? ? ? HTH_3 HTH_3 HTH_26 HTH_26 PBECR3 T4_baseplate ? HTH_19 ? Bro-N ? Resolvase Recombinase Zn_ribbon_recom

WP_123054442.1
NZ_CP033465.1:2224892-2236593
Clostridium sp. JN-1

SpoOE-like ? B12-binding Radical_SAM NUDIX Rieske DAO ? NUDIX PBECR3 PolyA_pol PolyA_pol_RNAbd tRNA_NucTran2_2 CheW Receiver ? Sulf_transp ? ? DUF3189

WP_124068485.1
NZ_CBCRXF010000030.1:6560-16286
Filibacter tadaridae

? PBECR4 ? ? Sigma70_r4_2 YvrJ HTH_16 PBECR3 Phage_integrase ? ? ? SocA_Panacea HTH_31 ? DUF2239

WP_125005083.1
NZ_BHYK01000032.1:37637-47571
Clostridium tagluense

Phage_integrase Phage_int_SAM_1 DUF4158 DDE_Tnp_Tn3 Pox_D5 DUF5906 D5_N Toprim HTH_3 HTH_3 RE_CfrBI HTH_17 PBECR3

WP_127702582.1
NZ_RPNM01000118.1:1-643
Sinorhizobium meliloti

? PBECR3

WP_129612526.1
NZ_SEHK01000003.1:541421-558583
Bacillus infantis

YomX_N YomX_C YomW YomV_2nd YomU_Tube YomO YomN Phage_int_SAM_4 PBECR3 RHH_3 ? HNH_repeat PH_gly_zipper PhageMin_Tail PhageMin_Tail SLT Peptidase_M23 DUF6558 DUF6558_C

WP_132220496.1
NZ_SMDD01000001.1:1253033-1266769
Laceyella sacchari

FR47 Phage_int_SAM_1 Phage_integrase NMO Orn_Arg_deC_N ? ABC_tran UvrA_DNA-bind Acetylase PBECR3 ParE-like_toxin Resolvase HTH_7 ? GIY-YIG ? ? HIRAN

WP_133969589.1
NZ_LR215983.1:4688712-4700412
Eubacterium limosum

? SPOUT_MTase Aldedh Aldedh Hydrolase ABC_tran SMC_N SUFBD_core EamA EamA PBECR3 DegV ? ? Peptidase_M73 Peptidase_M73 ? ?

WP_134170083.1
NZ_SODD01000026.1:22333-32209
Breznakia blatticola

ParE_toxin PhdYeFM_antitox HTH_11 Mga PRD PTS_IIB PTS_EIIA_2 PTS_EIIA_2 PTS_IIB EIIC-GAT SIS_2 PBECR3 Na_Pi_cotrans Na_Pi_cotrans PhoU PhoU Receiver Trans_reg_C ?

WP_134687547.1
NZ_SDKD01000049.1:84-6230
Brevibacillus migulae

? DUF3967 MerR_1 ? PBECR3 VbhA RHH_5 Phage_P22_Y8K6 Cu_amine_oxidN1 ?

WP_135157222.1
NZ_JACBYB010000007.1:93019-110179
Gemella sp. GL1

Baculo_PEP_C Mce4_CUP1 CCDC39 ? CBM_4_9 Peptidase_S74 DUF1366 ? Phage_holin_5_2 ? PBECR3 CHAP Amidase_3 CW_7 NikA-like RepB-RCR_reg Phage_pRha Rhomboid ? PTS_EIIB PTS_EIIC

WP_137100569.1
NZ_CP039865.1:3479693-3492897
Phreatobacter aquaticus

? Phage_TAC_7 PhageMin_Tail Phage_attach ? ? ? PBECR3 ? ? ? Phage_integrase Phage_int_M Arm-DNA-bind_3 DNA_photolyase FAD_binding_7 CMAS PALP

WP_138192710.1
NZ_VBWP01000019.1:1933-12569
Culicoidibacter larvae

IMS_C YolD ? DUF1451 PBECR3 ? LRRs LRRs LRRs Flg_new Gram_pos_anchor Pectate_lyase_5 Big Bact_surface_Ig-like Bact_surface_Ig-like IstB_IS21 Mu-transpos_C_2 rve_3 HTH_7 DUF1451

WP_139492564.1
NZ_VCNA01000016.1:25381-40534
Brevibacillus dissolubilis

Peptidase_A24 Flp1_like T2SSF T2SSF T2SSE ParA-Soj-NTPase RcpC PBECR3 DUF4097 ? Pan_kinase QRPTase_C QRPTase_N Succ_DH_flav_C FAD_binding_2 AbrB FUR

WP_141430877.1
NZ_VATK01000001.1:224895-239801
Bacillus sp. 03113

ThiW GGDEF_2 HTH_30 AlaDh_PNT_N AlaDh_PNT_C GyrI-like DUF2252 MPI_cupin_dom PMI_typeI_cat Gly_kinase PBECR3 Peptidase_M20 YkzB MerR_1 ThrE ThrE_2 Pyr_redox_2 Lipase_GDSL_2

WP_191767722.1
NZ_JACSRA010000004.1:27859-42988
Clostridium cibarium

BPD_transp_1 ABC_tran PhoU PhoU MMR_HSR1 RsgA_GTPase MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C Aminotran_4 PBECR3 Asparaginase Asparaginase_C YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C

WP_243132450.1
NZ_CABHNE010000013.1:6228-15321
Mediterraneibacter gnavus

DUF6751 Minor_capsid_2 ? ? DUF6673 Phage_Gp15 Bro-N PBECR2 Rv1352 Rv1352 Mucin ? ? Bro-N ? ?

WP_144995033.1
NZ_VNJK01000007.1:14929-25030
Paenibacillus agilis

? DUF3560 ? MazE_antitoxin ? ? LPD28 PBECR3 ? ? DUF6573 ? ? ? ?

WP_145950345.1
NZ_VDYU01000024.1:1-12417
Paenibacillus sp. Y412MC10

Cu_amine_oxidN1 ? ? Phage_integrase Phage_int_SAM_2 HTH_ParB ParB_N ? CENP-I PBECR3 ? ? DUF3139 Acyl_LnsA MarR_2

WP_146818656.1
NZ_BJYA01000027.1:1-13451
Alkalibacillus haloalkaliphilus

HTH_16 Sigma70_r4_2 YvrJ Rep3_C Rep3_N Phage_integrase ? HTH_3 Resolvase PBECR3 ? ParA-Soj-NTPase ? PG_binding_1

WP_270467099.1
NZ_JAQDCV010000004.1:182860-188696
Butyribacter intestini

? ? ? ? ? ? ? PBECR3 DUF5454 ? ? FN3_7 Phage_integrase Phage_integrase HTH_26 PemK_toxin

WP_308456485.1
NZ_JAJEQM010000010.1:20243-36357
Hominilimicola fabiformis

Phage_sheath_domII Phage_sheath_1 DUF2001 Phage_TAC_5 ? YtzH ? LysM PBECR2 YQBQ DUF2577 Sheath_initiator Baseplate_J Baseplate_J_M Baseplate_J_C Phage_Mu_Gp48 ? ?

WP_149233967.1
NZ_JALJXJ010000015.1:83744-94305
Azospirillum lipoferum

? ANT Phage_pRha ? ? AIPR ? ? PBECR3 Peptidase_S24 HTH_3 ? ? ? ? Zn_ribbon_DnaG DUF7146 Toprim_3 AAA_25

WP_194252071.1
NZ_WIJP01000025.1:1-11117
Streptococcus mitis

NikA-like HOOK TfoX_N HOOK NikA-like MazE_antitoxin PemK_toxin ? ? PBECR3 ? Phage_integrase Phage_integrase Replic_Relax ?

WP_153723922.1
NZ_CP045875.1:151424-167823
Heliorestis convoluta

Sigma70_r4_2 HTH_3 HTH_3 Peptidase_S24 HHH_3 Lactamase_B Cu_amine_oxidN1 ? DUF3006 DndB PBECR3 ? DUF2325 Golgin_A5 HATPase_c_3 IstB_IS21 Mu-transpos_C_2 Phage_integrase Phage_int_SAM_4 ?

WP_154609128.1
NZ_JABFQS010000017.1:258-5322
Streptococcus zhangguiae

NikA-like RepL ? ? Rep_trans FtsK_SpoIIIE PBECR3

WP_158363693.1
NZ_JAOQKC010000012.1:38888-50017
Laedolimicola ammoniilytica

? HTH_3 HTH_3 ? HTH_8 HTH_3 Peptidase_M78 PBECR3 ? Resolvase Recombinase DegV MarR_2 Aldose_epim Form_Nir_trans DUF7577

WP_158412723.1
NZ_CP077151.1:1-10656
Fusobacterium animalis

? ? ? Exonuc_VII_S Mob_Pre Mob_Pre ? Sigma70_r4_2 PBECR3 ? HTH_28 Resolvase ? ? PhdYeFM_antitox ? ?

WP_236570684.1
NZ_WTWX01000055.1:303-3119
Paenibacillus sp. An7

PBECR3 NAGPA ? HTH_16

WP_291501935.1
NZ_JAAWMX010000002.1:129414-143161
Acutalibacter sp.

Melibiase_C Melibiase_2 Melibiase_2 GFO_IDH_MocA_C3 GFO_IDH_MocA Ni_insertion Peptidase_M18 ? YbjQ_1 Xan_ur_permease PBECR2 HATPase_c_3 HSP90 HSP90 Rce1-like MafB19-deam DUF7537 DUF4430 ? VanZ PP-binding

WP_163411618.1
NZ_CP066204.1:1052033-1066540
Tuanshanicoccus yangjingiae

? Gp58 CBM_4_9 Gp58 Gp58 Gp58 Peptidase_S74 NNH3 ? CD1375-like ? Phage_holin_5_2 PBECR3 Glucosaminidase Amidase_5 SH3_5 DUF4300 Endonuclea_NS_2 Acetylase Glycos_transf_2 RNase_Y_N KH_1 HD HTH_18

WP_169103623.1
NZ_JABAGC010000002.1:2148-17440
Clostridium sp. SM-530-WT-3G

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_169295780.1
NZ_JABBNI010000001.1:63950-74154
Clostridium muellerianum

? SLFN_AlbA_2 ? cREC_REC ? ? ? PBECR3 HTH_19 ? ? Phage_ProQ_C_like ? ? DUF5659

WP_294403638.1
NZ_CAKSOA010000014.1:49239-66309
uncultured Ruminococcus sp.

? DUF3987 ? RepB-RCR_reg ParE_toxin ? UPF0150 PBECR3 Phosphorylase Alpha-amylase Dockerin_1 CTP_synth_N GATase Receiver Trans_reg_C HAMP His-Kinase His-Kinase RecR_C Toprim_4 RecR_ZnF RecR_HhH

WP_293726738.1
NZ_CABRJQ010000001.1:137710-151699
uncultured Phascolarctobacterium sp.

BPD_transp_1 ABC_tran ? FlK AMP-binding AMP-binding_C NAD_synthase Usp PBECR3 Amidohydro_3 Creatininase EamA EamA HTH_30 GGDEF_2 PucR DUF190 DUF697 FAD-depend_C Pyr_redox_2

WP_179942501.1
NZ_JACBYC010000083.1:76045-87732
Gemella sp. GH3

? Hydrolase_3 MarR Nitroreductase DUF1304 SSB SSB PBECR3 FolB HPPK Dus PK PK_C Phage_integrase Phage_int_SAM_3 Arm-DNA-bind_4 DNA-PKcs_N A_deaminase

WP_183564821.1
NZ_BAAIEK010000003.1:237682-254913
Methylorubrum rhodinum

DUF6111 PolyA_pol PolyA_pol_RNAbd ATP-synt_DE_N ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_N ATP-synt ATP-synt_ab_C ATP-synt_ab ATP-synt_ab_N OSCP PBECR3 Ferrochelatase Methyltransf_23 Calcineurin 5_nucleotid_C UDG ? Rotamase_3 SecA_DEAD P-loop_SecA SecA_SW SEC-C

WP_186908846.1
NZ_JACOPP010000035.1:1-9151
Lawsonibacter hominis

? Terminase_3 Terminase_3C Spc42p SPP1_portal Phage_min_cap2 PBECR2 Phage_scaffold SU10_MCP ? ? Minor_capsid_2 ? ?

WP_257916836.1
NZ_JAMDLQ010000032.1:97289-112048
Clostridium sp. LY3-2

DivIVA S4 PseudoU_synth_2 Pribosyltran RlmL_1st THUMP UPF0020 NFACT-R_1 NFACT_N DAP_epimerase DAP_epimerase DUF4969 Tricorn_2nd PBECR3 Asparaginase Asparaginase_C Xan_ur_permease SdpI GNAT_acetyltran ? Peptidase_M20 ?

WP_197265499.1
NZ_JACZDX010000018.1:314-11576
Fictibacillus sp. 23RED33

? Lin1244_N ? Phage_int_SAM_1 Phage_integrase DUF3967 MerR_1 PBECR3 ? ? ? HTH_17 MMS19_N ? ?

WP_292224932.1
NZ_JABAZW010000035.1:1-3084
Brevundimonas sp.

Bac_DnaA_C Receiver Trans_reg_C PBECR3 His-Kinase His-Kinase HAMP

WP_209301936.1
NZ_CAXSPF010000044.1:212-10193
Anaerostipes hadrus

Resolvase ? ? NOP19 SprT-like ParA-Soj-NTPase ? PBECR3 RH3_dom ? ? NUMOD4 HNH_3

WP_209782791.1
NZ_JAGINM010000038.1:28961-41985
Azospirillum agricola

HEPN_AbiU2 LPG_synthase_TM SBP_bac_3 ? SRAP ? HTH_36 PBECR3 Phage_integrase ? HTH_ParB ParB_N HTH_31 ? ? HTH_3 ?

WP_317243392.1
NZ_CP136628.1:3058573-3076940
Bradyrhizobium sp. NDS-1

DUF6894 DUF2798 ? ? tRNA_anti-like WcbI HARE-HTH PBECR3 RecA_N RecA_C ? GcvP2_C GDC-P GDC-P GCV_H GCV_T_C GCV_T tRNA-synt_2c tRNA_SAD DHHA1 Ion_trans cNMPBD

WP_249209526.1
NZ_JAFCKA010000001.1:141265-154580
Bradyrhizobium manausense

Fasciclin ? ? ? Glyco_hydro_19 Pantoate_ligase DpnII-MboI PBECR3 ? Polbeta HTH_37 ? ? NTP_transf_2 HTH_17

WP_212695543.1
NZ_CP058649.1:5355875-5369528
Vallitalea pronyensis

? KbaA ? ? ? HTH_3 DUF2726 PBECR3 Radical_SAM B12-binding ? HTH_3 Aminotran_3 Inositol_P ADH_N Glu_dehyd_C Aminotran_3

WP_216462406.1
NZ_CAKVJM010000005.1:4400-17940
uncultured Clostridium sp.

Zn_ribbon_8 tRNA_U5-meth_tr TRAM ? ? ? Glucosaminidase ChW ChW ChW ChW ChW SMI1_KNR4 PBECR3 PYNP_C Glycos_transf_3 Glycos_trans_3N PNPase PNPase Metalloenzyme Phage_integrase Phage_int_SAM_1 SpoIIM NUDIX NDPSase_N

WP_320075188.1
NZ_CP132136.1:4333531-4344425
Eubacterium callanderi

ArdA YodL ? ? ? ? ParA-Soj-NTPase PBECR3 ? ? Sortase SpaA GramPos_pilinD1 SpaA SpaA Toprim_2 DUF3991 Zn_ribbon_DnaG

WP_216563276.1
NZ_JAHLQE010000007.1:79-6165
Eubacterium sp. MSJ-13

Ribonuclease_3 SpoU_sub_bind SpoU_methylase PBECR2 Sigma70_r2 Sigma70_r4_2 HTH_3 DUF2007 GGDEF

WP_217959036.1
NZ_CAJTCQ010000005.1:368-12902
Acutalibacter muris

Resolvase Recombinase Zn_ribbon_recom ArdA T4SS_LegC2C7 ? PAPS_reduct DUF3849 YodL DUF3846 PBECR2 ? Fic ? ?

WP_226395690.1
NZ_CALKQL010000020.1:31432-44480
Lachnoclostridium phocaeense

HTH_18 Aldo_ket_red Aldo_ket_red FMN_red Flavodoxin_4 Glyco_transf_8 DUF4097 PBECR3 ? DLH BD-FAE LysR_substrate HTH_1 Receiver MarR_2 HEPN NTP_transf_2

WP_233055303.1
NZ_JAJTJP010000001.1:96296-108381
Parvimonas micra

SMC_ScpA SMC_ScpB S4 PseudoU_synth_2 SH3_14 rRNA_methylase CRCB ? PBECR3 ? Aminotran_1_2 Asp-Al_Ex TrkA_C Asp-Al_Ex ? TMEM100 MMR_HSR1 RNase_HII

WP_282924423.1
NZ_CATKPB010000001.1:297570-301064
Peptoniphilus genitalis

? ? XhlA NLPC_P60 SH3_3 ? Phage_holin_6_1 ? PBECR3

WP_240750356.1
NZ_JAKVTG010000015.1:1-14547
Planococcus halocryophilus

? MFS_1 Peptidase_M3 SEC-C AP_endonuc_2 SUKH_6 SMI1_KNR4 PBECR3 HTH_7 Resolvase Methyltransf_11 PRK ? Mrr_cat ? Gp49

WP_244833231.1
NZ_CP094971.1:1664688-1678740
Clostridium sp. BJN0001

BPD_transp_1 ABC_tran PhoU PhoU PDZ_6 Radical_SAM_2 DUF512 MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Xan_ur_permease MgtC YicC-like_N YicC-like_C RemA-like Guanylate_kin RNA_pol_Rpb6 Flavoprotein DFP

WP_257530922.1
NZ_JANKAS010000006.1:113274-129583
Irregularibacter muris

SpoVG NTP_transferase LbH_EIF2B GMPPB_C Hexapep Pribosyltran_N Pribosyl_synth Trypsin PGA_cap Pept_tRNA_hydro ? PBECR3 ? UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF SurA_N_3 Rotamase_2 MazE_antitoxin SpoVT_C Polysacc_synt MatE TP_methylase MazG MazG Bac_DNA_binding

WP_258886585.1
NZ_CP096972.1:415795-431000
Sphingomonas sp. SUN039

NAD_binding_1 FAD_binding_1 Flavodoxin_1 p450 Aminotran_1_2 OmpA OMP_b-brl DUF2793 Rcc01698_C Phage-tail_3 DjlA_TM NLPC_P60 Phage_BR0599 Phage_phiJL001_Gp84_N PBECR3 DUF2460 TIM-barrel_NCTSP NCTSP_N ? Phage_TAC_6 Phage_TAC_11 Phage_tail_2 ? HCP

WP_261830534.1
NZ_BROD01000001.1:4510896-4526416
Inconstantimicrobium mannanitabidum

SIS SIS GATase_6 PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I ? SHOCT-like_DUF2089-C SHOCT-like_DUF2089-C DUF2089 Zn_ribbon_DUF2089 Bactofilin Bactofilin DUF4004 PBECR3 Trp_repressor AIM24 DUF3784 Acetate_kinase PTA_PTB PTA_PTB FAD-depend_C HI0933_like DUF8436 YbbR YbbR YbbR YbbR

WP_262575055.1
NZ_JAOQKJ010000008.1:29942-48428
Suilimivivens aceti

Peptidase_M20 ? DUF5058 ? HTH_18 AraC_binding MatE MatE Hydrolase_3 PBECR2 His_Phos_1 MatE MatE ? Transpeptidase Transgly Impact_N DUF1949 MgtE_N CBS MgtE BipA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU

WP_265473903.1
NZ_CP022012.1:1474143-1484796
Mergibacter septicus

DUF2644 DUF2681 ? ? DUF1804 ? Portal_Mu Phage_Mu_F HI_1570 PBECR3 Phage_tail_S Mu-like_Pro Mu-like_gpT FluMu_N Phage_Mu_Gp36 Phage_Mu_Gp37 DUF2635

WP_265474013.1
NZ_CP022012.1:1609322-1633520
Mergibacter septicus

oligo_HPY ABC_tran oligo_HPY ABC_tran BPD_transp_1 OppC_N BPD_transp_1 BPD_transp_1_N SBP_bac_5 PRODH Pro_dh-DNA_bdg Pro_dh Aldedh SSF PBECR3 Phage_tail_S Cas_Csy4 Cas_Csy3 Androglobin_V Cas_Csy3 Cas_Csy2 Cas_Csy1 Cas3-like_C Cas3-like_C_2 Cas3_HD Cas3_I-F_Cas2 Cas_Cas1

WP_268041984.1
NZ_JAPQER010000007.1:81992-102104
Clostridium aestuarii

? Cas_TM1802 Cas_Cas7 Cas_Cas5d Cas3_HD DEAD Cas3-like_C_2 Cas6b_N Cas6b_C ? PBECR3 ? RVT_1 GIIM PDDEXK_2 ? ? CARF_Card1 Card1_endonuc CARF_Card1

WP_268069333.1
NZ_JAPPRQ010000004.1:144-10874
Clostridium sp. ZS2-4

? HTH_7 ? Synaptojanin N6_Mtase PBECR3 Eco57I TaqI_C DUF1659 ?

WP_269920357.1
NZ_JANUVP010000081.1:40-7857
Caldifermentibacillus hisashii

? Rubis-subs-bind ? PBECR3 MA3 MerR_1 DUF3967 TnpV RHH_3 DUF5511 DUF8208 ?

WP_270942953.1
NZ_JAPNNV010000015.1:41722-65912
Romboutsia lituseburensis

GDP_Man_Dehyd YkuD Choline_bind_3 Choline_bind_1 Choline_bind_3 Choline_bind_3 Choline_bind_3 CW_binding_2 CW_binding_2 CW_binding_2 DUF2231 Sulfatase TIG CW_binding_2 CW_binding_2 CW_binding_2 CW_binding_2 CW_binding_2 CW_binding_2 CW_binding_2 CW_binding_2 CW_binding_2 PIG-L PBECR3 CW_binding_2 CW_binding_2 CW_binding_2 DUF6080 UbiA HAD GtrA_DPMS_TM VTT_dom Na_Ala_symp Choline_bind_1 Choline_bind_3 Choline_bind_2 CAP

WP_272435957.1
NZ_JAMQKB010000004.1:128840-148699
Terrihalobacillus insolitus

HTH_1 LysR_substrate TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C CRWN_N AA_permease_2 Xan_ur_permease AAA_33 ? PBECR3 Bflower DDE_Tnp_1 DDE_Tnp_1_assoc DruA PAP2 RlmN_N Radical_SAM Fer2_4 NADH-G_4Fe-4S_3 Fer4_7 Molybdop_Fe4S4 Molybdopterin Molydop_binding DUF1641

WP_274397719.1
NZ_JARAJZ010000001.1:466153-476645
Sporolactobacillus sp. CQH2019

? ? N6_Mtase ? ? ? ? PBECR3 DUF6558 DUF6558_C UFC1 HNH Terminase_4 TerL_ATPase TerL_nuclease Phage_portal Peptidase_S78

WP_274776283.1
NZ_JARBFX010000006.1:151180-170130
Aquabacter sediminis

DnaA_N Bac_DnaA Bac_DnaA_C DNA_pol3_beta DNA_pol3_beta_2 DNA_pol3_beta_3 SMC_N YkuD LMWPc LysE His-Kinase DNA_gyraseB Toprim GyrB_hook GyrB_insert DNA_gyraseB_C PBECR3 ? DAO ABC_membrane ABC_tran Peptidase_S11 cobW CobW_C HI0933_like GGDEF

WP_278838137.1
NZ_JABUTL010000019.1:14193-24338
Fusobacterium varium

? ArdcN MPTase-PolyVal HTH_28 ? ? DUF7343 PBECR3 Phage_integrase ? ? ? ? Rep3_C Rep3_N TraD

WP_280789114.1
NZ_JARXWM010000036.1:5094-13429
Paenibacillus sp. PastF-3

? ? PBECR3 ? Glyco_trans_A_1 PKD_4 ? ? ? ?

WP_281115076.1
NZ_CP123966.1:3702499-3722199
Lysinibacillus sp. 1 U-2021

? HTH_26 HTH_26 ? YomO DUF4379 DUF4379 DUF4379 DUF3800 PBECR3 Seryl_tRNA_N PhageMin_Tail ? ? PhageMin_Tail Peptidase_M23 Phage_lysozyme2 ? ? ?

WP_285252811.1
NZ_JASPDG010000006.1:87-11762
Granulicatella sp. UMB5615A

? ? Receiver LytTR HATPase_c_5 ABC_tran ABC2_membrane ? PBECR3 ABC_tran CbiQ PDGLE CbiM Acetyltransf_CG MTS Sortase ZT_dimer Cation_efflux

WP_288222529.1
NZ_CABQTX010000022.1:4244-13851
uncultured Clostridium sp.

HTH_17 GIY-YIG ? Resolvase ? ? HTH_36 PBECR3 Phage_terminase Phage_integrase ? DnaJ HTH_23 DUF5381

WP_288775734.1
NZ_CAKRIM010000042.1:7623-18092
uncultured Eubacterium sp.

Big_3_4 ? Terminase_2 ? Peptidase_M13 Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F PBECR2 RepB-RCR_reg ? AntA ORF6C Phage_scaffold Phage_capsid_4 ? Phage_HCP

WP_296118380.1
NZ_CAKRAS010000047.1:166-19761
uncultured Eubacterium sp.

dCache_1 GGDEF dCache_1 GGDEF Receiver HD_5 GGDEF DAPG_hydrolase HTH_1 LysR_substrate PBECR2 CitMHS PrpR_N PAS NtrC-AAA AAA_lid_14 HTH_7 HTH_18 LeuA_dimer HMGL-like VanZ CoA_transf_3 HMGL-like

WP_289119977.1
NZ_CASYWU010000111.1:59-2611
uncultured Bacteroides sp.

ABC_tran ? PBECR3 Phage_integrase

WP_289336719.1
NZ_JAUCEZ010000003.1:3440-12256
Peribacillus simplex

Phage_integrase HTH_36 PBECR3 MobA_MobL DUF3847 ? ? MobA_MobL MobA_MobL 60KD_IMP

WP_291635143.1
NZ_JAENWL010000147.1:792-7809
Clostridium sp.

Rep3_N Rep3_C HTH_36 PBECR3 ParA-Soj-NTPase ? ? ? Resolvase HTH_11 ?

WP_291637520.1
NZ_JAENWL010000364.1:577-3662
Clostridium sp.

PBECR3 Phage_integrase Phage_int_SAM_1 SHOCT

WP_293658926.1
NZ_JAOART010000124.1:1-9393
Anaeromicrobium sp.

? DUF6892 DUF1963 Imm17 NADase_transloc AAA_18 DUF2500 PBECR3 DUF4357 OST-HTH NYN ? VanZ GIIM DUF4261

WP_314796428.1
NZ_CAUVJE010000009.1:24014-41747
uncultured Selenomonas sp.

NAD_synthase tRNA_edit Thr_synth_N PALP THR4_C PEPCK_ATP RecG_wedge DEAD Helicase_C RecG_dom3_C DapB_N DapB_C Semialdhyde_dh Semialdhyde_dhC PBECR2 MraZ MraZ Methyltransf_5 DivIC Mur_ligase Mur_ligase_M Mur_ligase_C Mur_ligase Mur_ligase_M Mur_ligase_C MraY_sig1 Glycos_transf_4 MurD-like_N Mur_ligase_M Mur_ligase_C

WP_294350809.1
NZ_CAKVLD010000012.1:18334-32034
uncultured Clostridium sp.

ABC_tran PhoU PhoU ? MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C Aminotran_4 PBECR3 Asparaginase Asparaginase_C Xan_ur_permease ? Acetylase HTH_18 Cass2 dCMP_cyt_deam_1 DUF1653

WP_294409267.1
NZ_CAKSNX010000010.1:38642-57359
uncultured Ruminococcus sp.

Ac81 XhlA Glyco_hydro_25 LysM Abi_2 Mn_catalase PBECR3 PBECR2 PBECR2 ? S-methyl_trans Pterin_bind B12-binding_2 B12-binding ? ADDB_N PDDEXK_1 UvrD-helicase UvrD_C PDDEXK_1 Radical_SAM_N2 Radical_SAM

WP_295760689.1
NZ_CAKVLF010000126.1:49-1158
uncultured Holdemanella sp.

? PBECR3 ?

WP_311538060.1
NZ_CAUPMI010000015.1:42986-54240
uncultured Anaerococcus sp.

Molybdop_Fe4S4 DUF1667 Peptidase_S8 MIP Fer2_BFD DAO Zn_Ribbon_1 Alpha-amylase Alpha-amylase Alpha-amylase_C_2 ? PBECR3

WP_297571044.1
NZ_CAMVFT010000124.1:1541-5771
uncultured Anaerovibrio sp.

? ? ? ? HicB HTH_3 HTH_3 PBECR3 Trypsin

WP_299447358.1
NZ_CALIFJ010000017.1:44561-55562
uncultured Phascolarctobacterium sp.

Ala_racemase_N Ala_racemase_C TMP-TENI BATS Radical_SAM ThiG ThiF ThiS SLH PBECR2 ? HepT-like Polbeta SCIFF ? Pro_isomerase GIY-YIG

WP_300207914.1
NZ_JAQVBV010000054.1:8023-19860
Bacteroides sp.

PBECR3 ? JetA DUF4194 AAA_29 SbcC_Walker_B DUF3322 JetD_C DUF499 Fn3_assoc DUF3780

WP_300383133.1
NZ_JARIDN010000204.1:1-11917
Clostridium sp.

BPD_transp_1 BPD_transp_1 ABC_tran PhoU PhoU MMR_HSR1 MMR_HSR1 KH_dom-like NAD_Gly3P_dh_N NAD_Gly3P_dh_C SpoIVA_ATPase SpoIVA_middle SpoIVA_C PBECR3 Asparaginase Asparaginase_C Xan_ur_permease YicC-like_N YicC-like_C RemA-like

WP_302488304.1
NZ_CAUCIZ010000062.1:6494-8883
uncultured Ruminococcus sp.

? ? ? ? ? HIRAN HTH_26 PBECR3

WP_304987676.1
NZ_BAAACR010000002.1:75307-89078
Selenomonas dianae

? Zn_ribbon_Brz ? DUF134 Nitro_FeMo-Co Flavodoxin_3 Radical_SAM HemN_C NUDIX_4 HhH-GPD PBECR2 Seryl_tRNA_N tRNA-synt_2b Flavoprotein PagL Cauli_VI RNase_H MlaE ABC_tran MlaD Helical_needle Aq_493_lipo

WP_306978411.1
NZ_JAUSTQ010000021.1:1-16833
Alkalibacillus salilacus

MazE_antitoxin rve_2 ? Phage_integrase ? PBECR3 Nbl1_Borealin_N ? Trypsin RVT_1 Phage_integrase ? Rep3_C Rep3_N Ninein_long_CC

WP_307254914.1
NZ_JAUSUV010000021.1:11000-20843
Croceifilum oryzae

? ? PBECR3 HTH_31 ? ? Asn_synthase GATase_7 HTH_1 LysR_substrate ? Peptidase_M6 InhA-like_VEG

WP_312613786.1
NZ_DAMBXZ010000011.1:15468-30770
Oscillibacter sp.

? ? NrS1-1_pol-like_HBD VapE-like_dom ? ? DUF4355 S-l_SbsC_C PBECR3 Peripla_BP_3 LacI DctM DUF3394 NMT1_3 PrpF CoA_transf_3 HMGL-like Aconitase Aconitase

WP_312814842.1
NZ_DAMBRP010000012.1:88740-104702
Sedimentibacter sp.

HisKA_3 His-Kinase Receiver GerE Hol_Tox ? ABC_tran DrrA1-3_C ABC2_membrane_3 ABC2_membrane_3 PBECR3 RNA_helicase LonC_helical AAA_32 LonB_AAA-LID Lon_C CinA CinA_KH MoCF_biosynth Peptidase_M14 DUF308 SLH SLH

WP_314407412.1
NZ_CAURMG010000035.1:1-6958
uncultured Gemella sp.

SSSPR-51 LZ_Tnp_IS66 PBECR3 Lactamase_B NfeD_membrane NfeD FloA FUT8_N_cat ? ? Glycos_trans_3N Glycos_transf_3 PYNP_C

WP_314349324.1
NZ_CAUQVI010000019.1:14816-33067
Granulicatella elegans

ABC_membrane ABC_tran PDDEXK_2 Tex_N Tex_central_region Tex_YqgF HHH_3 HHH_9 S1 DUF1507 FTSW_RODA_SPOVE Biotin_carb_N CPSase_L_D2 Biotin_carb_C HMGL-like PYC_OADA Biotin_lipoyl DUF2129 PBECR3 Cons_hypoth95 CTP_transf_like PDZ_2 Lon_C SLBB HHH_3 Competence DNA_pol3_delta DNA_pol3_delta_C

WP_315451645.1
NZ_CAUVVR010000002.1:142973-165306
uncultured Selenomonas sp.

RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 RNA_pol_Rpb1_5 BioY SDF RmlD_sub_bind His-Kinase His-Kinase HAMP Trans_reg_C Receiver TP_methylase PBECR2 tRNA-synt_1g tRNA-synt_1g Anticodon_3 TatD_DNase DUF348 3D MM_CoA_mutase MM_CoA_mutase B12-binding MeaB PMSR

WP_315006547.1
NZ_CAUPZT010000002.1:59192-73803
uncultured Granulicatella sp.

DHFR_1 UDG Sigma54_activ_2 Endonuclea_NS_2 Adhesin_P1_N Gram_pos_anchor ABC_tran ? DUF7330 PBECR3 ? LicD LicD Polysacc_synt Glycos_transf_2 ABC_membrane ABC_tran HicA_toxin

WP_317982488.1
NZ_BTCL01000051.1:1233-14473
Paenibacillus glycanilyticus

DUF3006 ? ? ? Peptidase_M23 DUF8128 HTH_36 PBECR3 MerR_1 ? ? TIR DNA_ligase_A_C DNA_ligase_A_M YhjD GP3_package

WP_319566900.1
NZ_OY762533.1:137603-153332
Cohaesibacter marisflavi

rve DUF1153 DUF4440 COesterase COesterase DUF4172 Fic DIP2311-like_C ? Resolvase HTH_7 PBECR3 SbcC_Walker_B AAA_23 SbcD_C Calcineurin WHD_BrxR WYL BrxR_CTD WGR ? Xre_MbcA_ParS_C Xre-like-HTH HipA_C Couple_hipA

WP_321076102.1
NZ_JAXFDL010000031.1:137-7734
Gemmiger sp.

Terminase_6N SPP1_portal VG15 PBECR2 Phage_scaffold Phage_capsid_4 ? ? ? ? Phage_TTP_17

WP_321033314.1
NZ_JAXFJB010000004.1:1-4624
Lentihominibacter sp.

HTH_3 PBECR3 ? ? ? ? PT-HINT TEN_YD-shell

WP_321123232.1
NZ_JAXGAM010000126.1:245-11933
Treponema sp.

? Cytidylate_kin2 FliG_N GidB FliG_C ? ? PBECR3 ? SSNA1 MYPU_1760 LacI Peripla_BP_3 GH94_b-supersand Glyco_hydro_94

WP_323415753.1
NZ_JAYFQK010000054.1:7150-25075
Desulfitobacterium sp.

EAL HDOD 4HB_MCP_1 HAMP MCPsignal Cytochrome_C7 Cytochrom_C552 NIT HAMP MCPsignal Fe_dep_repr_C Fe_dep_repress Nramp Cation_efflux ZT_dimer PBECR3 Nitroreductase Nitroreductase Glycos_trans_3N Glycos_transf_3 IGPS PRAI PRAI PALP Trp_syntA Anth_synt_I_N Chorismate_bind

WP_323432696.1
NZ_JAYFPJ010000108.1:73-4689
Lutispora sp.

DUF4160 DUF2442 PBECR3 DUF3795 ? rve_3 rve HTH_21 HTH_Tnp_1

WP_326907874.1
NZ_CP142396.1:2335676-2350050
Sedimentibacter sp. MB31-C6

GGDEF DUF2284 PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran Hydrolase_3 PBECR3 SBP_bac_3 PAS His-Kinase His-Kinase Receiver NtrC-AAA AAA_lid_14 HTH_8 Na_H_antiporter Na_H_antiporter OrtA PALP ? ?